

The SPELL Expression Archive: Publications and Links

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The SPELL archive distributed by the *Saccharomyces* Genome Database holds 603 yeast expression studies, one publication each. Table 5 lists every one with its DOI, PubMed and PubMed Central links and its GEO series. Its first 48 rows are the studies that add a perturbed genotype to the built Kemmeren 2014 and Sameith 2015 datasets, most new genotypes first; the rest follow, most conditions first. Table 2 ranks the 107 studies with a gene deletion in their condition labels by the number of genes deleted. 88 studies name no GEO series, and Table 4 lists those separately because their data location has to be found in the paper.

Tables generated by `experiments/015-spell/scripts/spell_publication_tables.py`

Manifest: `experiments/015-spell/results/spell_publications.csv`

Deletion calls: `experiments/015-spell/results/spell_knockout_conditions.csv`

Working notes: `notes/experiments.015-spell.publications.md`

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 What the archive holds ✗

The archive is one file, `all_spell_datasets.tar.gz`, from [the SGD archive server](#). It unpacks to 603 study folders named for the first author, the year and the PMID, and every PMID is distinct. Each folder holds a README and one or more PCL files, a PCL file being a tab-separated matrix of genes by conditions. Across the archive there are 778 PCL files and 16,160 conditions, and the publication years run from 1997 to 2023 (Table 1).

Table 1: Studies in the SPELL archive by whether the study README names a GEO series. *PCL datasets* counts the dataset rows in the READMEs, one per PCL matrix file. *Conditions* sums the # `conditions` field over those rows. *DOI* and *PMC id* count the studies whose PubMed record carries one. *Years* is the range of publication years in the folder names.

	studies	PCL datasets	conditions	DOI	PMC id	years
GEO accession in README	515	648	13,919	514	367	2000–2023
no GEO accession (GEO ID: N/A)	88	130	2,241	88	42	1997–2006
all	603	778	16,160	602	409	1997–2023

1.1 Where each field comes from ✗

The PMID, the GEO series and the dataset table are read from the study README. The README carries no DOI, so the DOI, the PMC id and the journal citation come from the PubMed record for that PMID, retrieved through the NCBI E-utilities `esummary` endpoint and stored beside the manifest in `experiments/015-spell/results/spell_pubmed_esummary.json`.

602 of the 603 PubMed records carry a DOI. The remaining 1 is linked by its PubMed record alone. 409 records carry a PMC id, which means PubMed Central holds a full text. Whether that copy is in the open-access subset served by the PMC OA API was not checked.

The GEO field of a README holds either a bare series id or a PCL filename that begins with one, and the series id is read from the start of the field. For 37 studies the PCL filenames name series beyond the one in the GEO field, and Table 5 lists those as well.

2 Gene-deletion coverage ✗

A SPELL condition is described only by the free text of its column header, so deletion coverage is estimated from that text. A header counts as a **deletion condition** when a deletion marker is attached to a token that resolves to a yeast gene. The markers are `-del`, a delta sign, the words **delta**, **deletion**, **knockout**, **null** and **KO**, and a lowercase gene name followed by a capital D. A token resolves when it is a systematic name, a standard name, or an alias of exactly one gene in the SGD R64-4-1 annotation.

By that rule 107 of the 603 studies have at least one deletion condition, 2,349 of the 16,160 conditions are deletion conditions, and 685 distinct genes are deleted across the archive (Table 2). Coverage is concentrated: 9 studies delete ten or more genes. Of the 685 genes, 53 are absent from the 1,484 mutants of Kemmeren 2014 Table S1, the deletion compendium already built, and they are spread over 36 studies.

The estimate errs in both directions. It misses a deletion written without a marker, such as **ptc1 vs wt**, and it counts a marked allele that is not a full deletion, such as the stabilized **SIC1Δ3P**. Every call is listed with its header in `experiments/015-spell/results/spell_knockout_conditions.csv`. No call was checked against its paper.

2.1 Genotypes new to the built datasets ✗

Kemmeren 2014 (1,484 single deletions) and Sameith 2015 (single and double deletions) are the built expression datasets with a gene perturbation. A single-gene genotype counts as covered when its gene is in Kemmeren 2014 Table S1, and any genotype counts as covered when it occurs in the Sameith 2015 folder of the archive, which stands in for the built Sameith datasets. A genotype that is neither is **new**.

48 studies carry at least one new genotype, for 472 genotypes in all. 97 of those are deletion genotypes spread over 44 studies, and 375 are gene-named genotypes (Sec. 2.2), nearly all in Hughes 2000 and Mnaimneh 2004. Only 4 studies add five or more. The same genotype can be counted in two studies, so the per-study numbers sum to more than the number of distinct new genotypes.

Table 2: Studies with at least one deletion call, most deleted genes first. # is the row number in Table 5. *KO genes* is the number of distinct genes a deletion marker is attached to in the study’s column headers. *genotypes* counts distinct sets of deleted genes, so a double mutant is one genotype, and *new* is how many of those genotypes are in neither Kemmeren 2014 nor Sameith 2015 (Sec. 2.1); a dash is zero. *KO cond.* is the number of conditions with a call and *cond.* the study’s total. The calls are read from header text and are estimates (Sec. 2).

#	Publication	KO genes	new	genotypes	KO cond.	cond.	GEO	link
4	Hu 2007 Genetic reconstruction of a functional transcriptional regulatory network. <i>Nat Genet</i> 2007.	269	14	269	269	269	GSE4654	doi
51	Lenstra 2011 The specificity and topology of chromatin interaction pathways in yeast. <i>Mol Cell</i> 2011.	162	–	162	313	313	GSE25909	doi
3	van Wageningen 2010 Functional overlap and regulatory links shape genetic interactions between signaling pathways. <i>Cell</i> 2010.	133	14	145	290	377	GSE25644	doi
8	Apweiler 2012 Yeast glucose pathways converge on the transcriptional regulation of trehalose biosynthesis. <i>BMC Genomics</i> 2012.	91	3	93	183	275	GSE33098 GSE33097	doi
52	Sameith 2015 A high-resolution gene expression atlas of epistasis between gene-specific transcription factors exposes potential mechanisms for genetic interactions. <i>BMC Biol</i> 2015.	82	–	154	287	287	GSE42536	doi
53	Chua 2006 Identifying transcription factor functions and targets by phenotypic activation. <i>Proc Natl Acad Sci U S A</i> 2006.	51	–	51	102	270	GSE5499	doi
128	van de Pasch 2013 Centromere binding and a conserved role in chromosome stability for SUMO-dependent ubiquitin ligases. <i>PLoS One</i> 2013.	17	–	17	34	34	GSE33929	doi
132	Garcia-Oliver 2013 A novel role for Sem1 and TREX-2 in transcription involves their impact on recruitment and H2B deubiquitylation activity of SAGA. <i>Nucleic Acids Res</i> 2013.	16	–	16	32	32	GSE39861	doi
29	Huang 2002 Dissection of a complex phenotype by functional genomics reveals roles for the yeast cyclin-dependent protein kinase Pho85 in stress adaptation and cell integrity. <i>Mol Cell Biol</i> 2002.	12	1	12	12	20	none	doi
188	Lenstra 2013 The role of Ctk1 kinase in termination of small non-coding RNAs. <i>PLoS One</i> 2013.	7	–	7	16	22	GSE40254	doi
25	Munding 2010 Integration of a splicing regulatory network within the meiotic gene expression program of <i>Saccharomyces cerevisiae</i> . <i>Genes Dev</i> 2010.	6	1	6	20	48	GSE24675 GSE24681	doi
15	Gitter 2013 Linking the signaling cascades and dynamic regulatory networks controlling stress responses. <i>Genome Res</i> 2013.	6	2	6	12	16	GSE28213	doi
5	Zhou 2011 Integrated approaches reveal determinants of genome-wide binding and function of the transcription factor Pho4. <i>Mol Cell</i> 2011.	5	4	5	14	66	GSE23580	doi
355	Wan 2011 Transcriptome profiling reveals a novel role for trichostatin A in antagonizing histone chaperone Chz1 mediated telomere anti-silencing. <i>FEBS Lett</i> 2011.	5	–	5	10	11	GSE30228	doi
13	Cho 2019 The cell-cycle transcriptional network generates and transmits a pulse of transcription once each cell cycle. <i>Cell Cycle</i> 2019.	4	2	2	96	270	GSE75694	doi

Table 2, continued

#	Publication	KO genes	new	genotypes	KO cond.	cond.	GEO	link
6	Braberg 2013 From structure to systems: high-resolution, quantitative genetic analysis of RNA polymerase II. <i>Cell</i> 2013.	4	4	4	55	57	GSE47429	doi
9	Morillo-Huesca 2010 The SWR1 histone replacement complex causes genetic instability and genome-wide transcription misregulation in the absence of H2A.Z. <i>PLoS One</i> 2010.	4	3	7	21	24	GSE21571	doi
16	Li 2010 Thiamine biosynthesis in <i>Saccharomyces cerevisiae</i> is regulated by the NAD ⁺ -dependent histone deacetylase Hst1. <i>Mol Cell Biol</i> 2010.	4	2	4	8	14	GSE18488	doi
525	Baetz 2001 Transcriptional coregulation by the cell integrity mitogen-activated protein kinase Slt2 and the cell cycle regulator Swi4. <i>Mol Cell Biol</i> 2001.	4	–	4	5	5	none	doi
368	Moxley 2009 Linking high-resolution metabolic flux phenotypes and transcriptional regulation in yeast modulated by the global regulator Gcn4p. <i>Proc Natl Acad Sci U S A</i> 2009.	4	–	4	4	10	GSE4709 GSE7369	doi
76	Capaldi 2008 Structure and function of a transcriptional network activated by the MAPK Hog1. <i>Nat Genet</i> 2008.	3	–	3	38	85	GSE12270	doi
7	Verzijlbergen 2011 A barcode screen for epigenetic regulators reveals a role for the NuB4/HAT-B histone acetyltransferase complex in histone turnover. <i>PLoS Genet</i> 2011.	3	4	7	24	36	GSE30168	doi
19	Mutiu 2007 The role of histone ubiquitylation and deubiquitylation in gene expression as determined by the analysis of an HTB1(K123R) <i>Saccharomyces cerevisiae</i> strain. <i>Mol Genet Genomics</i> 2007.	3	2	4	9	9	GSE6316	doi
260	Santos-Pereira 2014 A genome-wide function of THSC/TREX-2 at active genes prevents transcription-replication collisions. <i>Nucleic Acids Res</i> 2014.	3	–	3	9	15	GSE56702	doi
10	Thibault 2011 The unfolded protein response supports cellular robustness as a broad-spectrum compensatory pathway. <i>Proc Natl Acad Sci U S A</i> 2011.	3	3	3	9	15	GSE33844	doi
30	Thibault 2012 The membrane stress response buffers lethal effects of lipid disequilibrium by reprogramming the protein homeostasis network. <i>Mol Cell</i> 2012.	3	1	3	9	16	GSE39419	doi
11	Gomez-Gonzalez 2011 Genome-wide function of THO/TREX in active genes prevents R-loop-dependent replication obstacles. <i>EMBO J</i> 2011.	3	3	3	7	10	GSE24802	doi
197	Smith 2007 Transcriptional responses to fatty acid are coordinated by combinatorial control. <i>Mol Syst Biol</i> 2007.	3	–	3	6	20	GSE5862	doi
463	Van de Vosse 2013 A role for the nucleoporin Nup170p in chromatin structure and gene silencing. <i>Cell</i> 2013.	3	–	3	3	6	GSE36793	doi
110	Gardner 2005 Ubp10/Dot4p regulates the persistence of ubiquitinated histone H2B: distinct roles in telomeric silencing and general chromatin. <i>Mol Cell Biol</i> 2005.	2	–	2	18	42	GSE2330 GSE2329	doi

Table 2, continued

#	Publication	KO genes	new	genotypes	KO cond.	cond.	GEO	link
163	Zhu 2009 High-resolution DNA-binding specificity analysis of yeast transcription factors. <i>Genome Res</i> 2009.	2	–	3	18	24	GSE13684	doi
27	Ansari 2012 Distinct role of Mediator tail module in regulation of SAGA-dependent, TATA-containing genes in yeast. <i>EMBO J</i> 2012.	2	1	2	12	21	GSE31774	doi
406	Dowell 2010 Chromatin-dependent binding of the <i>S. cerevisiae</i> HMGB protein Nhp6A affects nucleosome dynamics and transcription. <i>Genes Dev</i> 2010.	2	–	2	8	8	GSE23607	doi
28	Borde 2009 Histone H3 lysine 4 trimethylation marks meiotic recombination initiation sites. <i>EMBO J</i> 2009.	2	1	1	7	21	GSE10948 GSE10944 GSE10947	doi
20	Belli 2004 <i>Saccharomyces cerevisiae</i> glutaredoxin 5-deficient cells subjected to continuous oxidizing conditions are affected in the expression of specific sets of genes. <i>J Biol Chem</i> 2004.	2	2	2	6	9	none	doi
12	Cain 2012 A conserved transcriptional regulator governs fungal morphology in widely diverged species. <i>Genetics</i> 2012.	2	3	3	6	8	GSE32550	doi
297	Carroll 2001 Chemical inhibition of the Pho85 cyclin-dependent kinase reveals a role in the environmental stress response. <i>Proc Natl Acad Sci U S A</i> 2001.	2	–	2	6	12	none	doi
191	Kunkel 2019 Integrated TORC1 and PKA signaling control the temporal activation of glucose-induced gene expression in yeast. <i>Nat Commun</i> 2019.	2	–	1	6	21	GSE133591	doi
21	Chen 2017 HAL2 overexpression induces iron acquisition in <i>bdf1Δ</i> cells and enhances their salt resistance. <i>Curr Genet</i> 2017.	2	2	2	5	5	GSE75828	doi
38	Leadsham 2010 cAMP/PKA signaling balances respiratory activity with mitochondria dependent apoptosis via transcriptional regulation. <i>BMC Cell Biol</i> 2010.	2	1	2	5	8	GSE25541	doi
450	Chen 2013 Stabilization of the promoter nucleosomes in nucleosome-free regions by the yeast Cyc8-Tup1 corepressor. <i>Genome Res</i> 2013.	2	–	2	4	7	GSE37466	doi
407	Kugou 2007 Mre11 mediates gene regulation in yeast spore development. <i>Genes Genet Syst</i> 2007.	2	–	2	4	8	GSE6620	doi
46	Lai 2013 Yeast hEST1A/B (SMG5/6)-like proteins contribute to environment-sensing adaptive gene expression responses. <i>G3 (Bethesda)</i> 2013.	2	1	1	4	4	GSE48956	doi
298	Navlakha 2012 A network-based approach for predicting missing pathway interactions. <i>PLoS Comput Biol</i> 2012.	2	–	2	4	12	GSE28213	doi
39	Revilleza 2022 Regulation of CLB6 expression by the cytoplasmic deadenylase Ccr4 through its coding and 3' UTR regions. <i>PLoS One</i> 2022.	2	1	2	4	6	GSE198743	doi
464	Tamaki 2005 Glucose-dependent cell size is regulated by a G protein-coupled receptor system in yeast <i>Saccharomyces cerevisiae</i> . <i>Genes Cells</i> 2005.	2	–	2	4	6	GSE970	doi

Table 2, continued

#	Publication	KO genes	new	genotypes	KO cond.	cond.	GEO	link
18	Hazelwood 2008 The Ehrlich pathway for fusel alcohol production: a century of research on <i>Saccharomyces cerevisiae</i> metabolism. <i>Appl Environ Microbiol</i> 2008.	2	2	2	3	12	GSE9590	doi
356	Penheiter 2005 A posttranscriptional role for the yeast Paf1-RNA polymerase II complex is revealed by identification of primary targets. <i>Mol Cell</i> 2005.	2	–	2	3	11	GSE3200	doi
573	Byeon 2013 The ATP-dependent chromatin remodeling enzyme Fun30 represses transcription by sliding promoter-proximal nucleosomes. <i>J Biol Chem</i> 2013.	2	–	2	2	3	GSE48570	doi
22	Johnson 2014 Methionine restriction activates the retrograde response and confers both stress tolerance and lifespan extension to yeast, mouse and human cells. <i>PLoS One</i> 2014.	2	2	2	2	3	GSE56641	doi
574	Kamata 2023 Spt3 and Spt8 Are Involved in the Formation of a Silencing Boundary by Interacting with TATA-Binding Protein. <i>Biomolecules</i> 2023.	2	–	2	2	3	GSE220290	doi
534	Orlandi 2004 Transcriptional profiling of ubp10 null mutant reveals altered subtelomeric gene expression and insurgence of oxidative stress response. <i>J Biol Chem</i> 2004.	2	–	2	2	4	none	doi
575	Uluisek 2011 Boron stress activates the general amino acid control mechanism and inhibits protein synthesis. <i>PLoS One</i> 2011.	2	–	2	2	3	GSE29887	doi
23	Huisinga 2007 A TATA binding protein regulatory network that governs transcription complex assembly. <i>Genome Biol</i> 2007.	1	1	1	63	149	GSE7385	doi
111	Johansson 2007 Association of yeast Upf1p with direct substrates of the NMD pathway. <i>Proc Natl Acad Sci U S A</i> 2007.	1	–	1	20	40	GSE9482	doi
31	Pitkanen 2004 Excess mannose limits the growth of phosphomannose isomerase PMI40 deletion strain of <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2004.	1	1	1	15	15	none	doi
154	Cipollina 2008 <i>Saccharomyces cerevisiae</i> SFP1: at the crossroads of central metabolism and ribosome biogenesis. <i>Microbiology (Reading)</i> 2008.	1	–	1	13	26	GSE9644	doi
32	Barbara 2007 The transcription factor Gcr1 stimulates cell growth by participating in nutrient-responsive gene expression on a global level. <i>Mol Genet Genomics</i> 2007.	1	1	1	12	12	GSE5027	doi
33	Crosas 2015 The yeast ζ -crystallin/NADPH:quinone oxidoreductase (Zta1p) is under nutritional control by the target of rapamycin pathway and is involved in the regulation of argininosuccinate lyase mRNA half-life. <i>FEBS J</i> 2015.	1	1	1	12	12	GSE45297	doi
138	Zhou 2015 UV induced ubiquitination of the yeast Rad4-Rad23 complex promotes survival by regulating cellular dNTP pools. <i>Nucleic Acids Res</i> 2015.	1	–	1	12	30	GSE23204	doi
24	Levy 2007 Strategy of transcription regulation in the budding yeast. <i>PLoS One</i> 2007.	1	1	1	10	95	GSE6302	doi

Table 2, continued

#	Publication	KO genes	new	genotypes	KO cond.	cond.	GEO	link
299	Boer 2005 Contribution of the <i>Saccharomyces cerevisiae</i> transcriptional regulator Leu3p to physiology and gene expression in nitrogen- and carbon-limited chemostat cultures. <i>FEMS Yeast Res</i> 2005.	1	–	1	6	12	none	doi
208	Busti 2012 Overexpression of Far1, a cyclin-dependent kinase inhibitor, induces a large transcriptional reprogramming in which RNA synthesis senses Far1 in a Sfp1-mediated way. <i>Biotechnol Adv</i> 2012.	1	–	1	6	18	GSE31143	doi
301	Cipollina 2008 Revisiting the role of yeast Sfp1 in ribosome biogenesis and cell size control: a chemostat study. <i>Microbiology (Reading)</i> 2008.	1	–	1	6	12	GSE5238	doi
41	Gaillard 2019 The Nup84 complex coordinates the DNA damage response to warrant genome integrity. <i>Nucleic Acids Res</i> 2019.	1	1	1	6	6	GSE113118	doi
357	Guan 2012 Cellular memory of acquired stress resistance in <i>Saccharomyces cerevisiae</i> . <i>Genetics</i> 2012.	1	–	1	6	11	GSE32196	doi
303	Hou 2014 Management of the endoplasmic reticulum stress by activation of the heat shock response in yeast. <i>FEMS Yeast Res</i> 2014.	1	–	1	6	12	GSE39311	doi
304	Pedersen 2012 DNA Topoisomerases maintain promoters in a state competent for transcriptional activation in <i>Saccharomyces cerevisiae</i> . <i>PLoS Genet</i> 2012.	1	–	1	6	12	GSE22809	doi
306	Rossmann 2011 A common telomeric gene silencing assay is affected by nucleotide metabolism. <i>Mol Cell</i> 2011.	1	–	1	6	12	GSE27222	doi
165	Sapra 2004 Genome-wide analysis of pre-mRNA splicing: intron features govern the requirement for the second-step factor, Prp17 in <i>Saccharomyces cerevisiae</i> and <i>Schizosaccharomyces pombe</i> . <i>J Biol Chem</i> 2004.	1	–	1	6	24	none	doi
205	Shivaswamy 2008 Stress-dependent dynamics of global chromatin remodeling in yeast: dual role for SWI/SNF in the heat shock stress response. <i>Mol Cell Biol</i> 2008.	1	–	1	6	19	GSE7665	doi
369	Dong 2007 YRA1 autoregulation requires nuclear export and cytoplasmic Edc3p-mediated degradation of its pre-mRNA. <i>Mol Cell</i> 2007.	1	–	1	5	10	GSE6647	doi
164	Aristizabal 2013 High-throughput genetic and gene expression analysis of the RNAPII-CTD reveals unexpected connections to SRB10/CDK8. <i>PLoS Genet</i> 2013.	1	–	1	4	24	GSE43120	doi
300	Chen 2006 Coordinate regulation of multiple and distinct biosynthetic pathways by TOR and PKA kinases in <i>S. cerevisiae</i> . <i>Curr Genet</i> 2006.	1	–	1	4	12	GSE3160	doi
302	Drozdova 2014 Transcriptional response to the [ISP(+)] prion of <i>Saccharomyces cerevisiae</i> differs from that induced by the deletion of its structural gene, SFP1. <i>FEMS Yeast Res</i> 2014.	1	–	1	4	12	GSE52189	doi
409	Klopf 2017 INO80 represses osmostress induced gene expression by resetting promoter proximal nucleosomes. <i>Nucleic Acids Res</i> 2017.	1	–	1	4	8	GSE78766	doi

Table 2, continued

#	Publication	KO genes	new	genotypes	KO cond.	cond.	GEO	link
36	Lee 2005 Multiple RNA surveillance pathways limit aberrant expression of iron uptake mRNAs and prevent iron toxicity in <i>S. cerevisiae</i> . <i>Mol Cell</i> 2005.	1	1	1	4	9	none	doi
410	Liko 2007 Stb3 binds to ribosomal RNA processing element motifs that control transcriptional responses to growth in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2007.	1	–	1	4	8	GSE8379	doi
35	Alkim 2013 Mechanisms other than activation of the iron regulon account for the hyper-resistance to cobalt of a <i>Saccharomyces cerevisiae</i> strain obtained by evolutionary engineering. <i>Metalomics</i> 2013.	1	1	1	3	9	GSE39185	doi
408	Angus-Hill 2001 A Rsc3/Rsc30 zinc cluster dimer reveals novel roles for the chromatin remodeler RSC in gene expression and cell cycle control. <i>Mol Cell</i> 2001.	1	–	1	3	8	none	doi
275	Arana 2012 Transcriptional responses to loss of RNase H2 in <i>Saccharomyces cerevisiae</i> . <i>DNA Repair (Amst)</i> 2012.	1	–	1	3	14	GSE34668	doi
451	Chia 2018 Knockout of the Hmt1p Arginine Methyltransferase in <i>Saccharomyces cerevisiae</i> Leads to the Dysregulation of Phosphate-associated Genes and Processes. <i>Mol Cell Proteomics</i> 2018.	1	–	1	3	7	GSE99869	doi
40	Eriksson 2005 Global regulation by the yeast Spt10 protein is mediated through chromatin structure and the histone upstream activating sequence elements. <i>Mol Cell Biol</i> 2005.	1	1	1	3	6	none	doi
42	Garca-Rodriguez 2015 Manganese redistribution by calcium-stimulated vesicle trafficking bypasses the need for P-type ATPase function. <i>J Biol Chem</i> 2015.	1	1	1	3	6	GSE62161	doi
43	Garcia-Rodriguez 2012 Impaired manganese metabolism causes mitotic misregulation. <i>J Biol Chem</i> 2012.	1	1	1	3	6	GSE29420	doi
44	Gavalda 2016 Excess of Yra1 RNA-Binding Factor Causes Transcription-Dependent Genome Instability, Replication Impairment and Telomere Shortening. <i>PLoS Genet</i> 2016.	1	1	1	3	6	GSE68487	doi
465	Hickman 2007 Heme levels switch the function of Hap1 of <i>Saccharomyces cerevisiae</i> between transcriptional activator and transcriptional repressor. <i>Mol Cell Biol</i> 2007.	1	–	1	3	6	GSE8613	doi
34	Hopfler 2019 Slx5/Slx8-dependent ubiquitin hotspots on chromatin contribute to stress tolerance. <i>EMBO J</i> 2019.	1	1	1	3	12	GSE118817	doi
261	Lanza 2012 Linking yeast Gcn5p catalytic function and gene regulation using a quantitative, graded dominant mutant approach. <i>PLoS One</i> 2012.	1	–	1	3	15	GSE26923	doi
37	Mendiratta 2006 The DNA-binding domain of the yeast Spt10p activator includes a zinc finger that is homologous to foamy virus integrase. <i>J Biol Chem</i> 2006.	1	1	1	3	9	GSE3354	doi
466	Rodriguez-Navarro 2004 Sus1, a functional component of the SAGA histone acetylase complex and the nuclear pore-associated mRNA export machinery. <i>Cell</i> 2004.	1	–	1	3	6	none	doi

Table 2, continued

#	Publication	KO genes	new	genotypes	KO cond.	cond.	GEO	link
467	Santos-Pereira 2013 The Npl3 hnRNP prevents R-loop-mediated transcription-replication conflicts and genome instability. <i>Genes Dev</i> 2013.	1	–	1	3	6	GSE50186	doi
588	Costelloe 2012 The yeast Fun30 and human SMARCAD1 chromatin remodellers promote DNA end resection. <i>Nature</i> 2012.	1	–	1	2	2	GSE38735	doi
26	Garcia-Martinez 2016 The cellular growth rate controls overall mRNA turnover, and modulates either transcription or degradation rates of particular gene regulons. <i>Nucleic Acids Res</i> 2016.	1	1	1	2	32	GSE63729 GSE57467 GSE65225	doi
48	Kang 2014 A novel protein, Pho92, has a conserved YTH domain and regulates phosphate metabolism by decreasing the mRNA stability of PHO4 in <i>Saccharomyces cerevisiae</i> . <i>Biochem J</i> 2014.	1	1	1	2	2	GSE50086	doi
576	Levy 2008 Yeast linker histone Hho1p is required for efficient RNA polymerase I processivity and transcriptional silencing at the ribosomal DNA. <i>Proc Natl Acad Sci U S A</i> 2008.	1	–	1	2	3	GSE9095	doi
526	Lu 2005 The YJR127C/ZMS1 gene product is involved in glycerol-based respiratory growth of the yeast <i>Saccharomyces cerevisiae</i> . <i>Curr Genet</i> 2005.	1	–	1	2	5	GSE29530	doi
370	Lu 2022 A balancing act: interactions within NuA4/TIP60 regulate picNuA4 function in <i>Saccharomyces cerevisiae</i> and humans. <i>Genetics</i> 2022.	1	–	1	2	10	GSE137261	doi
589	Radman-Livaja 2012 A key role for Chd1 in histone H3 dynamics at the 3' ends of long genes in yeast. <i>PLoS Genet</i> 2012.	1	–	1	2	2	GSE38496	doi
305	Rodriguez-Colman 2010 The forkhead transcription factor Hcm1 promotes mitochondrial biogenesis and stress resistance in yeast. <i>J Biol Chem</i> 2010.	1	–	1	2	12	GSE20420	doi
45	Rudra 2005 Central role of Ifh1p-Fhl1p interaction in the synthesis of yeast ribosomal proteins. <i>EMBO J</i> 2005.	1	1	1	2	6	none	doi
209	Sabet 2004 Genome-wide analysis of the relationship between transcriptional regulation by Rpd3p and the histone H3 and H4 amino termini in budding yeast. <i>Mol Cell Biol</i> 2004.	1	–	1	2	18	none	doi
276	Saint 2014 The TAF9 C-terminal conserved region domain is required for SAGA and TFIID promoter occupancy to promote transcriptional activation. <i>Mol Cell Biol</i> 2014.	1	–	1	2	14	GSE44544	doi
47	Sariki 2016 Sen1, the homolog of human Senataxin, is critical for cell survival through regulation of redox homeostasis, mitochondrial function, and the TOR pathway in <i>Saccharomyces cerevisiae</i> . <i>FEBS J</i> 2016.	1	1	1	2	4	GSE75447	doi
599	Davis 2006 Accumulation of unstable promoter-associated transcripts upon loss of the nuclear exosome subunit Rrp6p in <i>Saccharomyces cerevisiae</i> . <i>Proc Natl Acad Sci U S A</i> 2006.	1	–	1	1	1	GSE3813	doi

Table 2, continued

#	Publication	KO genes	new	genotypes	KO cond.	cond.	GEO	link
242	Hughes Hallett 2014 State transitions in the TORC1 signaling pathway and information processing in <i>Saccharomyces cerevisiae</i> . <i>Genetics</i> 2014.	1	–	1	1	16	GSE58992	doi
535	Sen 2011 A new, highly conserved domain in Swi2/Snf2 is required for SWI/SNF remodeling. <i>Nucleic Acids Res</i> 2011.	1	–	1	1	4	GSE30675	doi

2.2 Conditions labeled by gene name alone ✗

7 studies label conditions with nothing but gene names, for 522 conditions and 496 distinct genes (Table 3). These name a perturbed gene without saying how it was perturbed, so they are kept out of the deletion count. Two studies hold almost all of them. The headers of Hughes 2000 mix lowercase names with entries marked (**tet promoter**), and those of Mnaimneh 2004 pair a standard name with its systematic name. Hypothesis (untested against the papers): the lowercase Hughes 2000 names are deletion mutants and the Mnaimneh 2004 entries are tetracycline-repressible promoter alleles of essential genes.

Table 3: Studies with conditions labeled by gene name alone. # is the row number in Table 5. *genes* counts the distinct genes named, *new* the gene-named genotypes in neither Kemmeren 2014 nor Sameith 2015, *named cond.* the conditions labeled that way and *cond.* the study’s total. The header does not say how the gene was perturbed, so these are not counted as deletions.

#	study	genes	new	named cond.	cond.	GEO	link
2	Hughes 2000	276	157	284	300	none	doi
1	Mnaimneh 2004	215	214	215	215	none	doi
212	Bernstein 2000	3	–	6	18	none	doi
176	Meneghini 2003	3	–	12	24	none	doi
17	Hong 2012	2	2	2	14	GSE36118	doi
14	Zheng 2010	2	2	2	48	GSE19634	doi
384	Travers 2000	1	–	1	10	none	doi

3 Studies with no GEO accession ✗

88 READMEs read GEO ID: N/A (Table 4). These studies were published between 1997 and 2006 and account for 130 PCL files and 2,241 conditions. None of their PCL filenames contains a GEO series id, so the archive gives no pointer to the authors’ released data. The data location has to be recovered from the paper: its methods, its supplementary files, or a laboratory or repository address it names. 42 of these studies have a PMC id and 46 do not. 85 of the 88 are two-channel microarray studies. 14 carry a deletion call, and 5 of the 7 gene-named studies of Table 3, Hughes 2000 and Mnaimneh 2004 among them, are in this group.

All 130 of their PCL filenames begin with 2010. and carry suffixes such as **flt**, **knn**, **avg**, **div** and **log**, which the READMEs do not define. Hypothesis (untested): the suffixes name processing steps applied after the authors’ release, such as filtering, nearest-neighbor imputation, replicate averaging, division by a reference and log transformation. If so, the PCL values for these studies are further from the source measurements than a GEO series would be.

Table 4: The studies whose README reads GEO ID: N/A, in the order of Table 5. # is the row number in Table 5 and *KO genes* the estimate of Sec. 2. *PCL files in the archive* lists each matrix SPELL distributes for the study, which is the only data location the archive gives for these rows. *SPELL tags* are the topic tags in the README dataset table. A row with a PMC link has a full text in PubMed Central to search for the data location; a row without one does not.

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
1	Mnaimneh 2004 Mnaimneh S, Davierwala AP, Haynes J, et al., Hughes TR. Exploration of essential gene functions via titratable promoter alleles. <i>Cell</i> 2004;118(1):31–44.	doi:10.1016/j.cell.2004.06.013 PMID 15242642	2010.EssentialGenes.pcl	chemical stimulus; stress	215	–	2
2	Hughes 2000 Hughes TR, Marton MJ, Jones AR, et al., Friend SH. Functional discovery via a compendium of expression profiles. <i>Cell</i> 2000;102(1):109–26.	doi:10.1016/s0092-8674(00)00015-5 PMID 10929718	2010.Hughes00.flt.knn.avg.pcl	chemical stimulus	300	–	2
20	Belli 2004 Bellí G, Molina MM, García-Martínez J, Pérez-Ortín JE, Herrero E. Saccharomyces cerevisiae glutaredoxin 5-deficient cells subjected to continuous oxidizing conditions are affected in the expression of specific sets of genes. <i>J Biol Chem</i> 2004;279(13):12386–95.	doi:10.1074/jbc.M311879200 PMID 14722110	2010.Belli04.filter.flt.knn.avg.div.log.pcl	oxidative stress; stress	9	2	2
29	Huang 2002 Huang D, Moffat J, Andrews B. Dissection of a complex phenotype by functional genomics reveals roles for the yeast cyclin-dependent protein kinase Pho85 in stress adaptation and cell integrity. <i>Mol Cell Biol</i> 2002;22(14):5076–88.	doi:10.1128/MCB.22.14.5076-5088.2002 PMID 12077337 PMC139770	2010.Huang02.flt.knn.avg.pcl	protein phosphorylation	20	12	2
31	Pitkanen 2004 Pitkänen JP, Törmä A, Alff S, Huopaniemi L, Mattila P, Renkonen R. Excess mannose limits the growth of phosphomannose isomerase PMI40 deletion strain of <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2004;279(53):55737–43.	doi:10.1074/jbc.M410619200 PMID 15520001	2010.Pitkanen04.filter.flt.knn.avg.div.log.pcl	carbon utilization; protein glycosylation	15	1	2
36	Lee 2005 Lee A, Henras AK, Chanfreau G. Multiple RNA surveillance pathways limit aberrant expression of iron uptake mRNAs and prevent iron toxicity in <i>S. cerevisiae</i> . <i>Mol Cell</i> 2005;19(1):39–51.	doi:10.1016/j.molcel.2005.05.021 PMID 15989963	2010.Lee05.filter.flt.knn.avg.div.log.pcl	RNA catabolism	9	1	2
40	Eriksson 2005 Eriksson PR, Mendiratta G, McLaughlin NB, et al., Clark DJ. Global regulation by the yeast Spt10 protein is mediated through chromatin structure and the histone upstream activating sequence elements. <i>Mol Cell Biol</i> 2005;25(20):9127–37.	doi:10.1128/MCB.25.20.9127-9137.2005 PMID 16199888 PMC1265784	2010.Eriksson05.filter.flt.knn.avg.div.log.pcl	chromatin organization; histone modification	6	1	2
45	Rudra 2005 Rudra D, Zhao Y, Warner JR. Central role of Ifh1p-Fhl1p interaction in the synthesis of yeast ribosomal proteins. <i>EMBO J</i> 2005;24(3):533–42.	doi:10.1038/sj.emboj.7600553 PMID 15692568 PMC548658	2010.Rudra05.filter.flt.knn.avg.div.log.pcl	transcription	6	1	1

Table 4, continued

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
63	Gasch 2000 Gasch AP, Spellman PT, Kao CM, et al., Brown PO. Genomic expression programs in the response of yeast cells to environmental changes. <i>Mol Biol Cell</i> 2000;11(12):4241–57.	doi:10.1091/mbc.11.12.4241 PMID 11102521 PMC15070	2010.Gasch00_hyper-osmotic.flt.knn.avg.pcl 2010.Gasch00_hypo-osmotic.flt.knn.avg.pcl 2010.Gasch00_menadione.flt.knn.avg.pcl 2010.Gasch00_Ndepletion.flt.knn.avg.pcl 2010.Gasch00_stationaryPhase(y12).flt.knn.avg.pcl 2010.Gasch00_stationaryPhase(y14).flt.knn.avg.pcl 2010.Gasch00_steadyState(y13).flt.knn.avg.pcl 2010.Gasch00_steadyState(y14).flt.knn.avg.pcl 2010.Gasch00_HS29-33.flt.knn.avg.pcl 2010.Gasch00_HS30-37.flt.knn.avg.pcl 2010.Gasch00_HS37-25.flt.knn.avg.pcl 2010.Gasch00_HSmild.flt.knn.avg.pcl 2010.Gasch00_HSto37.flt.knn.avg.pcl 2010.Gasch00_DTT(y13).flt.knn.avg.pcl 2010.Gasch00_DTT(y14).flt.knn.avg.pcl 2010.Gasch00_H0timeCourse.flt.knn.avg.pcl 2010.Gasch00_HS25-37.flt.knn.avg.pcl 2010.Gasch00_carbonSources.flt.knn.avg.pcl 2010.Gasch00_diamideTreatment.flt.knn.avg.pcl 2010.Gasch00_adenineStarvation.flt.knn.avg.pcl	amino acid utilization; carbon utilization; chemical stimulus; heat shock; nitrogen utilization; osmotic stress; oxidative stress; response to unfolded protein; starvation; stationary phase entry; stress	138	—	2
65	O’Rourke 2004 O’Rourke SM, Herskowitz I. Unique and redundant roles for HOG MAPK pathway components as revealed by whole-genome expression analysis. <i>Mol Biol Cell</i> 2004;15(2):532–42.	doi:10.1091/mbc.e03-07-0521 PMID 14595107 PMC329229	2010.0Rourke03.flt.knn.avg.pcl	osmotic stress; protein phosphorylation; signaling; stress	133	—	2
66	Brem 2005 Brem RB, Kruglyak L. The landscape of genetic complexity across 5,700 gene expression traits in yeast. <i>Proc Natl Acad Sci U S A</i> 2005;102(5):1572–7.	doi:10.1073/pnas.0408709102 PMID 15659551 PMC547855	2010.Brem05_orig.flt.knn.avg.pcl	QTLs	131	—	2
69	Saldanha 2004 Saldanha AJ, Brauer MJ, Botstein D. Nutritional homeostasis in batch and steady-state culture of yeast. <i>Mol Biol Cell</i> 2004;15(9):4089–104.	doi:10.1091/mbc.e04-04-0306 PMID 15240820 PMC515343	2010.Saldanha04_UracilBatchChem.flt.knn.avg.pcl 2010.Saldanha04_UraSulPhoLeuComp.flt.knn.avg.pcl 2010.Saldanha04_SulfateBatchChem.flt.knn.avg.pcl 2010.Saldanha04_LeucineBatchChem.flt.knn.avg.pcl 2010.Saldanha04_PhosphateBatchChem.flt.knn.avg.pcl	amino acid utilization; nutrient utilization; phosphorus utilization; sulfur utilization	124	—	2
82	Shapira 2004 Shapira M, Segal E, Botstein D. Disruption of yeast forkhead-associated cell cycle transcription by oxidative stress. <i>Mol Biol Cell</i> 2004;15(12):5659–69.	doi:10.1091/mbc.e04-04-0340 PMID 15371544 PMC532044	2010.Shapira04.flt.knn.avg.pcl	chemical stimulus; mitotic cell cycle; oxidative stress; stress	70	—	2

Table 4, continued

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
95	Spellman 1998 Spellman PT, Sherlock G, Zhang MQ, et al., Fitcher B. Comprehensive identification of cell cycle-regulated genes of the yeast <i>Saccharomyces cerevisiae</i> by microarray hybridization. <i>Mol Biol Cell</i> 1998;9(12):3273–97.	doi:10.1091/mbc.9.12.3273 PMID 9843569 PMC25624	2010.Spellman98_cdc15.flt.knn.avg.pcl 2010.Spellman98_elutriation.flt.knn.avg.pcl 2010.Spellman98_alphaFactor.flt.knn.avg.pcl	mitotic cell cycle	55	–	2
97	Gasch 2001 Gasch AP, Huang M, Metzner S, Botstein D, Elledge SJ, Brown PO. Genomic expression responses to DNA-damaging agents and the regulatory role of the yeast ATR homolog Mec1p. <i>Mol Biol Cell</i> 2001;12(10):2987–3003.	doi:10.1091/mbc.12.10.2987 PMID 11598186 PMC60150	2010.DNA Damage.pcl	DNA damage stimulus; radiation; signaling; stress	52	–	2
100	Lai 2006 Lai LC, Kosorukoff AL, Burke PV, Kwast KE. Metabolic-state-dependent remodeling of the transcriptome in response to anoxia and subsequent reoxygenation in <i>Saccharomyces cerevisiae</i> . <i>Eukaryot Cell</i> 2006;5(9):1468–89.	doi:10.1128/EC.00107-06 PMID 16963631 PMC1563586	2010.glucose_anoxia_reoxygenation.pcl 2010.galactose_anoxia_reoxygenation.pcl	carbon utilization; oxygen level alteration; stress	48	–	2
105	Causton 2001 Causton HC, Ren B, Koh SS, et al., Young RA. Remodeling of yeast genome expression in response to environmental changes. <i>Mol Biol Cell</i> 2001;12(2):323–37.	doi:10.1091/mbc.12.2.323 PMID 11179418 PMC30946	2010.Causton01_peroxide.filter.flt.knn.avg. div.log.pcl 2010.Causton01_sorbitol.filter.flt.knn.avg. div.log.pcl 2010.Causton01_acid.filter.flt.knn.avg.div. log.pcl 2010.Causton01_alkali.filter.flt.knn.avg.div. log.pcl 2010.Causton01_heat.filter.flt.knn.avg.div. log.pcl 2010.Causton01_NaCl.filter.flt.knn.avg.div. log.pcl	chemical stimulus; heat shock; osmotic stress; oxidative stress; response to unfolded protein; stress	45	–	2
115	Yoshimoto 2002 Yoshimoto H, Saltsman K, Gasch AP, et al., Cyert MS. Genome-wide analysis of gene expression regulated by the calcineurin/Crz1p signaling pathway in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2002;277(34):31079–88.	doi:10.1074/jbc.M202718200 PMID 12058033	2010.Yoshimoto02_Ca.flt.knn.avg.pcl 2010.Yoshimoto02_Na.flt.knn.avg.pcl	cellular ion homeostasis; chemical stimulus; signaling	40	–	2
137	Segal 2003 Segal E, Shapira M, Regev A, et al., Friedman N. Module networks: identifying regulatory modules and their condition-specific regulators from gene expression data. <i>Nat Genet</i> 2003;34(2):166–76.	doi:10.1038/ng1165 PMID 12740579	2010.Segal03_HSkin82.flt.knn.avg.pcl 2010.Segal03_hypo-osmoticPpt1.flt.knn.avg.pcl 2010.Segal03_stationaryPhaseYPL230W.flt.knn. avg.pcl	heat shock; osmotic stress; protein dephosphorylation; protein phosphorylation; stationary phase entry; stress	32	–	2
140	Duvel 2003 Duvel K, Santhanam A, Garrett S, Schneper L, Broach JR. Multiple roles of Tap42 in mediating rapamycin-induced transcriptional changes in yeast. <i>Mol Cell</i> 2003;11(6):1467–78.	doi:10.1016/s1097-2765(03)00228-4 PMID 12820961	2010.Duvel03_delayedRap.flt.knn.avg.pcl 2010.Duvel03_immediateRap.flt.knn.avg.pcl	chemical stimulus; heat shock; protein dephosphorylation; signaling	30	–	2
141	Fleming 2002 Fleming JA, Lightcap ES, Sadis S, Thoroddsen V, Bulawa CE, Blackman RK. Complementary whole-genome technologies reveal the cellular response to proteasome inhibition by PS-341. <i>Proc Natl Acad Sci U S A</i> 2002;99(3):1461–6.	doi:10.1073/pnas.032516399 PMID 11830665 PMC122213	2010.Fleming99.log.flt.knn.avg.pcl	chemical stimulus; proteolysis	30	–	2

Table 4, continued

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
165	Sapra 2004 Sapra AK, Arava Y, Khandelia P, Vijayraghavan U. Genome-wide analysis of pre-mRNA splicing: intron features govern the requirement for the second-step factor, Prp17 in <i>Saccharomyces cerevisiae</i> and <i>Schizosaccharomyces pombe</i> . <i>J Biol Chem</i> 2004;279(50):52437–46.	doi:10.1074/jbc.M408815200 PMID 15452114	2010.Sapra04.flt.knn.avg.pcl	mRNA processing	24	1	2
168	Caba 2005 Caba E, Dickinson DA, Warnes GR, Aubrecht J. Differentiating mechanisms of toxicity using global gene expression analysis in <i>Saccharomyces cerevisiae</i> . <i>Mutat Res</i> 2005;575(1–2):34–46.	doi:10.1016/j.mrfmmm.2005.02.005 PMID 15878181	2010.Caba05.filter.flt.knn.avg.div.log.pcl	DNA damage stimulus; osmotic stress; stress	24	–	1
176	Meneghini 2003 Meneghini MD, Wu M, Madhani HD. Conserved histone variant H2A.Z protects euchromatin from the ectopic spread of silent heterochromatin. <i>Cell</i> 2003;112(5):725–36.	doi:10.1016/s0092-8674(03)00123-5 PMID 12628191	2010.swr1_htz1_hmr_sir2.pcl	chromatin organization	24	–	2
177	Primig 2000 Primig M, Williams RM, Winzeler EA, et al., Esposito RE. The core meiotic transcriptome in budding yeasts. <i>Nat Genet</i> 2000;26(4):415–23.	doi:10.1038/82539 PMID 11101837	2010.Primig00.filter.flt.knn.avg.div.log.pcl	sporulation	24	–	2
178	Schawaldner 2004 Schawaldner SB, Kabani M, Howald I, Choudhury U, Werner M, Shore D. Growth-regulated recruitment of the essential yeast ribosomal protein gene activator Ifh1. <i>Nature</i> 2004;432(7020):1058–61.	doi:10.1038/nature03200 PMID 15616569	2010.Schawaldner04.filter.flt.knn.avg.div.log.pcl	transcription	24	–	2
180	Tai 2005 Tai SL, Boer VM, Daran-Lapujade P, et al., Pronk JT. Two-dimensional transcriptome analysis in chemostat cultures. Combinatorial effects of oxygen availability and macronutrient limitation in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2005;280(1):437–47.	doi:10.1074/jbc.M410573200 PMID 15496405	2010.Tai05.filter.flt.knn.avg.div.log.pcl	carbon utilization; fermentation; nitrogen utilization; oxygen level alteration; phosphorus utilization; respiration; sulfur utilization	24	–	2
194	Ideker 2001 Ideker T, Thorsson V, Ranish JA, et al., Hood L. Integrated genomic and proteomic analyses of a systematically perturbed metabolic network. <i>Science</i> 2001;292(5518):929–34.	doi:10.1126/science.292.5518.929 PMID 11340206	2010.Ideker01.flt.knn.avg.pcl	carbon utilization	21	–	2
198	Brauer 2005 Brauer MJ, Saldanha AJ, Dolinski K, Botstein D. Homeostatic adjustment and metabolic remodeling in glucose-limited yeast cultures. <i>Mol Biol Cell</i> 2005;16(5):2503–17.	doi:10.1091/mbc.e04-11-0968 PMID 15758028 PMC1087253	2010.Brauer05_batch1.flt.knn.avg.pcl 2010.Brauer05_batch2.flt.knn.avg.pcl	carbon utilization; diauxic shift; fermentation; respiration	20	–	2
200	Chitikila 2002 Chitikila C, Huisinga KL, Irvin JD, Basehoar AD, Pugh BF. Interplay of TBP inhibitors in global transcriptional control. <i>Mol Cell</i> 2002;10(4):871–82.	doi:10.1016/s1097-2765(02)00683-4 PMID 12419230	2010.Chitikila02.flt.knn.avg.pcl	transcription	20	–	2

Table 4, continued

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
209	Sabet 2004 Sabet N, Volo S, Yu C, Madigan JP, Morse RH. Genome-wide analysis of the relationship between transcriptional regulation by Rpd3p and the histone H3 and H4 amino termini in budding yeast. <i>Mol Cell Biol</i> 2004;24(20):8823–33.	doi:10.1128/MCB.24.20.8823-8833.2004 PMID 15456858 PMC517902	2010.Sabet04.filter.flt.knn.avg.div.log.pcl	histone modification	18	1	2
212	Bernstein 2000 Bernstein BE, Tong JK, Schreiber SL. Genomewide studies of histone deacetylase function in yeast. <i>Proc Natl Acad Sci U S A</i> 2000;97(25):13708–13.	doi:10.1073/pnas.250477697 PMID 11095743 PMC17640	2010.Bernstein00_HDACrpd3sin3hda1.log.flt.knn.avg.pcl 2010.Bernstein00_HDACsin3sap30ume6hda1hos2hos3.filter.flt.knn.avg.div.log.pcl 2010.Bernstein00_TSA.filter.flt.knn.avg.div.log.pcl	chromatin organization; histone modification	18	–	2
233	Cho 1998 Cho RJ, Campbell MJ, Winzeler EA, et al., Davis RW. A genome-wide transcriptional analysis of the mitotic cell cycle. <i>Mol Cell</i> 1998;2(1):65–73.	doi:10.1016/s1097-2765(00)80114-8 PMID 9702192	2010.Cho98.filter.flt.knn.avg.div.log.pcl	mitotic cell cycle	17	–	2
234	Clark 2002 Clark TA, Sugnet CW, Ares M Jr. Genomewide analysis of mRNA processing in yeast using splicing-specific microarrays. <i>Science</i> 2002;296(5569):907–10.	doi:10.1126/science.1069415 PMID 11988574	2010.Clark02_orig.flt.knn.avg.pcl	mRNA processing	17	–	2
237	Haugen 2004 Haugen AC, Kelley R, Collins JB, et al., Van Houten B. Integrating phenotypic and expression profiles to map arsenic-response networks. <i>Genome Biol</i> 2004;5(12):R95.	doi:10.1186/gb-2004-5-12-r95 PMID 15575969 PMC545798	2010.arsenic_time_dose.pcl 2010.arsenic_deletions.pcl	chemical stimulus; metal or metalloid ion stress	17	–	2
240	Lopez 2000 López MC, Baker HV. Understanding the growth phenotype of the yeast gcr1 mutant in terms of global genomic expression patterns. <i>J Bacteriol</i> 2000;182(17):4970–8.	doi:10.1128/JB.182.17.4970-4978.2000 PMID 10940042 PMC111378	2010.Lopez00.filter.flt.knn.avg.div.log.pcl	carbon utilization; fermentation; respiration	17	–	2
264	Carmel-Harel 2001 Carmel-Harel O, Stearman R, Gasch AP, Botstein D, Brown PO, Storz G. Role of thioredoxin reductase in the Yap1p-dependent response to oxidative stress in <i>Saccharomyces cerevisiae</i> . <i>Mol Microbiol</i> 2001;39(3):595–605.	doi:10.1046/j.1365-2958.2001.02255.x PMID 11169101	2010.CarmelHarel01.flt.knn.avg.pcl	chemical stimulus; oxidative stress; stress	15	–	2
270	Roberts 2000 Roberts CJ, Nelson B, Marton MJ, et al., Friend SH. Signaling and circuitry of multiple MAPK pathways revealed by a matrix of global gene expression profiles. <i>Science</i> 2000;287(5454):873–80.	doi:10.1126/science.287.5454.873 PMID 10657304	2010.alpha_time.pcl 2010.alpha_conc.pcl	cell morphogenesis; mating; signaling	15	–	2
281	Hardwick 1999 Hardwick JS, Kuruvilla FG, Tong JK, Shamji AF, Schreiber SL. Rapamycin-modulated transcription defines the subset of nutrient-sensitive signaling pathways directly controlled by the Tor proteins. <i>Proc Natl Acad Sci U S A</i> 1999;96(26):14866–70.	doi:10.1073/pnas.96.26.14866 PMID 10611304 PMC24739	2010.Hardwick00.flt.knn.avg.pcl	amino acid utilization; chemical stimulus; signaling	14	–	2
293	Leber 2004 Leber JH, Bernales S, Walter P. IRE1-independent gain control of the unfolded protein response. <i>PLoS Biol</i> 2004;2(8):E235.	doi:10.1371/journal.pbio.0020235 PMID 15314654 PMC509300	2010.Leber04.flt.knn.avg.pcl	chemical stimulus; response to unfolded protein; stress	13	–	1

Table 4, continued

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
297	Carroll 2001 Carroll AS, Bishop AC, DeRisi JL, Shokat KM, O'Shea EK. Chemical inhibition of the Pho85 cyclin-dependent kinase reveals a role in the environmental stress response. <i>Proc Natl Acad Sci U S A</i> 2001;98(22):12578–83.	doi:10.1073/pnas.211195798 PMID 11675494 PMC60096	2010.Carroll101.log.flt.knn.avg.pcl	protein phosphorylation; stress	12	2	2
299	Boer 2005 Boer VM, Daran JM, Almering MJ, de Winde JH, Pronk JT. Contribution of the <i>Saccharomyces cerevisiae</i> transcriptional regulator Leu3p to physiology and gene expression in nitrogen- and carbon-limited chemostat cultures. <i>FEMS Yeast Res</i> 2005;5(10):885–97.	doi:10.1016/j.femsyr.2005.04.003 PMID 15949974	2010.Boer05.filter.flt.knn.avg.div.log.pcl	carbon utilization; nitrogen utilization; transcription	12	1	2
321	De Nadal 2004 De Nadal E, Zapater M, Alepuz PM, Sumoy L, Mas G, Posas F. The MAPK Hog1 recruits Rpd3 histone deacetylase to activate osmoresponsive genes. <i>Nature</i> 2004;427(6972):370–4.	doi:10.1038/nature02258 PMID 14737171	2010.deNadal04.flt.knn.avg.pcl	histone modification; osmotic stress; stress	12	–	2
334	Martin 2004 Martin DE, Demougin P, Hall MN, Bellis M. Rank Difference Analysis of Microarrays (RDAM), a novel approach to statistical analysis of microarray expression profiling data. <i>BMC Bioinformatics</i> 2004;5:148.	doi:10.1186/1471-2105-5-148 PMID 15476558 PMC526220	2010.Martin04.filter.flt.knn.avg.div.log.pcl	protein phosphorylation; signaling	12	–	2
337	Olesen 2002 Olesen K, Felding T, Gjermansen C, Hansen J. The dynamics of the <i>Saccharomyces carlsbergensis</i> brewing yeast transcriptome during a production-scale lager beer fermentation. <i>FEMS Yeast Res</i> 2002;2(4):563–73.	doi:10.1111/j.1567-1364.2002.tb00123.x PMID 12702272	2010.Olesen02.filter.flt.knn.avg.div.log.pcl	carbon utilization; fermentation	12	–	2
342	Robertson 2000 Robertson LS, Causton HC, Young RA, Fink GR. The yeast A kinases differentially regulate iron uptake and respiratory function. <i>Proc Natl Acad Sci U S A</i> 2000;97(11):5984–8.	doi:10.1073/pnas.100113397 PMID 10811893 PMC18545	2010.Robertson00.filter.flt.knn.avg.div.log.pcl	protein phosphorylation; signaling	12	–	2
347	Sudarsanam 2000 Sudarsanam P, Iyer VR, Brown PO, Winston F. Whole-genome expression analysis of snf/swi mutants of <i>Saccharomyces cerevisiae</i> . <i>Proc Natl Acad Sci U S A</i> 2000;97(7):3364–9.	doi:10.1073/pnas.97.7.3364 PMID 10725359 PMC16245	2010.snf_swi_mutants.pcl	chromatin organization	12	–	2
350	Takagi 2005 Takagi Y, Masuda CA, Chang WH, et al., Kornberg RD. Ubiquitin ligase activity of TFIIF and the transcriptional response to DNA damage. <i>Mol Cell</i> 2005;18(2):237–43.	doi:10.1016/j.molcel.2005.03.007 PMID 15837426	2010.Takagi05.filter.flt.knn.avg.div.log.pcl	DNA damage stimulus; transcription; ubiquitin or ULP modification	12	–	2
360	Bulik 2003 Bulik DA, Olczak M, Lucero HA, Osmond BC, Robbins PW, Specht CA. Chitin synthesis in <i>Saccharomyces cerevisiae</i> in response to supplementation of growth medium with glucosamine and cell wall stress. <i>Eukaryot Cell</i> 2003;2(5):886–900.	doi:10.1128/EC.2.5.886-900.2003 PMID 14555471 PMC219353	2010.Bulik03.filter.flt.knn.avg.div.log.pcl	cell wall organization; chemical stimulus; stress	11	–	2

Table 4, continued

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
361	Cohen 2002 Cohen BA, Pilpel Y, Mitra RD, Church GM. Discrimination between paralogs using microarray analysis: application to the Yap1p and Yap2p transcriptional networks. <i>Mol Biol Cell</i> 2002;13(5):1608–14.	doi:10.1091/mbc.01-10-0472 PMID 12006656 PMC111130	2010.Cohen02.filter.flt.knn.avg.div.log.pcl	chemical stimulus; oxidative stress; stress; transcription	11	–	2
362	Epstein 2001 Epstein CB, Waddle JA, Hale W 4th, et al., Butow RA. Genome-wide responses to mitochondrial dysfunction. <i>Mol Biol Cell</i> 2001;12(2):297–308.	doi:10.1091/mbc.12.2.297 PMID 11179416 PMC30944	2010.Epstein00.flt.knn.avg.pcl	carbon utilization; respiration	11	–	2
364	Galitski 1999 Galitski T, Saldanha AJ, Styles CA, Lander ES, Fink GR. Ploidy regulation of gene expression. <i>Science</i> 1999;285(5425):251–4.	doi:10.1126/science.285.5425.251 PMID 10398601	2010.ploidy.pcl	cell morphogenesis; mitotic cell cycle	11	–	2
366	Madhani 1999 Madhani HD, Galitski T, Lander ES, Fink GR. Effectors of a developmental mitogen-activated protein kinase cascade revealed by expression signatures of signaling mutants. <i>Proc Natl Acad Sci U S A</i> 1999;96(22):12530–5.	doi:10.1073/pnas.96.22.12530 PMID 10535956 PMC22972	2010.Madhani99.filter.flt.knn.avg.div.log.pcl	filamentous growth; signaling	11	–	2
378	Huang 2004 Huang J, Zhu H, Haggarty SJ, et al., Schreiber SL. Finding new components of the target of rapamycin (TOR) signaling network through chemical genetics and proteome chips. <i>Proc Natl Acad Sci U S A</i> 2004;101(47):16594–9.	doi:10.1073/pnas.0407117101 PMID 15539461 PMC527135	2010.rapamycin.pcl	chemical stimulus; signaling	10	–	2
382	Prinz 2004 Prinz S, Avila-Campillo I, Aldridge C, et al., Galitski T. Control of yeast filamentous-form growth by modules in an integrated molecular network. <i>Genome Res</i> 2004;14(3):380–90.	doi:10.1101/gr.2020604 PMID 14993204 PMC353223	2010.Prinz04.flt.knn.avg.pcl	filamentous growth	10	–	2
384	Travers 2000 Travers KJ, Patil CK, Wodicka L, Lockhart DJ, Weissman JS, Walter P. Functional and genomic analyses reveal an essential coordination between the unfolded protein response and ER-associated degradation. <i>Cell</i> 2000;101(3):249–58.	doi:10.1016/s0092-8674(00)80835-1 PMID 10847680	2010.unfolded_protein_response.pcl	chemical stimulus; response to unfolded protein; stress	10	–	2
399	Lyons 2000 Lyons TJ, Gasch AP, Gaither LA, Botstein D, Brown PO, Eide DJ. Genome-wide characterization of the Zap1p zinc-responsive regulon in yeast. <i>Proc Natl Acad Sci U S A</i> 2000;97(14):7957–62.	doi:10.1073/pnas.97.14.7957 PMID 10884426 PMC16652	2010.Lyons00.flt.knn.avg.pcl	cellular ion homeostasis; metal or metalloid ion stress	9	–	2
408	Angus-Hill 2001 Angus-Hill ML, Schlichter A, Roberts D, Erdjument-Bromage H, Tempst P, Cairns BR. A Rsc3/Rsc30 zinc cluster dimer reveals novel roles for the chromatin remodeler RSC in gene expression and cell cycle control. <i>Mol Cell</i> 2001;7(4):741–51.	doi:10.1016/s1097-2765(01)00219-2 PMID 11336698	2010.Angus01.flt.knn.avg.pcl	chromatin organization	8	1	2
421	Fry 2003 Fry RC, Sambandan TG, Rha C. DNA damage and stress transcripts in <i>Saccharomyces cerevisiae</i> mutant sgs1. <i>Mech Ageing Dev</i> 2003;124(7):839–46.	doi:10.1016/s0047-6374(03)00144-1 PMID 12875747	2010.Fry03.filter.flt.knn.avg.div.log.pcl	DNA damage stimulus; stress	8	–	2

Table 4, continued

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
428	Lee 2000 Lee SE, Pellicoli A, Demeter J, et al., Haber JE. Arrest, adaptation, and recovery following a chromosome double-strand break in <i>Saccharomyces cerevisiae</i> . <i>Cold Spring Harb Symp Quant Biol</i> 2000;65:303–14.	doi:10.1101/sqb.2000.65.303 PMID 12760044	2010.H0_endonuclease.pcl	DNA damage stimulus	8	—	2
433	Ogawa 2000 Ogawa N, DeRisi J, Brown PO. New components of a system for phosphate accumulation and polyphosphate metabolism in <i>Saccharomyces cerevisiae</i> revealed by genomic expression analysis. <i>Mol Biol Cell</i> 2000;11(12):4309–21.	doi:10.1091/mbc.11.12.4309 PMID 11102525 PMC15074	2010.Ogawa00_set1.flt.knn.avg.pcl 2010.Ogawa00_set2.flt.knn.avg.pcl	phosphorus utilization; signaling; starvation	8	—	2
443	Smith 2002 Smith JJ, Marelli M, Christmas RH, et al., Aitchison JD. Transcriptome profiling to identify genes involved in peroxisome assembly and function. <i>J Cell Biol</i> 2002;158(2):259–71.	doi:10.1083/jcb.200204059 PMID 12135984 PMC2173120	2010.peroxisome.pcl	carbon utilization; lipid metabolism	8	—	2
447	Williams 2002 Williams RM, Primig M, Washburn BK, et al., Esposito RE. The Ume6 regulon coordinates metabolic and meiotic gene expression in yeast. <i>Proc Natl Acad Sci U S A</i> 2002;99(21):13431–6.	doi:10.1073/pnas.202495299 PMID 12370439 PMC129690	2010.Williams02.filter.flt.knn.avg.div.log.pcl	carbon utilization; sporulation	8	—	2
452	Bedalov 2001 Bedalov A, Gatbonton T, Irvine WP, Gottschling DE, Simon JA. Identification of a small molecule inhibitor of Sir2p. <i>Proc Natl Acad Sci U S A</i> 2001;98(26):15113–8.	doi:10.1073/pnas.261574398 PMID 11752457 PMC64992	2010.Bedalov01.flt.knn.avg.pcl	chemical stimulus; chromatin organization; histone modification	7	—	2
453	Bro 2003 Bro C, Regenber B, Lagniel G, Labarre J, Montero-Lomeli M, Nielsen J. Transcriptional, proteomic, and metabolic responses to lithium in galactose-grown yeast cells. <i>J Biol Chem</i> 2003;278(34):32141–9.	doi:10.1074/jbc.M304478200 PMID 12791685	2010.Bro03.filter.flt.knn.avg.div.log.pcl	chemical stimulus; metal or metalloid ion stress	7	—	2
455	Chu 1998 Chu S, DeRisi J, Eisen M, et al., Herskowitz I. The transcriptional program of sporulation in budding yeast. <i>Science</i> 1998;282(5389):699–705.	doi:10.1126/science.282.5389.699 PMID 9784122	2010.Chu98.flt.knn.avg.pcl	sporulation	7	—	2
456	DeRisi 1997 DeRisi JL, Iyer VR, Brown PO. Exploring the metabolic and genetic control of gene expression on a genomic scale. <i>Science</i> 1997;278(5338):680–6.	doi:10.1126/science.278.5338.680 PMID 9381177	2010.diauxic.pcl	carbon utilization; diauxic shift; fermentation; respiration	7	—	2
460	Marton 1998 Marton MJ, DeRisi JL, Bennett HA, et al., Friend SH. Drug target validation and identification of secondary drug target effects using DNA microarrays. <i>Nat Med</i> 1998;4(11):1293–301.	doi:10.1038/3282 PMID 9809554	2010.Marton98.log.flt.knn.avg.pcl	chemical stimulus; protein dephosphorylation; signaling	7	—	2
461	Wyrick 1999 Wyrick JJ, Holstege FC, Jennings EG, et al., Young RA. Chromosomal landscape of nucleosome-dependent gene expression and silencing in yeast. <i>Nature</i> 1999;402(6760):418–21.	doi:10.1038/46567 PMID 10586882	2010.histone.pcl	chromatin organization	7	—	2

Table 4, continued

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
462	Yeang 2005 Yeang CH, Mak HC, McCuine S, Workman C, Jaakkola T, Ideker T. Validation and refinement of gene-regulatory pathways on a network of physical interactions. <i>Genome Biol</i> 2005;6(7):R62.	doi:10.1186/gb-2005-6-7-r62 PMID 15998451 PMC1175993	2010.Yeang05.flt.knn.avg.pcl	transcription	7	—	2
466	Rodriguez-Navarro 2004 Rodríguez-Navarro S, Fischer T, Luo MJ, et al., Hurt E. Sus1, a functional component of the SAGA histone acetylase complex and the nuclear pore-associated mRNA export machinery. <i>Cell</i> 2004;116(1):75–86.	doi:10.1016/s0092-8674(03)01025-0 PMID 14718168	2010.RodriguezNavarro04.filter.flt.knn.avg.div.log.pcl	histone modification	6	1	2
479	Gross 2000 Gross C, Kelleher M, Iyer VR, Brown PO, Winge DR. Identification of the copper regulon in <i>Saccharomyces cerevisiae</i> by DNA microarrays. <i>J Biol Chem</i> 2000;275(41):32310–6.	doi:10.1074/jbc.M005946200 PMID 10922376	2010.Gross00.flt.knn.avg.pcl	cellular ion homeostasis; metal or metalloid ion stress	6	—	2
487	Jin 2004 Jin YS, Laplaza JM, Jeffries TW. <i>Saccharomyces cerevisiae</i> engineered for xylose metabolism exhibits a respiratory response. <i>Appl Environ Microbiol</i> 2004;70(11):6816–25.	doi:10.1128/AEM.70.11.6816-6825.2004 PMID 15528549 PMC525251	2010.Jin04.filter.flt.knn.avg.div.log.pcl	carbon utilization; oxygen level alteration; respiration	6	—	2
510	Shakoury-Elizeh 2004 Shakoury-Elizeh M, Tiedeman J, Rashford J, et al., Philpott CC. Transcriptional remodeling in response to iron deprivation in <i>Saccharomyces cerevisiae</i> . <i>Mol Biol Cell</i> 2004;15(3):1233–43.	doi:10.1091/mbc.e03-09-0642 PMID 14668481 PMC363115	2010.ShakouryElizeh04.flt.knn.avg.pcl	metal or metalloid ion stress; starvation	6	—	2
525	Baetz 2001 Baetz K, Moffat J, Haynes J, Chang M, Andrews B. Transcriptional coregulation by the cell integrity mitogen-activated protein kinase Slt2 and the cell cycle regulator Swi4. <i>Mol Cell Biol</i> 2001;21(19):6515–28.	doi:10.1128/MCB.21.19.6515-6528.2001 PMID 11533240 PMC99798	2010.Baetz01.flt.knn.avg.pcl	mitotic cell cycle; signaling	5	4	2
529	Jones 2003 Jones DL, Petty J, Hoyle DC, et al., Stateva LI. Transcriptome profiling of a <i>Saccharomyces cerevisiae</i> mutant with a constitutively activated Ras/cAMP pathway. <i>Physiol Genomics</i> 2003;16(1):107–18.	doi:10.1152/physiolgenomics.00139.2003 PMID 14570984	2010.Jones03_orig.flt.knn.avg.pcl	signaling	5	—	2
531	Roberts 2006 Roberts GG, Hudson AP. Transcriptome profiling of <i>Saccharomyces cerevisiae</i> during a transition from fermentative to glycerol-based respiratory growth reveals extensive metabolic and structural remodeling. <i>Mol Genet Genomics</i> 2006;276(2):170–86.	doi:10.1007/s00438-006-0133-9 PMID 16741729	2010.glucose_glycerol_transition.pcl	carbon utilization; fermentation; respiration	5	—	2
533	van Attikum 2004 van Attikum H, Fritsch O, Hohn B, Gasser SM. Recruitment of the INO80 complex by H2A phosphorylation links ATP-dependent chromatin remodeling with DNA double-strand break repair. <i>Cell</i> 2004;119(6):777–88.	doi:10.1016/j.cell.2004.11.033 PMID 15607975	2010.Ino80Arp8.pcl 2010.MMSresponse.pcl	DNA damage stimulus; chromatin organization	5	—	2

Table 4, continued

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
534	Orlandi 2004 Orlandi I, Bettiga M, Alberghina L, Vai M. Transcriptional profiling of ubp10 null mutant reveals altered subtelomeric gene expression and insurgence of oxidative stress response. <i>J Biol Chem</i> 2004;279(8):6414–25.	doi:10.1074/jbc.M306464200 PMID 14623890	2010.Orlandi04.filter.flt.knn.avg.div.log.pcl	stress; ubiquitin or ULP modification	4	2	2
545	Ferea 1999 Ferea TL, Botstein D, Brown PO, Rosenzweig RF. Systematic changes in gene expression patterns following adaptive evolution in yeast. <i>Proc Natl Acad Sci U S A</i> 1999;96(17):9721–6.	doi:10.1073/pnas.96.17.9721 PMID 10449761 PMC22277	2010.evolution.pcl	evolution	4	–	2
551	Keller 2001 Keller G, Ray E, Brown PO, Winge DR. Haa1, a protein homologous to the copper-regulated transcription factor Ace1, is a novel transcriptional activator. <i>J Biol Chem</i> 2001;276(42):38697–702.	doi:10.1074/jbc.M107131200 PMID 11504737	2010.Keller01.flt.knn.avg.pcl	metal or metalloid ion stress; starvation; stress; transcription	4	–	2
554	Miyake 2004 Miyake T, Reese J, Loch CM, Auble DT, Li R. Genome-wide analysis of ARS (autonomously replicating sequence) binding factor 1 (Abflp)-mediated transcriptional regulation in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2004;279(33):34865–72.	doi:10.1074/jbc.M405156200 PMID 15192094	2010.Miyake04.flt.knn.avg.pcl	transcription	4	–	2
557	Protchenko 2001 Protchenko O, Ferea T, Rashford J, et al., Philpott CC. Three cell wall mannoproteins facilitate the uptake of iron in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2001;276(52):49244–50.	doi:10.1074/jbc.M109220200 PMID 11673473	2010.Protchenko01.flt.knn.avg.pcl	metal or metalloid ion stress	4	–	2
570	Yamamoto 2005 Yamamoto A, Mizukami Y, Sakurai H. Identification of a novel class of target genes and a novel type of binding sequence of heat shock transcription factor in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2005;280(12):11911–9.	doi:10.1074/jbc.M411256200 PMID 15647283	2010.Yamamoto05.filter.flt.knn.avg.div.log.pcl	heat shock; transcription	4	–	2
571	Yun 2000 Yun CW, Ferea T, Rashford J, et al., Philpott CC. Desferrioxamine-mediated iron uptake in <i>Saccharomyces cerevisiae</i> . Evidence for two pathways of iron uptake. <i>J Biol Chem</i> 2000;275(14):10709–15.	doi:10.1074/jbc.275.14.10709 PMID 10744769	2010.Yun00.flt.knn.avg.pcl	cellular ion homeostasis; metal or metalloid ion stress	4	–	2
582	Kuhn 2001 Kuhn KM, DeRisi JL, Brown PO, Sarnow P. Global and specific translational regulation in the genomic response of <i>Saccharomyces cerevisiae</i> to a rapid transfer from a fermentable to a nonfermentable carbon source. <i>Mol Cell Biol</i> 2001;21(3):916–27.	doi:10.1128/MCB.21.3.916-927.2001 PMID 11154278 PMC86682	2010.Kuhn01.flt.knn.avg.pcl	carbon utilization; fermentation; respiration	3	–	2
583	Mizuguchi 2004 Mizuguchi G, Shen X, Landry J, Wu WH, Sen S, Wu C. ATP-driven exchange of histone H2AZ variant catalyzed by SWR1 chromatin remodeling complex. <i>Science</i> 2004;303(5656):343–8.	doi:10.1126/science.1090701 PMID 14645854	2010.swr1_htz1_ino80.pcl	chromatin organization	3	–	2

Table 4, continued

#	Publication	Links	PCL files in the archive	SPELL tags	cond.	KO genes	ch.
590	Cullen 2004 Cullen PJ, Sabbagh W Jr, Graham E, et al., Sprague GF Jr. A signaling mucin at the head of the Cdc42- and MAPK-dependent filamentous growth pathway in yeast. <i>Genes Dev</i> 2004;18(14):1695–708.	doi:10.1101/gad.1178604 PMID 15256499 PMC478191	2010.glycosylation.pcl	filamentous growth; protein glycosylation; signaling	2	–	2

4 All publications

Table 5 has one row per study folder. The 48 studies with a new genotype (Sec. 2.1) come first, ordered by the number of new genotypes. The remaining studies are ordered by the number of conditions, which is the number of expression profiles a study contributes. Authors are given in full up to six names; a longer list is cut to the first three and the last.

Table 5: Every study in the SPELL archive. The studies that add a perturbed genotype come first, most new genotypes first; the rest follow, most conditions first. *new* is the number of genotypes, deletion or gene-named, found in neither Kemmeren 2014 nor Sameith 2015 (Sec. 2.1); a dash is zero. The bold label is the folder’s first author and year. *Links* gives the DOI, the PubMed record and, where one exists, the PubMed Central copy. *GEO* is the series named in the README, followed by any further series that appear in the study’s PCL filenames; **none** marks a README with **GEO ID: N/A**, and those rows are repeated in Table 4. *cond.* is the number of conditions summed over the study’s PCL matrices. *KO cond.* and *KO genes* are the deletion conditions and distinct deleted genes estimated from the column headers (Sec. 2), and *named* is the number of genes in headers that are bare gene names; a dash is zero. *PCL* is the number of dataset matrices and *ch.* the microarray channel count the README reports. Every link is clickable.

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
1	Mnaimneh 2004 Mnaimneh S, Davierwala AP, Haynes J, et al., Hughes TR. Exploration of essential gene functions via titratable promoter alleles. <i>Cell</i> 2004;118(1):31–44.	doi:10.1016/j.cell.2004.06.013 PMID 15242642	none	214	215	–	–	215	1	2
2	Hughes 2000 Hughes TR, Marton MJ, Jones AR, et al., Friend SH. Functional discovery via a compendium of expression profiles. <i>Cell</i> 2000;102(1):109–26.	doi:10.1016/s0092-8674(00)00015-5 PMID 10929718	none	157	300	–	–	276	1	2
3	van Wageningen 2010 van Wageningen S, Kemmeren P, Lijnzaad P, et al., Holstege FC. Functional overlap and regulatory links shape genetic interactions between signaling pathways. <i>Cell</i> 2010;143(6):991–1004.	doi:10.1016/j.cell.2010.11.021 PMID 21145464 PMC3073509	GSE25644	14	377	290	133	–	1	2
4	Hu 2007 Hu Z, Killion PJ, Iyer VR. Genetic reconstruction of a functional transcriptional regulatory network. <i>Nat Genet</i> 2007;39(5):683–7.	doi:10.1038/ng2012 PMID 17417638	GSE4654	14	269	269	269	–	1	2
5	Zhou 2011 Zhou X, O’Shea EK. Integrated approaches reveal determinants of genome-wide binding and function of the transcription factor Pho4. <i>Mol Cell</i> 2011;42(6):826–36.	doi:10.1016/j.molcel.2011.05.025 PMID 21700227 PMC3127084	GSE23580	4	66	14	5	–	1	2
6	Braberg 2013 Braberg H, Jin H, Moehle EA, et al., Krogan NJ. From structure to systems: high-resolution, quantitative genetic analysis of RNA polymerase II. <i>Cell</i> 2013;154(4):775–88.	doi:10.1016/j.cell.2013.07.033 PMID 23932120 PMC3932829	GSE47429	4	57	55	4	–	1	2
7	Verzijlbergen 2011 Verzijlbergen KF, van Welsom T, Sie D, et al., van Leeuwen F. A barcode screen for epigenetic regulators reveals a role for the NuB4/HAT-B histone acetyltransferase complex in histone turnover. <i>PLoS Genet</i> 2011;7(10):e1002284.	doi:10.1371/journal.pgen.1002284 PMID 21998594 PMC3188528	GSE30168	4	36	24	3	–	1	2
8	Apweiler 2012 Apweiler E, Sameith K, Margaritis T, et al., Kemmeren P. Yeast glucose pathways converge on the transcriptional regulation of trehalose biosynthesis. <i>BMC Genomics</i> 2012;13:239.	doi:10.1186/1471-2164-13-239 PMID 22697265 PMC3472246	GSE33098 GSE33097	3	275	183	91	–	2	2
9	Morillo-Huesca 2010 Morillo-Huesca M, Clemente-Ruiz M, Andújar E, Prado F. The SWR1 histone replacement complex causes genetic instability and genome-wide transcription misregulation in the absence of H2A.Z. <i>PLoS One</i> 2010;5(8):e12143.	doi:10.1371/journal.pone.0012143 PMID 20711347 PMC2920830	GSE21571	3	24	21	4	–	1	1
10	Thibault 2011 Thibault G, Ismail N, Ng DT. The unfolded protein response supports cellular robustness as a broad-spectrum compensatory pathway. <i>Proc Natl Acad Sci U S A</i> 2011;108(51):20597–602.	doi:10.1073/pnas.1117184109 PMID 22143797 PMC3251055	GSE33844	3	15	9	3	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
11	Gomez-Gonzalez 2011 Gómez-González B, García-Rubio M, Bermejo R, et al., Aguilera A. Genome-wide function of THO/TREX in active genes prevents R-loop-dependent replication obstacles. <i>EMBO J</i> 2011;30(15):3106–19.	doi:10.1038/emboj.2011.206 PMID 21701562 PMC3160181	GSE24802	3	10	7	3	–	1	1
12	Cain 2012 Cain CW, Lohse MB, Homann OR, Sil A, Johnson AD. A conserved transcriptional regulator governs fungal morphology in widely diverged species. <i>Genetics</i> 2012;190(2):511–21.	doi:10.1534/genetics.111.134080 PMID 22095082 PMC3276625	GSE32550	3	8	6	2	–	1	2
13	Cho 2019 Cho CY, Kelliher CM, Haase SB. The cell-cycle transcriptional network generates and transmits a pulse of transcription once each cell cycle. <i>Cell Cycle</i> 2019;18(4):363–378.	doi:10.1080/15384101.2019.1570655 PMID 30668223 PMC6422481	GSE75694	2	270	96	4	–	1	1
14	Zheng 2010 Zheng W, Zhao H, Mancera E, Steinmetz LM, Snyder M. Genetic analysis of variation in transcription factor binding in yeast. <i>Nature</i> 2010;464(7292):1187–91.	doi:10.1038/nature08934 PMID 20237471 PMC2941147	GSE19634	2	48	–	–	2	1	1
15	Gitter 2013 Gitter A, Carmi M, Barkai N, Bar-Joseph Z. Linking the signaling cascades and dynamic regulatory networks controlling stress responses. <i>Genome Res</i> 2013;23(2):365–76.	doi:10.1101/gr.138628.112 PMID 23064748 PMC3561877	GSE28213	2	16	12	6	–	1	1
16	Li 2010 Li M, Petteys BJ, McClure JM, et al., Smith JS. Thiamine biosynthesis in <i>Saccharomyces cerevisiae</i> is regulated by the NAD ⁺ -dependent histone deacetylase Hst1. <i>Mol Cell Biol</i> 2010;30(13):3329–41.	doi:10.1128/MCB.01590-09 PMID 20439498 PMC2897578	GSE18488	2	14	8	4	–	1	1
17	Hong 2012 Hong KK, Nielsen J. Recovery of phenotypes obtained by adaptive evolution through inverse metabolic engineering. <i>Appl Environ Microbiol</i> 2012;78(21):7579–86.	doi:10.1128/AEM.01444-12 PMID 22904057 PMC3485731	GSE36118	2	14	–	–	2	1	1
18	Hazelwood 2008 Hazelwood LA, Daran JM, van Maris AJ, Pronk JT, Dickinson JR. The Ehrlich pathway for fusel alcohol production: a century of research on <i>Saccharomyces cerevisiae</i> metabolism. <i>Appl Environ Microbiol</i> 2008;74(8):2259–66.	doi:10.1128/AEM.02625-07 PMID 18281432 PMC2293160	GSE9590	2	12	3	2	–	1	1
19	Mutiu 2007 Mutiu AI, Hoke SM, Genereaux J, Liang G, Brandl CJ. The role of histone ubiquitylation and deubiquitylation in gene expression as determined by the analysis of an HTB1(K123R) <i>Saccharomyces cerevisiae</i> strain. <i>Mol Genet Genomics</i> 2007;277(5):491–506.	doi:10.1007/s00438-007-0212-6 PMID 17447102	GSE6316	2	9	9	3	–	1	2
20	Belli 2004 Bellí G, Molina MM, García-Martínez J, Pérez-Ortín JE, Herrero E. <i>Saccharomyces cerevisiae</i> glutaredoxin 5-deficient cells subjected to continuous oxidizing conditions are affected in the expression of specific sets of genes. <i>J Biol Chem</i> 2004;279(13):12386–95.	doi:10.1074/jbc.M311879200 PMID 14722110	none	2	9	6	2	–	1	2
21	Chen 2017 Chen L, Wang M, Hou J, et al., Bao X. HAL2 overexpression induces iron acquisition in <i>bdf1Δ</i> cells and enhances their salt resistance. <i>Curr Genet</i> 2017;63(2):229–239.	doi:10.1007/s00294-016-0628-9 PMID 27387517	GSE75828	2	5	5	2	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
22	Johnson 2014 Johnson JE, Johnson FB. Methionine restriction activates the retrograde response and confers both stress tolerance and lifespan extension to yeast, mouse and human cells. <i>PLoS One</i> 2014;9(5):e97729.	doi:10.1371/journal.pone.0097729 PMID 24830393 PMC4022668	GSE56641	2	3	2	2	–	1	1
23	Huisinga 2007 Huisinga KL, Pugh BF. A TATA binding protein regulatory network that governs transcription complex assembly. <i>Genome Biol</i> 2007;8(4):R46.	doi:10.1186/gb-2007-8-4-r46 PMID 17407552 PMC1896006	GSE7385	1	149	63	1	–	1	2
24	Levy 2007 Levy S, Ihmels J, Carmi M, Weinberger A, Friedlander G, Barkai N. Strategy of transcription regulation in the budding yeast. <i>PLoS One</i> 2007;2(2):e250.	doi:10.1371/journal.pone.0000250 PMID 17327914 PMC1803021	GSE6302	1	95	10	1	–	11	2
25	Munding 2010 Munding EM, Igel AH, Shiue L, Dorigi KM, Treviño LR, Ares M Jr. Integration of a splicing regulatory network within the meiotic gene expression program of <i>Saccharomyces cerevisiae</i> . <i>Genes Dev</i> 2010;24(23):2693–704.	doi:10.1101/gad.1977410 PMID 21123654 PMC2994042	GSE24675 GSE24681	1	48	20	6	–	2	2
26	Garcia-Martínez 2016 García-Martínez J, Delgado-Ramos L, Ayala G, et al., Pérez-Ortín JE. The cellular growth rate controls overall mRNA turnover, and modulates either transcription or degradation rates of particular gene regulons. <i>Nucleic Acids Res</i> 2016;44(8):3643–58.	doi:10.1093/nar/gkv1512 PMID 26717982 PMC4856968	GSE63729 GSE57467 GSE65225	1	32	2	1	–	3	1, 2
27	Ansari 2012 Ansari SA, Ganapathi M, Benschop JJ, Holstege FC, Wade JT, Morse RH. Distinct role of Mediator tail module in regulation of SAGA-dependent, TATA-containing genes in yeast. <i>EMBO J</i> 2012;31(1):44–57.	doi:10.1038/emboj.2011.362 PMID 21971086 PMC3252575	GSE31774	1	21	12	2	–	1	2
28	Borde 2009 Borde V, Robine N, Lin W, Bonfils S, Géli V, Nicolas A. Histone H3 lysine 4 trimethylation marks meiotic recombination initiation sites. <i>EMBO J</i> 2009;28(2):99–111.	doi:10.1038/emboj.2008.257 PMID 19078966 PMC2634730	GSE10948 GSE10944 GSE10947	1	21	7	2	–	3	2
29	Huang 2002 Huang D, Moffat J, Andrews B. Dissection of a complex phenotype by functional genomics reveals roles for the yeast cyclin-dependent protein kinase Pho85 in stress adaptation and cell integrity. <i>Mol Cell Biol</i> 2002;22(14):5076–88.	doi:10.1128/MCB.22.14.5076-5088.2002 PMID 12077337 PMC139770	none	1	20	12	12	–	1	2
30	Thibault 2012 Thibault G, Shui G, Kim W, et al., Ng DT. The membrane stress response buffers lethal effects of lipid disequilibrium by reprogramming the protein homeostasis network. <i>Mol Cell</i> 2012;48(1):16–27.	doi:10.1016/j.molcel.2012.08.016 PMID 23000174 PMC3496426	GSE39419	1	16	9	3	–	1	2
31	Pitkanen 2004 Pitkänen JP, Törmä A, Alf S, Huopaniemi L, Mattila P, Renkonen R. Excess mannose limits the growth of phosphomannose isomerase PMI40 deletion strain of <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2004;279(53):55737–43.	doi:10.1074/jbc.M410619200 PMID 15520001	none	1	15	15	1	–	1	2
32	Barbara 2007 Barbara KE, Haley TM, Willis KA, Santangelo GM. The transcription factor Gcr1 stimulates cell growth by participating in nutrient-responsive gene expression on a global level. <i>Mol Genet Genomics</i> 2007;277(2):171–88.	doi:10.1007/s00438-006-0182-0 PMID 17124610	GSE5027	1	12	12	1	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
33	Crosas 2015 Crosas E, Sumoy L, González E, et al., Fernández MR. The yeast ζ -crystallin/NADPH:quinone oxidoreductase (Zta1p) is under nutritional control by the target of rapamycin pathway and is involved in the regulation of argininosuccinate lyase mRNA half-life. <i>FEBS J</i> 2015;282(10):1953–64.	doi:10.1111/febs.13246 PMID 25715111	GSE45297	1	12	12	1	–	1	2
34	Höpfler 2019 Höpfler M, Kern MJ, Straub T, et al., Jentsch S. Slx5/Slx8-dependent ubiquitin hotspots on chromatin contribute to stress tolerance. <i>EMBO J</i> 2019;38(11).	doi:10.15252/embj.2018100368 PMID 31015336 PMC6545562	GSE118817	1	12	3	1	–	1	1
35	Alkim 2013 Alkim C, Benbadis L, Yilmaz U, Cakar ZP, François JM. Mechanisms other than activation of the iron regulon account for the hyper-resistance to cobalt of a <i>Saccharomyces cerevisiae</i> strain obtained by evolutionary engineering. <i>Metalomics</i> 2013;5(8):1043–60.	doi:10.1039/c3mt00107e PMID 23864114	GSE39185	1	9	3	1	–	1	1
36	Lee 2005 Lee A, Henras AK, Chanfreau G. Multiple RNA surveillance pathways limit aberrant expression of iron uptake mRNAs and prevent iron toxicity in <i>S. cerevisiae</i> . <i>Mol Cell</i> 2005;19(1):39–51.	doi:10.1016/j.molcel.2005.05.021 PMID 15989963	none	1	9	4	1	–	1	2
37	Mendiratta 2006 Mendiratta G, Eriksson PR, Shen CH, Clark DJ. The DNA-binding domain of the yeast Spt10p activator includes a zinc finger that is homologous to foamy virus integrase. <i>J Biol Chem</i> 2006;281(11):7040–8.	doi:10.1074/jbc.M511416200 PMID 16415340	GSE3354	1	9	3	1	–	1	1
38	Leadsham 2010 Leadsham JE, Gourlay CW. cAMP/PKA signaling balances respiratory activity with mitochondria dependent apoptosis via transcriptional regulation. <i>BMC Cell Biol</i> 2010;11:92.	doi:10.1186/1471-2121-11-92 PMID 21108829 PMC3001716	GSE25541	1	8	5	2	–	1	1
39	Revilleza 2022 Revilleza JEC, Sato M, Irie K, Suda Y, Mizuno T, Irie K. Regulation of CLB6 expression by the cytoplasmic deadenylase Ccr4 through its coding and 3' UTR regions. <i>PLoS One</i> 2022;17(5):e0268283.	doi:10.1371/journal.pone.0268283 PMID 35522675 PMC9075657	GSE198743	1	6	4	2	–	1	1
40	Eriksson 2005 Eriksson PR, Mendiratta G, McLaughlin NB, et al., Clark DJ. Global regulation by the yeast Spt10 protein is mediated through chromatin structure and the histone upstream activating sequence elements. <i>Mol Cell Biol</i> 2005;25(20):9127–37.	doi:10.1128/MCB.25.20.9127-9137.2005 PMID 16199888 PMC1265784	none	1	6	3	1	–	1	2
41	Gaillard 2019 Gaillard H, Santos-Pereira JM, Aguilera A. The Nup84 complex coordinates the DNA damage response to warrant genome integrity. <i>Nucleic Acids Res</i> 2019;47(8):4054–4067.	doi:10.1093/nar/gkz066 PMID 30715474 PMC6486642	GSE113118	1	6	6	1	–	1	1
42	Garca-Rodriguez 2015 García-Rodríguez N, Manzano-López J, Muñoz-Bravo M, Fernández-García E, Muñoz M, Wellinger RE. Manganese redistribution by calcium-stimulated vesicle trafficking bypasses the need for P-type ATPase function. <i>J Biol Chem</i> 2015;290(15):9335–47.	doi:10.1074/jbc.M114.616334 PMID 25713143 PMC4392241	GSE62161	1	6	3	1	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
43	Garcia-Rodriguez 2012 García-Rodríguez N, Díaz de la Loza Mdel C, Andreson B, Monje-Casas F, Rothstein R, Wellinger RE. Impaired manganese metabolism causes mitotic misregulation. <i>J Biol Chem</i> 2012;287(22):18717–29.	doi:10.1074/jbc.M112.358309 PMID 22493290 PMC3365726	GSE29420	1	6	3	1	–	1	1
44	Gavalda 2016 Gavaldá S, Santos-Pereira JM, García-Rubio ML, Luna R, Aguilera A. Excess of Yra1 RNA-Binding Factor Causes Transcription-Dependent Genome Instability, Replication Impairment and Telomere Shortening. <i>PLoS Genet</i> 2016;12(4):e1005966.	doi:10.1371/journal.pgen.1005966 PMID 27035147 PMC4818039	GSE68487	1	6	3	1	–	1	1
45	Rudra 2005 Rudra D, Zhao Y, Warner JR. Central role of Ifh1p-Fhl1p interaction in the synthesis of yeast ribosomal proteins. <i>EMBO J</i> 2005;24(3):533–42.	doi:10.1038/sj.emboj.7600553 PMID 15692568 PMC548658	none	1	6	2	1	–	1	1
46	Lai 2013 Lai X, Beilharz T, Au WC, et al., Heierhorst J. Yeast hEST1A/B (SMG5/6)-like proteins contribute to environment-sensing adaptive gene expression responses. <i>G3 (Bethesda)</i> 2013;3(10):1649–59.	doi:10.1534/g3.113.006924 PMID 23893744 PMC3789790	GSE48956	1	4	4	2	–	1	2
47	Sariki 2016 Sariki SK, Sahu PK, Golla U, Singh V, Azad GK, Tomar RS. Sen1, the homolog of human Senataxin, is critical for cell survival through regulation of redox homeostasis, mitochondrial function, and the TOR pathway in <i>Saccharomyces cerevisiae</i> . <i>FEBS J</i> 2016;283(22):4056–4083.	doi:10.1111/febs.13917 PMID 27718307	GSE75447	1	4	2	1	–	1	1
48	Kang 2014 Kang HJ, Jeong SJ, Kim KN, et al., Yun CW. A novel protein, Pho92, has a conserved YTH domain and regulates phosphate metabolism by decreasing the mRNA stability of PHO4 in <i>Saccharomyces cerevisiae</i> . <i>Biochem J</i> 2014;457(3):391–400.	doi:10.1042/BJ20130862 PMID 24206186	GSE50086	1	2	2	1	–	1	2
49	Kemmeren 2014 Kemmeren P, Sameith K, van de Pasch LA, et al., Holstege FC. Large-scale genetic perturbations reveal regulatory networks and an abundance of gene-specific repressors. <i>Cell</i> 2014;157(3):740–52.	doi:10.1016/j.cell.2014.02.054 PMID 24766815	GSE42241 GSE42215 GSE42217 GSE42240 GSE45115	–	672	–	–	–	5	2
50	Aragon 2008 Aragon AD, Rodriguez AL, Meirelles O, et al., Werner-Washburne M. Characterization of differentiated quiescent and nonquiescent cells in yeast stationary-phase cultures. <i>Mol Biol Cell</i> 2008;19(3):1271–80.	doi:10.1091/mbc.e07-07-0666 PMID 18199684 PMC2262958	GSE8624	–	414	–	–	–	3	2
51	Lenstra 2011 Lenstra TL, Benschop JJ, Kim T, et al., Holstege FC. The specificity and topology of chromatin interaction pathways in yeast. <i>Mol Cell</i> 2011;42(4):536–49.	doi:10.1016/j.molcel.2011.03.026 PMID 21596317 PMC4435841	GSE25909	–	313	313	162	–	1	2
52	Sameith 2015 Sameith K, Amini S, Groot Koerkamp MJ, et al., Kemmeren P. A high-resolution gene expression atlas of epistasis between gene-specific transcription factors exposes potential mechanisms for genetic interactions. <i>BMC Biol</i> 2015;13:112.	doi:10.1186/s12915-015-0222-5 PMID 26700642 PMC4690272	GSE42536	–	287	287	82	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
53	Chua 2006 Chua G, Morris QD, Sopko R, et al., Hughes TR. Identifying transcription factor functions and targets by phenotypic activation. <i>Proc Natl Acad Sci U S A</i> 2006;103(32):12045–50.	doi:10.1073/pnas.0605140103 PMID 16880382 PMC1567694	GSE5499	–	270	102	51	–	2	2
54	Smith 2008 Smith EN, Kruglyak L. Gene-environment interaction in yeast gene expression. <i>PLoS Biol</i> 2008;6(4):e83.	doi:10.1371/journal.pbio.0060083 PMID 18416601 PMC2292755	GSE9376	–	246	–	–	–	1	2
55	Lai 2008 Lai LC, Kissinger MT, Burke PV, Kwast KE. Comparison of the transcriptomic "stress response" evoked by antimycin A and oxygen deprivation in <i>Saccharomyces cerevisiae</i> . <i>BMC Genomics</i> 2008;9:627.	doi:10.1186/1471-2164-9-627 PMID 19105839 PMC2637875	GSE3706	–	227	–	–	–	1	2
56	Aragon 2006 Aragon AD, Quiñones GA, Thomas EV, Roy S, Werner-Washburne M. Release of extraction-resistant mRNA in stationary phase <i>Saccharomyces cerevisiae</i> produces a massive increase in transcript abundance in response to stress. <i>Genome Biol</i> 2006;7(2):R9.	doi:10.1186/gb-2006-7-2-r9 PMID 16507144 PMC1431719	GSE3729	–	219	–	–	–	7	1
57	Lewis 2014 Lewis JA, Broman AT, Will J, Gasch AP. Genetic architecture of ethanol-responsive transcriptome variation in <i>Saccharomyces cerevisiae</i> strains. <i>Genetics</i> 2014;198(1):369–82.	doi:10.1534/genetics.114.167429 PMID 24970865 PMC4174948	GSE54196	–	196	–	–	–	1	2
58	Simola 2010 Simola DF, Francis C, Sniegowski PD, Kim J. Heterochronic evolution reveals modular timing changes in budding yeast transcriptomes. <i>Genome Biol</i> 2010;11(10):R105.	doi:10.1186/gb-2010-11-10-r105 PMID 20969771 PMC3218661	GSE24771 GSE24770	–	196	–	–	–	2	2
59	Park 2023 Park ZM, Remillard M, Belnap E, Rose MD. Kar4 is required for the normal pattern of meiotic gene expression. <i>PLoS Genet</i> 2023;19(8):e1010898.	doi:10.1371/journal.pgen.1010898 PMID 37639444 PMC10491391	GSE220125	–	175	–	–	–	1	2
60	Knijnenburg 2009 Knijnenburg TA, Daran JM, van den Broek MA, et al., Wessels LF. Combinatorial effects of environmental parameters on transcriptional regulation in <i>Saccharomyces cerevisiae</i> : a quantitative analysis of a compendium of chemostat-based transcriptome data. <i>BMC Genomics</i> 2009;10:53.	doi:10.1186/1471-2164-10-53 PMID 19173729 PMC2640415	GSE11452	–	170	–	–	–	1	1
61	Bristow 2014 Bristow SL, Leman AR, Simmons Kovacs LA, Deckard A, Harer J, Haase SB. Checkpoints couple transcription network oscillator dynamics to cell-cycle progression. <i>Genome Biol</i> 2014;15(9):446.	doi:10.1186/s13059-014-0446-7 PMID 25200947 PMC4180952	GSE49650	–	160	–	–	–	1	1
62	Cho 2017 Cho CY, Motta FC, Kelliher CM, Deckard A, Haase SB. Reconciling conflicting models for global control of cell-cycle transcription. <i>Cell Cycle</i> 2017;16(20):1965–1978.	doi:10.1080/15384101.2017.1367073 PMID 28934013 PMC5638368	GSE96997	–	144	–	–	–	1	1
63	Gasch 2000 Gasch AP, Spellman PT, Kao CM, et al., Brown PO. Genomic expression programs in the response of yeast cells to environmental changes. <i>Mol Biol Cell</i> 2000;11(12):4241–57.	doi:10.1091/mbc.11.12.4241 PMID 11102521 PMC15070	none	–	138	–	–	–	20	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
64	Wenger 2010 Wenger JW, Schwartz K, Sherlock G. Bulk segregant analysis by high-throughput sequencing reveals a novel xylose utilization gene from <i>Saccharomyces cerevisiae</i> . <i>PLoS Genet</i> 2010;6(5):e1000942.	doi:10.1371/journal.pgen.1000942 PMID 20485559 PMC2869308	GSE19116	—	135	—	—	—	1	2
65	O'Rourke 2004 O'Rourke SM, Herskowitz I. Unique and redundant roles for HOG MAPK pathway components as revealed by whole-genome expression analysis. <i>Mol Biol Cell</i> 2004;15(2):532–42.	doi:10.1091/mbc.e03-07-0521 PMID 14595107 PMC329229	none	—	133	—	—	—	1	2
66	Brem 2005 Brem RB, Kruglyak L. The landscape of genetic complexity across 5,700 gene expression traits in yeast. <i>Proc Natl Acad Sci U S A</i> 2005;102(5):1572–7.	doi:10.1073/pnas.0408709102 PMID 15659551 PMC547855	none	—	131	—	—	—	1	2
67	Friedlander 2006 Friedlander G, Joseph-Strauss D, Carmi M, Zenvirth D, Simchen G, Barkai N. Modulation of the transcription regulatory program in yeast cells committed to sporulation. <i>Genome Biol</i> 2006;7(3):R20.	doi:10.1186/gb-2006-7-3-r20 PMID 16542486 PMC1557749	GSE3820 GSE3814 GSE3815 GSE3816	—	129	—	—	—	7	2
68	Jung 2015 Jung I, Seo J, Lee HS, Stanton LW, Kim D, Choi JK. Global mapping of the regulatory interactions of histone residues. <i>FEBS Lett</i> 2015;589(24 Pt B):4061–70.	doi:10.1016/j.febslet.2015.11.016 PMID 26602082	GSE29059	—	127	—	—	—	1	1
69	Saldanha 2004 Saldanha AJ, Brauer MJ, Botstein D. Nutritional homeostasis in batch and steady-state culture of yeast. <i>Mol Biol Cell</i> 2004;15(9):4089–104.	doi:10.1091/mbc.e04-04-0306 PMID 15240820 PMC515343	none	—	124	—	—	—	5	2
70	Alejandro-Orsorio 2009 Alejandro-Orsorio AL, Huebert DJ, Porcaro DT, et al., Gasch AP. The histone deacetylase Rpd3p is required for transient changes in genomic expression in response to stress. <i>Genome Biol</i> 2009;10(5):R57.	doi:10.1186/gb-2009-10-5-r57 PMID 19470158 PMC2718523	GSE9108	—	109	—	—	—	1	2
71	Carter 2012 Carter GW, Hays M, Sherman A, Galitski T. Use of pleiotropy to model genetic interactions in a population. <i>PLoS Genet</i> 2012;8(10):e1003010.	doi:10.1371/journal.pgen.1003010 PMID 23071457 PMC3469415	GSE34787	—	96	—	—	—	1	1
72	Guan 2006 Guan Q, Zheng W, Tang S, et al., Culbertson MR. Impact of nonsense-mediated mRNA decay on the global expression profile of budding yeast. <i>PLoS Genet</i> 2006;2(11):e203.	doi:10.1371/journal.pgen.0020203 PMID 17166056 PMC1657058	GSE3076	—	96	—	—	—	1	1
73	Lu 2016 Lu YJ, Swamy KB, Leu JY. Experimental Evolution Reveals Interplay between Sch9 and Polyploid Stability in Yeast. <i>PLoS Genet</i> 2016;12(11):e1006409.	doi:10.1371/journal.pgen.1006409 PMID 27812096 PMC5094715	GSE76596 GSE76597	—	93	—	—	—	2	2
74	Landry 2006 Landry CR, Oh J, Hartl DL, Cavalieri D. Genome-wide scan reveals that genetic variation for transcriptional plasticity in yeast is biased towards multi-copy and dispensable genes. <i>Gene</i> 2006;366(2):343–51.	doi:10.1016/j.gene.2005.10.042 PMID 16427747	GSE3021	—	91	—	—	—	1	2
75	Carter 2007 Carter GW, Prinz S, Neou C, et al., Galitski T. Prediction of phenotype and gene expression for combinations of mutations. <i>Mol Syst Biol</i> 2007;3:96.	doi:10.1038/msb4100137 PMID 17389876 PMC1847951	GSE5938	—	90	—	—	—	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
76	Capaldi 2008 Capaldi AP, Kaplan T, Liu Y, et al., O'Shea EK. Structure and function of a transcriptional network activated by the MAPK Hog1. <i>Nat Genet</i> 2008;40(11):1300–6.	doi:10.1038/ng.235 PMID 18931682 PMC2825711	GSE12270	–	85	38	3	–	1	2
77	Chin 2012 Chin SL, Marcus IM, Klevecz RR, Li CM. Dynamics of oscillatory phenotypes in <i>Saccharomyces cerevisiae</i> reveal a network of genome-wide transcriptional oscillators. <i>FEBS J</i> 2012;279(6):1119–30.	doi:10.1111/j.1742-4658.2012.08508.x PMID 22289124 PMC3368069	GSE30052 GSE30051	–	81	–	–	–	2	1
78	Berry 2008 Berry DB, Gasch AP. Stress-activated genomic expression changes serve a preparative role for impending stress in yeast. <i>Mol Biol Cell</i> 2008;19(11):4580–7.	doi:10.1091/mbc.e07-07-0680 PMID 18753408 PMC2575158	GSE8335	–	77	–	–	–	14	2
79	Chechik 2008 Chechik G, Oh E, Rando O, Weissman J, Regev A, Koller D. Activity motifs reveal principles of timing in transcriptional control of the yeast metabolic network. <i>Nat Biotechnol</i> 2008;26(11):1251–9.	doi:10.1038/nbt.1499 PMID 18953355 PMC2651818	GSE13219	–	72	–	–	–	13	2
80	Duc 2017 Duc C, Pradal M, Sanchez I, Noble J, Tesnière C, Blondin B. A set of nutrient limitations trigger yeast cell death in a nitrogen-dependent manner during wine alcoholic fermentation. <i>PLoS One</i> 2017;12(9):e0184838.	doi:10.1371/journal.pone.0184838 PMID 28922393 PMC5602661	GSE95152	–	72	–	–	–	1	1
81	Jaehnig 2013 Jaehnig EJ, Kuo D, Hombauer H, Ideker TG, Kolodner RD. Checkpoint kinases regulate a global network of transcription factors in response to DNA damage. <i>Cell Rep</i> 2013;4(1):174–88.	doi:10.1016/j.celrep.2013.05.041 PMID 23810556 PMC3855057	GSE40351	–	72	–	–	–	1	2
82	Shapira 2004 Shapira M, Segal E, Botstein D. Disruption of yeast forkhead-associated cell cycle transcription by oxidative stress. <i>Mol Biol Cell</i> 2004;15(12):5659–69.	doi:10.1091/mbc.e04-04-0340 PMID 15371544 PMC532044	none	–	70	–	–	–	1	2
83	Joseph-Strauss 2007 Joseph-Strauss D, Zenvirth D, Simchen G, Barkai N. Spore germination in <i>Saccharomyces cerevisiae</i> : global gene expression patterns and cell cycle landmarks. <i>Genome Biol</i> 2007;8(11):R241.	doi:10.1186/gb-2007-8-11-r241 PMID 17999778 PMC2258198	GSE7393	–	69	–	–	–	6	2
84	Baptista 2017 Baptista T, Grünberg S, Minoungou N, et al., Tora L. SAGA Is a General Cofactor for RNA Polymerase II Transcription. <i>Mol Cell</i> 2017;68(1):130–143. e5.	doi:10.1016/j.molcel.2017.08.016 PMID 28918903 PMC5632562	GSE96849	–	68	–	–	–	1	1
85	Singh 2005 Singh J, Kumar D, Ramakrishnan N, et al., Helm RF. Transcriptional response of <i>Saccharomyces cerevisiae</i> to desiccation and rehydration. <i>Appl Environ Microbiol</i> 2005;71(12):8752–63.	doi:10.1128/AEM.71.12.8752-8763.2005 PMID 16332871 PMC1317403	GSE1313 GSE1311 GSE1312	–	66	–	–	–	3	1
86	Gresham 2008 Gresham D, Desai MM, Tucker CM, et al., Dunham MJ. The repertoire and dynamics of evolutionary adaptations to controlled nutrient-limited environments in yeast. <i>PLoS Genet</i> 2008;4(12):e1000303.	doi:10.1371/journal.pgen.1000303 PMID 19079573 PMC2586090	GSE13435	–	63	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
87	Torres 2007 Torres EM, Sokolsky T, Tucker CM, et al., Amon A. Effects of aneuploidy on cellular physiology and cell division in haploid yeast. <i>Science</i> 2007;317(5840):916–24.	doi:10.1126/science.1142210 PMID 17702937	GSE7812	–	62	–	–	–	2	2
88	Travesa 2012 Travesa A, Kuo D, de Bruin RA, et al., Wittenberg C. DNA replication stress differentially regulates G1/S genes via Rad53-dependent inactivation of Nrm1. <i>EMBO J</i> 2012;31(7):1811–22.	doi:10.1038/emboj.2012.28 PMID 22333915 PMC3321207	GSE33695	–	61	–	–	–	1	2
89	Orlando 2008 Orlando DA, Lin CY, Bernard A, et al., Haase SB. Global control of cell-cycle transcription by coupled CDK and network oscillators. <i>Nature</i> 2008;453(7197):944–7.	doi:10.1038/nature06955 PMID 18463633 PMC2736871	GSE8799	–	60	–	–	–	1	1
90	Shalem 2011 Shalem O, Groisman B, Choder M, Dahan O, Pilpel Y. Transcriptome kinetics is governed by a genome-wide coupling of mRNA production and degradation: a role for RNA Pol II. <i>PLoS Genet</i> 2011;7(9):e1002273.	doi:10.1371/journal.pgen.1002273 PMID 21931566 PMC3169527	GSE26829	–	58	–	–	–	1	1
91	Wang 2002 Wang Y, Liu CL, Storey JD, Tibshirani RJ, Herschlag D, Brown PO. Precision and functional specificity in mRNA decay. <i>Proc Natl Acad Sci U S A</i> 2002;99(9):5860–5.	doi:10.1073/pnas.092538799 PMID 11972065 PMC122867	GSE2986	–	58	–	–	–	1	2
92	Brion 2013 Brion C, Ambroset C, Sanchez I, Legras JL, Blondin B. Differential adaptation to multi-stressed conditions of wine fermentation revealed by variations in yeast regulatory networks. <i>BMC Genomics</i> 2013;14:681.	doi:10.1186/1471-2164-14-681 PMID 24094006 PMC3870980	GSE41025	–	56	–	–	–	1	1
93	Hao 2012 Hao N, O’Shea EK. Signal-dependent dynamics of transcription factor translocation controls gene expression. <i>Nat Struct Mol Biol</i> 2011;19(1):31–9.	doi:10.1038/nsmb.2192 PMID 22179789 PMC3936303	GSE32703	–	56	–	–	–	1	2
94	Dion 2005 Dion MF, Altschuler SJ, Wu LF, Rando OJ. Genomic characterization reveals a simple histone H4 acetylation code. <i>Proc Natl Acad Sci U S A</i> 2005;102(15):5501–6.	doi:10.1073/pnas.0500136102 PMID 15795371 PMC555684	GSE2324	–	55	–	–	–	1	2
95	Spellman 1998 Spellman PT, Sherlock G, Zhang MQ, et al., Fitcher B. Comprehensive identification of cell cycle-regulated genes of the yeast <i>Saccharomyces cerevisiae</i> by microarray hybridization. <i>Mol Biol Cell</i> 1998;9(12):3273–97.	doi:10.1091/mbc.9.12.3273 PMID 9843569 PMC25624	none	–	55	–	–	–	3	2
96	Worley 2013 Worley J, Luo X, Capaldi AP. Inositol pyrophosphates regulate cell growth and the environmental stress response by activating the HDAC Rpd3L. <i>Cell Rep</i> 2013;3(5):1476–82.	doi:10.1016/j.celrep.2013.03.043 PMID 23643537 PMC3672359	GSE45370	–	55	–	–	–	1	2
97	Gasch 2001 Gasch AP, Huang M, Metzner S, Botstein D, Elledge SJ, Brown PO. Genomic expression responses to DNA-damaging agents and the regulatory role of the yeast ATR homolog Mec1p. <i>Mol Biol Cell</i> 2001;12(10):2987–3003.	doi:10.1091/mbc.12.10.2987 PMID 11598186 PMC60150	none	–	52	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
98	Pramila 2006 Pramila T, Wu W, Miles S, Noble WS, Breeden LL. The Forkhead transcription factor Hcm1 regulates chromosome segregation genes and fills the S-phase gap in the transcriptional circuitry of the cell cycle. <i>Genes Dev</i> 2006;20(16):2266–78.	doi:10.1101/gad.1450606 PMID 16912276 PMC1553209	GSE4987	–	50	–	–	–	1	2
99	Shalem 2008 Shalem O, Dahan O, Levo M, et al., Pilpel Y. Transient transcriptional responses to stress are generated by opposing effects of mRNA production and degradation. <i>Mol Syst Biol</i> 2008;4:223.	doi:10.1038/msb.2008.59 PMID 18854817 PMC2583085	GSE12222	–	49	–	–	–	6	1
100	Lai 2006 Lai LC, Kosorukoff AL, Burke PV, Kwast KE. Metabolic-state-dependent remodeling of the transcriptome in response to anoxia and subsequent reoxygenation in <i>Saccharomyces cerevisiae</i> . <i>Eukaryot Cell</i> 2006;5(9):1468–89.	doi:10.1128/EC.00107-06 PMID 16963631 PMC1563586	none	–	48	–	–	–	2	2
101	Li 2006 Li CM, Klevecz RR. A rapid genome-scale response of the transcriptional oscillator to perturbation reveals a period-doubling path to phenotypic change. <i>Proc Natl Acad Sci U S A</i> 2006;103(44):16254–9.	doi:10.1073/pnas.0604860103 PMID 17043222 PMC1613231	GSE9302	–	48	–	–	–	1	1
102	Sha 2013 Sha W, Martins AM, Laubenbacher R, Mendes P, Shulaev V. The genome-wide early temporal response of <i>Saccharomyces cerevisiae</i> to oxidative stress induced by cumene hydroperoxide. <i>PLoS One</i> 2013;8(9):e74939.	doi:10.1371/journal.pone.0074939 PMID 24073228 PMC3779239	GSE7645	–	48	–	–	–	1	1
103	Simmons Kovacs 2012 Simmons Kovacs LA, Mayhew MB, Orlando DA, et al., Haase SB. Cyclin-dependent kinases are regulators and effectors of oscillations driven by a transcription factor network. <i>Mol Cell</i> 2012;45(5):669–79.	doi:10.1016/j.molcel.2011.12.033 PMID 22306294 PMC3578314	GSE32974	–	48	–	–	–	1	1
104	Anderson 2009 Anderson JB, Sirjusingh C, Syed N, Lafayette S. Gene expression and evolution of antifungal drug resistance. <i>Antimicrob Agents Chemother</i> 2009;53(5):1931–6.	doi:10.1128/AAC.01315-08 PMID 19273689 PMC2681570	GSE12055	–	46	–	–	–	1	2
105	Causton 2001 Causton HC, Ren B, Koh SS, et al., Young RA. Remodeling of yeast genome expression in response to environmental changes. <i>Mol Biol Cell</i> 2001;12(2):323–37.	doi:10.1091/mbc.12.2.323 PMID 11179418 PMC30946	none	–	45	–	–	–	6	2
106	Lai 2005 Lai LC, Kosorukoff AL, Burke PV, Kwast KE. Dynamical remodeling of the transcriptome during short-term anaerobiosis in <i>Saccharomyces cerevisiae</i> : differential response and role of Msn2 and/or Msn4 and other factors in galactose and glucose media. <i>Mol Cell Biol</i> 2005;25(10):4075–91.	doi:10.1128/MCB.25.10.4075-4091.2005 PMID 15870279 PMC1087712	GSE1879	–	45	–	–	–	1	2
107	Thorsen 2007 Thorsen M, Lagniel G, Kristiansson E, et al., Tamás MJ. Quantitative transcriptome, proteome, and sulfur metabolite profiling of the <i>Saccharomyces cerevisiae</i> response to arsenite. <i>Physiol Genomics</i> 2007;30(1):35–43.	doi:10.1152/physiolgenomics.00236.2006 PMID 17327492	GSE6129 GSE6068	–	45	–	–	–	5	2
108	Oliveira 2015 Oliveira AP, Dimopoulos S, Busetto AG, et al., Stelling J. Inferring causal metabolic signals that regulate the dynamic TORC1-dependent transcriptome. <i>Mol Syst Biol</i> 2015;11(4):802.	doi:10.15252/msb.20145475 PMID 25888284 PMC4422559	GSE54850 GSE54844 GSE54851	–	43	–	–	–	3	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
109	Rossouw 2009 Rossouw D, Bauer FF. Comparing the transcriptomes of wine yeast strains: toward understanding the interaction between environment and transcriptome during fermentation. <i>Appl Microbiol Biotechnol</i> 2009;84(5):937–54.	doi:10.1007/s00253-009-2204-4 PMID 19711068	GSE11651	–	43	–	–	–	1	1
110	Gardner 2005 Gardner RG, Nelson ZW, Gottschling DE. Ubp10/Dot4p regulates the persistence of ubiquitinated histone H2B: distinct roles in telomeric silencing and general chromatin. <i>Mol Cell Biol</i> 2005;25(14):6123–39.	doi:10.1128/MCB.25.14.6123-6139.2005 PMID 15988024 PMC1168808	GSE2330 GSE2329	–	42	18	2	–	2	2
111	Johansson 2007 Johansson MJ, He F, Spatrick P, Li C, Jacobson A. Association of yeast Upf1p with direct substrates of the NMD pathway. <i>Proc Natl Acad Sci U S A</i> 2007;104(52):20872–7.	doi:10.1073/pnas.0709257105 PMID 18087042 PMC2409234	GSE9482	–	40	20	1	–	1	1
112	Ambroset 2011 Ambroset C, Petit M, Brion C, et al., Blondin B. Deciphering the molecular basis of wine yeast fermentation traits using a combined genetic and genomic approach. <i>G3 (Bethesda)</i> 2011;1(4):263–81.	doi:10.1534/g3.111.000422 PMID 22384338 PMC3276144	GSE26437	–	40	–	–	–	1	2
113	Dubacq 2006 Dubacq C, Chevalier A, Courbeyrette R, Petat C, Gidrol X, Mann C. Role of the iron mobilization and oxidative stress regulons in the genomic response of yeast to hydroxyurea. <i>Mol Genet Genomics</i> 2006;275(2):114–24.	doi:10.1007/s00438-005-0077-5 PMID 16328372	GSE1914 GSE1915	–	40	–	–	–	2	2
114	Yona 2012 Yona AH, Manor YS, Herbst RH, et al., Dahan O. Chromosomal duplication is a transient evolutionary solution to stress. <i>Proc Natl Acad Sci U S A</i> 2012;109(51):21010–5.	doi:10.1073/pnas.1211150109 PMID 23197825 PMC3529009	GSE40817	–	40	–	–	–	1	1
115	Yoshimoto 2002 Yoshimoto H, Saltsman K, Gasch AP, et al., Cyert MS. Genome-wide analysis of gene expression regulated by the calcineurin/Crz1p signaling pathway in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2002;277(34):31079–88.	doi:10.1074/jbc.M202718200 PMID 12058033	none	–	40	–	–	–	2	2
116	Mendes-Ferreira 2010 Mendes-Ferreira A, Sampaio-Marques B, Barbosa C, et al., Leão C. Accumulation of non-superoxide anion reactive oxygen species mediates nitrogen-limited alcoholic fermentation by <i>Saccharomyces cerevisiae</i> . <i>Appl Environ Microbiol</i> 2010;76(24):7918–24.	doi:10.1128/AEM.01535-10 PMID 20952643 PMC3008223	GSE5837 GSE5835 GSE5836	–	38	–	–	–	3	1
117	O'Duibhir 2014 O'Duibhir E, Lijnzaad P, Benschop JJ, et al., Holstege FC. Cell cycle population effects in perturbation studies. <i>Mol Syst Biol</i> 2014;10(6):732.	doi:10.15252/msb.20145172 PMID 24952590 PMC4265054	GSE54539 GSE54527 GSE54528	–	38	–	–	–	3	2
118	de Jonge 2017 de Jonge WJ, O'Duibhir E, Lijnzaad P, et al., Holstege FC. Molecular mechanisms that distinguish TFIID housekeeping from regulatable SAGA promoters. <i>EMBO J</i> 2017;36(3):274–290.	doi:10.15252/embj.201695621 PMID 27979920 PMC5286361	GSE81480 GSE81479	–	38	–	–	–	2	2
119	Bose 2012 Bose T, Lee KK, Lu S, et al., Gerton JL. Cohesin proteins promote ribosomal RNA production and protein translation in yeast and human cells. <i>PLoS Genet</i> 2012;8(6):e1002749.	doi:10.1371/journal.pgen.1002749 PMID 22719263 PMC3375231	GSE27235	–	36	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
120	Brauer 2008 Brauer MJ, Huttenhower C, Airoidi EM, et al., Botstein D. Coordination of growth rate, cell cycle, stress response, and metabolic activity in yeast. <i>Mol Biol Cell</i> 2008;19(1):352–67.	doi:10.1091/mbc.e07-08-0779 PMID 17959824 PMC2174172	GSE8825	–	36	–	–	–	6	2
121	Chu 2013 Chu HY, Hopper AK. Genome-wide investigation of the role of the tRNA nuclear-cytoplasmic trafficking pathway in regulation of the yeast <i>Saccharomyces cerevisiae</i> transcriptome and proteome. <i>Mol Cell Biol</i> 2013;33(21):4241–54.	doi:10.1128/MCB.00785-13 PMID 23979602 PMC3811888	GSE47713	–	36	–	–	–	1	1
122	Jaime 2012 Jaime MD, Lopez-Llorca LV, Conesa A, et al., Nislow C. Identification of yeast genes that confer resistance to chitosan oligosaccharide (COS) using chemogenomics. <i>BMC Genomics</i> 2012;13:267.	doi:10.1186/1471-2164-13-267 PMID 22727066 PMC3505485	GSE32888	–	36	–	–	–	1	1
123	Kuranda 2006 Kuranda K, Leberre V, Sokol S, Palamarczyk G, François J. Investigating the caffeine effects in the yeast <i>Saccharomyces cerevisiae</i> brings new insights into the connection between TOR, PKC and Ras/cAMP signalling pathways. <i>Mol Microbiol</i> 2006;61(5):1147–66.	doi:10.1111/j.1365-2958.2006.05300.x PMID 16925551	GSE4049	–	36	–	–	–	1	2
124	Mojzita 2014 Mojzita D, Oja M, Rintala E, Wiebe M, Penttilä M, Ruohonen L. Transcriptome of <i>Saccharomyces cerevisiae</i> during production of D-xylonate. <i>BMC Genomics</i> 2014;15(1):763.	doi:10.1186/1471-2164-15-763 PMID 25192596 PMC4176587	GSE52736	–	36	–	–	–	1	1
125	Molina-Navarro 2008 Molina-Navarro MM, Castells-Roca L, Bellí G, et al., Herrero E. Comprehensive transcriptional analysis of the oxidative response in yeast. <i>J Biol Chem</i> 2008;283(26):17908–18.	doi:10.1074/jbc.M800295200 PMID 18424442	GSE9663 GSE9645	–	36	–	–	–	2	1
126	Tu 2005 Tu BP, Kudlicki A, Rowicka M, McKnight SL. Logic of the yeast metabolic cycle: temporal compartmentalization of cellular processes. <i>Science</i> 2005;310(5751):1152–8.	doi:10.1126/science.1120499 PMID 16254148	GSE3431	–	36	–	–	–	1	2
127	Wenger 2011 Wenger JW, Piotrowski J, Nagarajan S, Chiotti K, Sherlock G, Rosenzweig F. Hunger artists: yeast adapted to carbon limitation show trade-offs under carbon sufficiency. <i>PLoS Genet</i> 2011;7(8):e1002202.	doi:10.1371/journal.pgen.1002202 PMID 21829391 PMC3150441	GSE25081	–	36	–	–	–	1	2
128	van de Pasch 2013 van de Pasch LA, Miles AJ, Nijenhuis W, et al., Holstege FC. Centromere binding and a conserved role in chromosome stability for SUMO-dependent ubiquitin ligases. <i>PLoS One</i> 2013;8(6):e65628.	doi:10.1371/journal.pone.0065628 PMID 23785440 PMC3681975	GSE33929	–	34	34	17	–	1	2
129	Kaplan 2008 Kaplan T, Liu CL, Erkmann JA, et al., Rando OJ. Cell cycle- and chaperone-mediated regulation of H3K56ac incorporation in yeast. <i>PLoS Genet</i> 2008;4(11):e1000270.	doi:10.1371/journal.pgen.1000270 PMID 19023413 PMC2581598	GSE12822	–	34	–	–	–	1	2
130	Chumnanpuen 2013 Chumnanpuen P, Nookaew I, Nielsen J. Integrated analysis, transcriptome-lipidome, reveals the effects of INO-level (INO2 and INO4) on lipid metabolism in yeast. <i>BMC Syst Biol</i> 2013;7 Suppl 3(Suppl 3):S7.	doi:10.1186/1752-0509-7-S3-S7 PMID 24456840 PMC3852131	GSE36298	–	33	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
131	Mendes-Ferreira 2007 Mendes-Ferreira A, del Olmo M, García-Martínez J, et al., Pérez-Ortín JE. Saccharomyces cerevisiae signature genes for predicting nitrogen deficiency during alcoholic fermentation. <i>Appl Environ Microbiol</i> 2007;73(16):5363–9.	doi:10.1128/AEM.01029-07 PMID 17601813 PMC1950961	GSE5842	–	33	–	–	–	1	1
132	García-Oliver 2013 García-Oliver E, Pascual-García P, García-Molinero V, Lenstra TL, Holstege FC, Rodríguez-Navarro S. A novel role for Sem1 and TREX-2 in transcription involves their impact on recruitment and H2B deubiquitylation activity of SAGA. <i>Nucleic Acids Res</i> 2013;41(11):5655–68.	doi:10.1093/nar/gkt272 PMID 23599000 PMC3675487	GSE39861	–	32	32	16	–	1	2
133	Eser 2011 Eser U, Falleur-Fettig M, Johnson A, Skotheim JM. Commitment to a cellular transition precedes genome-wide transcriptional change. <i>Mol Cell</i> 2011;43(4):515–27.	doi:10.1016/j.molcel.2011.06.024 PMID 21855792 PMC3160620	GSE29893 GSE29800 GSE29892	–	32	–	–	–	3	2
134	Klevecz 2004 Klevecz RR, Bolen J, Forrest G, Murray DB. A genomewide oscillation in transcription gates DNA replication and cell cycle. <i>Proc Natl Acad Sci U S A</i> 2004;101(5):1200–5.	doi:10.1073/pnas.0306490101 PMID 14734811 PMC337030	GSE2583	–	32	–	–	–	1	1
135	Landry 2014 Landry BD, Mapa CE, Arsenault HE, Poti KE, Benanti JA. Regulation of a transcription factor network by Cdk1 coordinates late cell cycle gene expression. <i>EMBO J</i> 2014;33(9):1044–60.	doi:10.1002/embj.201386877 PMID 24714560 PMC4193936	GSE55121	–	32	–	–	–	1	2
136	Roy 2013 Roy S, Wapinski I, Pfiffner J, et al., Regev A. Arboretum: reconstruction and analysis of the evolutionary history of condition-specific transcriptional modules. <i>Genome Res</i> 2013;23(6):1039–50.	doi:10.1101/gr.146233.112 PMID 23640720 PMC3668358	GSE38478	–	32	–	–	–	1	2
137	Segal 2003 Segal E, Shapira M, Regev A, et al., Friedman N. Module networks: identifying regulatory modules and their condition-specific regulators from gene expression data. <i>Nat Genet</i> 2003;34(2):166–76.	doi:10.1038/ng1165 PMID 12740579	none	–	32	–	–	–	3	2
138	Zhou 2015 Zhou Z, Humphries N, van Eijk P, et al., Reed SH. UV induced ubiquitination of the yeast Rad4-Rad23 complex promotes survival by regulating cellular dNTP pools. <i>Nucleic Acids Res</i> 2015;43(15):7360–70.	doi:10.1093/nar/gkv680 PMID 26150418 PMC4551923	GSE23204	–	30	12	1	–	1	1
139	Alff-Tuomala 2016 Alff-Tuomala S, Salusjärvi L, Barth D, et al., Jouhten P. Xylose-induced dynamic effects on metabolism and gene expression in engineered <i>Saccharomyces cerevisiae</i> in anaerobic glucose-xylose cultures. <i>Appl Microbiol Biotechnol</i> 2016;100(2):969–85.	doi:10.1007/s00253-015-7038-7 PMID 26454869	GSE69966	–	30	–	–	–	1	1
140	Düvel 2003 Düvel K, Santhanam A, Garrett S, Schneper L, Broach JR. Multiple roles of Tap42 in mediating rapamycin-induced transcriptional changes in yeast. <i>Mol Cell</i> 2003;11(6):1467–78.	doi:10.1016/s1097-2765(03)00228-4 PMID 12820961	none	–	30	–	–	–	2	2
141	Fleming 2002 Fleming JA, Lightcap ES, Sadis S, Thoroddsen V, Bulawa CE, Blackman RK. Complementary whole-genome technologies reveal the cellular response to proteasome inhibition by PS-341. <i>Proc Natl Acad Sci U S A</i> 2002;99(3):1461–6.	doi:10.1073/pnas.032516399 PMID 11830665 PMC122213	none	–	30	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
142	Gil 2018 Gil FN, Gonçalves AC, Becker JD, Viegas CA. Comparative analysis of transcriptomic responses to sub-lethal levels of six environmentally relevant pesticides in <i>Saccharomyces cerevisiae</i> . <i>Ecotoxicology</i> 2018;27(7):871–889.	doi:10.1007/s10646-018-1929-1 PMID 29611082	GSE53125	–	30	–	–	–	1	1
143	Huisinga 2004 Huisinga KL, Pugh BF. A genome-wide housekeeping role for TFIID and a highly regulated stress-related role for SAGA in <i>Saccharomyces cerevisiae</i> . <i>Mol Cell</i> 2004;13(4):573–85.	doi:10.1016/s1097-2765(04)00087-5 PMID 14992726	GSE885	–	30	–	–	–	1	2
144	Knijnenburg 2007 Knijnenburg TA, de Winde JH, Daran JM, et al., Wessels LF. Exploiting combinatorial cultivation conditions to infer transcriptional regulation. <i>BMC Genomics</i> 2007;8:25.	doi:10.1186/1471-2164-8-25 PMID 17241460 PMC1797021	GSE4807	–	30	–	–	–	1	1
145	Buck 2015 Buck TM, Jordan R, Lyons-Weiler J, et al., Brodsky JL. Expression of three topologically distinct membrane proteins elicits unique stress response pathways in the yeast <i>Saccharomyces cerevisiae</i> . <i>Physiol Genomics</i> 2015;47(6):198–214.	doi:10.1152/physiolgenomics.00101.2014 PMID 25759377 PMC4451388	GSE61244 GSE61243 GSE61245	–	29	–	–	–	3	1
146	Kao 2008 Kao KC, Sherlock G. Molecular characterization of clonal interference during adaptive evolution in asexual populations of <i>Saccharomyces cerevisiae</i> . <i>Nat Genet</i> 2008;40(12):1499–504.	doi:10.1038/ng.280 PMID 19029899 PMC2596280	GSE11071	–	29	–	–	–	1	2
147	Bester 2012 Bester MC, Jacobson D, Bauer FF. Many <i>Saccharomyces cerevisiae</i> Cell Wall Protein Encoding Genes Are Coregulated by Mss11, but Cellular Adhesion Phenotypes Appear Only Flo Protein Dependent. <i>G3 (Bethesda)</i> 2012;2(1):131–41.	doi:10.1534/g3.111.001644 PMID 22384390 PMC3276193	GSE29371 GSE17716	–	28	–	–	–	2	1
148	Rintala 2011 Rintala E, Jouhten P, Toivari M, et al., Ruohonen L. Transcriptional responses of <i>Saccharomyces cerevisiae</i> to shift from respiratory and respirofermentative to fully fermentative metabolism. <i>OMICS</i> 2011;15(7–8):461–76.	doi:10.1089/omi.2010.0082 PMID 21348598 PMC3146749	GSE22832	–	28	–	–	–	1	1
149	Ronald 2005 Ronald J, Akey JM, Whittle J, Smith EN, Yvert G, Kruglyak L. Simultaneous genotyping, gene-expression measurement, and detection of allele-specific expression with oligonucleotide arrays. <i>Genome Res</i> 2005;15(2):284–91.	doi:10.1101/gr.2850605 PMID 15687292 PMC546530	GSE1975	–	28	–	–	–	1	1
150	Kitagawa 2005 Kitagawa E, Akama K, Iwahashi H. Effects of iodine on global gene expression in <i>Saccharomyces cerevisiae</i> . <i>Biosci Biotechnol Biochem</i> 2005;69(12):2285–93.	doi:10.1271/bbb.69.2285 PMID 16377885	GSE9401	–	27	–	–	–	1	2
151	Lindgren 2014 Lindgren E, Hägg S, Giordano F, Björkegren J, Ström L. Inactivation of the budding yeast cohesin loader Scc2 alters gene expression both globally and in response to a single DNA double strand break. <i>Cell Cycle</i> 2014;13(23):3645–58.	doi:10.4161/15384101.2014.964108 PMID 25483075 PMC4612677	GSE61530	–	27	–	–	–	1	1
152	Lu 2014 Lu S, Lee KK, Harris B, et al., Gerton JL. The cohesin acetyltransferase Eco1 coordinates rDNA replication and transcription. <i>EMBO Rep</i> 2014;15(5):609–17.	doi:10.1002/embr.201337974 PMID 24631914 PMC4210108	GSE54741	–	27	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
153	Yiu 2008 Yiu G, McCord A, Wise A, et al., Mays Hoopes LL. Pathways change in expression during replicative aging in <i>Saccharomyces cerevisiae</i> . <i>J Gerontol A Biol Sci Med Sci</i> 2008;63(1):21–34.	doi:10.1093/gerona/63.1.21 PMID 18245757 PMC2562229	GSE10018	–	27	–	–	–	4	2
154	Cipollina 2008 Cipollina C, van den Brink J, Daran-Lapujade P, Pronk JT, Porro D, de Winde JH. <i>Saccharomyces cerevisiae</i> SFP1: at the crossroads of central metabolism and ribosome biogenesis. <i>Microbiology (Reading)</i> 2008;154(Pt 6):1686–1699.	doi:10.1099/mic.0.2008/017392-0 PMID 18524923	GSE9644	–	26	13	1	–	2	1
155	Anderson 2006 Anderson JB, Ricker N, Sirjusingh C. Antagonism between two mechanisms of antifungal drug resistance. <i>Eukaryot Cell</i> 2006;5(8):1243–51.	doi:10.1128/EC.00048-06 PMID 16896209 PMC1539145	GSE4261	–	26	–	–	–	1	2
156	Nagarajan 2014 Nagarajan S, Kruckeberg AL, Schmidt KH, et al., Rosenzweig F. Uncoupling reproduction from metabolism extends chronological lifespan in yeast. <i>Proc Natl Acad Sci U S A</i> 2014;111(15):E1538–47.	doi:10.1073/pnas.1323918111 PMID 24706810 PMC3992631	GSE21187	–	26	–	–	–	1	2
157	Pramila 2002 Pramila T, Miles S, GuhaThakurta D, Jemiolo D, Breeden LL. Conserved homeodomain proteins interact with MADS box protein Mcm1 to restrict ECB-dependent transcription to the M/G1 phase of the cell cycle. <i>Genes Dev</i> 2002;16(23):3034–45.	doi:10.1101/gad.1034302 PMID 12464633 PMC187489	GSE5283 GSE3635	–	26	–	–	–	2	2
158	Ronen 2006 Ronen M, Botstein D. Transcriptional response of steady-state yeast cultures to transient perturbations in carbon source. <i>Proc Natl Acad Sci U S A</i> 2006;103(2):389–94.	doi:10.1073/pnas.0509978103 PMID 16381818 PMC1326188	GSE4158	–	26	–	–	–	1	2
159	Zhu 2000 Zhu G, Spellman PT, Volpe T, et al., Futcher B. Two yeast forkhead genes regulate the cell cycle and pseudohyphal growth. <i>Nature</i> 2000;406(6791):90–4.	doi:10.1038/35017581 PMID 10894548	GSE3182	–	26	–	–	–	1	2
160	Azzouz 2009 Azzouz N, Panasencko OO, Deluen C, Hsieh J, Theiler G, Collart MA. Specific roles for the Ccr4-Not complex subunits in expression of the genome. <i>RNA</i> 2009;15(3):377–83.	doi:10.1261/rna.1348209 PMID 19155328 PMC2657018	GSE10521	–	25	–	–	–	1	1
161	Barbosa 2015 Barbosa C, García-Martínez J, Pérez-Ortín JE, Mendes-Ferreira A. Comparative transcriptomic analysis reveals similarities and dissimilarities in <i>Saccharomyces cerevisiae</i> wine strains response to nitrogen availability. <i>PLoS One</i> 2015;10(4):e0122709.	doi:10.1371/journal.pone.0122709 PMID 25884705 PMC4401569	GSE63187	–	25	–	–	–	1	1
162	Benton 2006 Benton MG, Somasundaram S, Glasner JD, Palecek SP. Analyzing the dose-dependence of the <i>Saccharomyces cerevisiae</i> global transcriptional response to methyl methanesulfonate and ionizing radiation. <i>BMC Genomics</i> 2006;7:305.	doi:10.1186/1471-2164-7-305 PMID 17140446 PMC1698923	GSE6018	–	25	–	–	–	2	1
163	Zhu 2009 Zhu C, Byers KJ, McCord RP, et al., Bulyk ML. High-resolution DNA-binding specificity analysis of yeast transcription factors. <i>Genome Res</i> 2009;19(4):556–66.	doi:10.1101/gr.090233.108 PMID 19158363 PMC2665775	GSE13684	–	24	18	2	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
164	Aristizabal 2013 Aristizabal MJ, Negri GL, Benschop JJ, Holstege FC, Krogan NJ, Kobor MS. High-throughput genetic and gene expression analysis of the RNAPII-CTD reveals unexpected connections to SRB10/CDK8. <i>PLoS Genet</i> 2013;9(8):e1003758.	doi:10.1371/journal.pgen.1003758 PMID 24009531 PMC3757075	GSE43120	–	24	4	1	–	1	2
165	Sapra 2004 Sapra AK, Arava Y, Khandelia P, Vijayraghavan U. Genome-wide analysis of pre-mRNA splicing: intron features govern the requirement for the second-step factor, Prp17 in <i>Saccharomyces cerevisiae</i> and <i>Schizosaccharomyces pombe</i> . <i>J Biol Chem</i> 2004;279(50):52437–46.	doi:10.1074/jbc.M408815200 PMID 15452114	none	–	24	6	1	–	1	2
166	Almario 2013 Almario MP, Reyes LH, Kao KC. Evolutionary engineering of <i>Saccharomyces cerevisiae</i> for enhanced tolerance to hydrolysates of lignocellulosic biomass. <i>Biotechnol Bioeng</i> 2013;110(10):2616–23.	doi:10.1002/bit.24938 PMID 23613173	GSE44085	–	24	–	–	–	1	2
167	Braun 2013 Braun KA, Parua PK, Dombek KM, Miner GE, Young ET. 14-3-3 (Bmh) proteins regulate combinatorial transcription following RNA polymerase II recruitment by binding at Adr1-dependent promoters in <i>Saccharomyces cerevisiae</i> . <i>Mol Cell Biol</i> 2013;33(4):712–24.	doi:10.1128/MCB.01226-12 PMID 23207903 PMC3571347	GSE40116	–	24	–	–	–	2	1
168	Caba 2005 Caba E, Dickinson DA, Warnes GR, Aubrecht J. Differentiating mechanisms of toxicity using global gene expression analysis in <i>Saccharomyces cerevisiae</i> . <i>Mutat Res</i> 2005;575(1–2):34–46.	doi:10.1016/j.mrfmmm.2005.02.005 PMID 15878181	none	–	24	–	–	–	1	1
169	Chen 2017 Chen X, Bisschops MMM, Agarwal NR, Ji B, Shanmugavel KP, Petranovic D. Interplay of Energetics and ER Stress Exacerbates Alzheimer's Amyloid- β (A β) Toxicity in Yeast. <i>Front Mol Neurosci</i> 2017;10:232.	doi:10.3389/fnmol.2017.00232 PMID 28798664 PMC5529408	GSE94793	–	24	–	–	–	1	1
170	Chumnanpuen 2012 Chumnanpuen P, Zhang J, Nookaew I, Nielsen J. Integrated analysis of transcriptome and lipid profiling reveals the co-influences of inositol-choline and Snf1 in controlling lipid biosynthesis in yeast. <i>Mol Genet Genomics</i> 2012;287(7):541–54.	doi:10.1007/s00438-012-0697-5 PMID 22622761	GSE32418	–	24	–	–	–	1	1
171	Godillot 2022 Godillot J, Sanchez I, Perez M, et al., Mouret JR. The Timing of Nitrogen Addition Impacts Yeast Genes Expression and the Production of Aroma Compounds During Wine Fermentation. <i>Front Microbiol</i> 2022;13:829786.	doi:10.3389/fmicb.2022.829786 PMID 35273585 PMC8902367	GSE189010	–	24	–	–	–	1	1
172	Klecker 2015 Klecker T, Wemmer M, Haag M, et al., Westermann B. Interaction of MDM33 with mitochondrial inner membrane homeostasis pathways in yeast. <i>Sci Rep</i> 2015;5:18344.	doi:10.1038/srep18344 PMID 26669658 PMC4680886	GSE49580	–	24	–	–	–	1	1
173	Komili 2007 Komili S, Farny NG, Roth FP, Silver PA. Functional specificity among ribosomal proteins regulates gene expression. <i>Cell</i> 2007;131(3):557–71.	doi:10.1016/j.cell.2007.08.037 PMID 17981122 PMC2443060	GSE8761	–	24	–	–	–	1	1
174	Kundaje 2008 Kundaje A, Xin X, Lan C, et al., Leslie C. A predictive model of the oxygen and heme regulatory network in yeast. <i>PLoS Comput Biol</i> 2008;4(11):e1000224.	doi:10.1371/journal.pcbi.1000224 PMID 19008939 PMC2573020	GSE8343	–	24	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
175	Medintz 2007 Medintz IL, Vora GJ, Rahbar AM, Thach DC. Transcript and proteomic analyses of wild-type and gpa2 mutant <i>Saccharomyces cerevisiae</i> strains suggest a role for glycolytic carbon source sensing in pseudohyphal differentiation. <i>Mol Biosyst</i> 2007;3(9):623–34.	doi:10.1039/b704199c PMID 17700863	GSE7820	–	24	–	–	–	3	1
176	Meneghini 2003 Meneghini MD, Wu M, Madhani HD. Conserved histone variant H2A.Z protects euchromatin from the ectopic spread of silent heterochromatin. <i>Cell</i> 2003;112(5):725–36.	doi:10.1016/s0092-8674(03)00123-5 PMID 12628191	none	–	24	–	–	3	1	2
177	Primig 2000 Primig M, Williams RM, Winzeler EA, et al., Esposito RE. The core meiotic transcriptome in budding yeasts. <i>Nat Genet</i> 2000;26(4):415–23.	doi:10.1038/82539 PMID 11101837	none	–	24	–	–	–	1	2
178	Schawalder 2004 Schawalder SB, Kabani M, Howald I, Choudhury U, Werner M, Shore D. Growth-regulated recruitment of the essential yeast ribosomal protein gene activator Ifh1. <i>Nature</i> 2004;432(7020):1058–61.	doi:10.1038/nature03200 PMID 15616569	none	–	24	–	–	–	1	2
179	Schosserer 2015 Schosserer M, Minois N, Angerer TB, et al., Grillari J. Methylation of ribosomal RNA by NSUN5 is a conserved mechanism modulating organismal lifespan. <i>Nat Commun</i> 2015;6:6158.	doi:10.1038/ncomms7158 PMID 25635753 PMC4317494	GSE63030	–	24	–	–	–	1	2
180	Tai 2005 Tai SL, Boer VM, Daran-Lapujade P, et al., Pronk JT. Two-dimensional transcriptome analysis in chemostat cultures. Combinatorial effects of oxygen availability and macronutrient limitation in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2005;280(1):437–47.	doi:10.1074/jbc.M410573200 PMID 15496405	none	–	24	–	–	–	1	2
181	Willis 2008 Willis IM, Chua G, Tong AH, et al., Moir RD. Genetic interactions of MAF1 identify a role for Med20 in transcriptional repression of ribosomal protein genes. <i>PLoS Genet</i> 2008;4(7):e1000112.	doi:10.1371/journal.pgen.1000112 PMID 18604275 PMC2435279	GSE11397	–	24	–	–	–	1	2
182	Zhang 2010 Zhang J, Olsson L, Nielsen J. The beta-subunits of the Snf1 kinase in <i>Saccharomyces cerevisiae</i> , Gal83 and Sip2, but not Sip1, are redundant in glucose derepression and regulation of sterol biosynthesis. <i>Mol Microbiol</i> 2010;77(2):371–83.	doi:10.1111/j.1365-2958.2010.07209.x PMID 20545859	GSE20591	–	24	–	–	–	1	1
183	Zhang 2011 Zhang J, Vaga S, Chumnanpue P, et al., Nielsen J. Mapping the interaction of Snf1 with TORC1 in <i>Saccharomyces cerevisiae</i> . <i>Mol Syst Biol</i> 2011;7:545.	doi:10.1038/msb.2011.80 PMID 22068328 PMC3261716	GSE24421	–	24	–	–	–	1	1
184	Fendt 2010 Fendt SM, Oliveira AP, Christen S, Picotti P, Dechant RC, Sauer U. Unraveling condition-dependent networks of transcription factors that control metabolic pathway activity in yeast. <i>Mol Syst Biol</i> 2010;6:432.	doi:10.1038/msb.2010.91 PMID 21119627 PMC3010106	GSE24056 GSE19569 GSE24053 GSE24054	–	23	–	–	–	4	1
185	Fowler 2011 Fowler DM, Cooper SJ, Stephany JJ, Hendon N, Nelson S, Fields S. Suppression of statin effectiveness by copper and zinc in yeast and human cells. <i>Mol Biosyst</i> 2011;7(2):533–44.	doi:10.1039/c0mb00166j PMID 21085730 PMC3138400	GSE25004	–	23	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
186	Kugler 2016 Kugler KG, Jandric Z, Beyer R, et al., Schüller C. Ribosome quality control is a central protection mechanism for yeast exposed to deoxynivalenol and trichothecin. <i>BMC Genomics</i> 2016;17:417.	doi:10.1186/s12864-016-2718-y PMID 27245696 PMC4888481	GSE75462	–	23	–	–	–	1	1
187	Oeser 2016 Oeser ML, Amen T, Nadel CM, et al., Gardner RG. Dynamic Sumoylation of a Conserved Transcription Corepressor Prevents Persistent Inclusion Formation during Hyperosmotic Stress. <i>PLoS Genet</i> 2016;12(1):e1005809.	doi:10.1371/journal.pgen.1005809 PMID 26800527 PMC4723248	GSE57476	–	23	–	–	–	1	2
188	Lenstra 2013 Lenstra TL, Tudek A, Clauder S, et al., Holstege FC. The role of Ctk1 kinase in termination of small non-coding RNAs. <i>PLoS One</i> 2013;8(12):e80495.	doi:10.1371/journal.pone.0080495 PMID 24324601 PMC3851182	GSE40254	–	22	16	7	–	1	2
189	Ng 2013 Ng TM, Lenstra TL, Duggan N, et al., Biggins S. Kinetochore function and chromosome segregation rely on critical residues in histones H3 and H4 in budding yeast. <i>Genetics</i> 2013;195(3):795–807.	doi:10.1534/genetics.113.152082 PMID 24037263 PMC3813865	GSE39903	–	22	–	–	–	1	2
190	Stern 2007 Stern S, Dror T, Stolovicki E, Brenner N, Braun E. Genome-wide transcriptional plasticity underlies cellular adaptation to novel challenge. <i>Mol Syst Biol</i> 2007;3:106.	doi:10.1038/msb4100147 PMID 17453047 PMC1865588	GSE6450	–	22	–	–	–	1	2
191	Kunkel 2019 Kunkel J, Luo X, Capaldi AP. Integrated TORC1 and PKA signaling control the temporal activation of glucose-induced gene expression in yeast. <i>Nat Commun</i> 2019;10(1):3558.	doi:10.1038/s41467-019-11540-y PMID 31395866 PMC6687784	GSE133591	–	21	6	2	–	1	2
192	Gonzalez-Aguilera 2011 González-Aguilera C, Tous C, Babiano R, de la Cruz J, Luna R, Aguilera A. Nab2 functions in the metabolism of RNA driven by polymerases II and III. <i>Mol Biol Cell</i> 2011;22(15):2729–40.	doi:10.1091/mbc.E11-01-0055 PMID 21680710 PMC3145548	GSE26303 GSE19302	–	21	–	–	–	2	1
193	Guo 2006 Guo Y, Breeden LL, Fan W, Zhao LP, Eaton DL, Zarbl H. Analysis of cellular responses to aflatoxin B(1) in yeast expressing human cytochrome P450 1A2 using cDNA microarrays. <i>Mutat Res</i> 2006;593(1–2):121–42.	doi:10.1016/j.mrfmmm.2005.07.001 PMID 16122766	GSE2420 GSE2419	–	21	–	–	–	2	2
194	Ideker 2001 Ideker T, Thorsson V, Ranish JA, et al., Hood L. Integrated genomic and proteomic analyses of a systematically perturbed metabolic network. <i>Science</i> 2001;292(5518):929–34.	doi:10.1126/science.292.5518.929 PMID 11340206	none	–	21	–	–	–	1	2
195	Marks 2008 Marks VD, Ho Sui SJ, Erasmus D, et al., van Vuuren HJ. Dynamics of the yeast transcriptome during wine fermentation reveals a novel fermentation stress response. <i>FEMS Yeast Res</i> 2008;8(1):35–52.	doi:10.1111/j.1567-1364.2007.00338.x PMID 18215224 PMC5065349	GSE8536	–	21	–	–	–	1	1
196	Montefusco 2013 Montefusco DJ, Chen L, Matmati N, et al., Lu X. Distinct signaling roles of ceramide species in yeast revealed through systematic perturbation and systems biology analyses. <i>Sci Signal</i> 2013;6(299):rs14.	doi:10.1126/scisignal.2004515 PMID 24170935 PMC3974757	GSE40399	–	21	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
197	Smith 2007 Smith JJ, Ramsey SA, Marelli M, et al., Aitchison JD. Transcriptional responses to fatty acid are coordinated by combinatorial control. <i>Mol Syst Biol</i> 2007;3:115.	doi:10.1038/msb4100157 PMID 17551510 PMC1911199	GSE5862	–	20	6	3	–	1	2
198	Brauer 2005 Brauer MJ, Saldanha AJ, Dolinski K, Botstein D. Homeostatic adjustment and metabolic remodeling in glucose-limited yeast cultures. <i>Mol Biol Cell</i> 2005;16(5):2503–17.	doi:10.1091/mbc.e04-11-0968 PMID 15758028 PMC1087253	none	–	20	–	–	–	2	2
199	Carter 2006 Carter GW, Rupp S, Fink GR, Galitski T. Disentangling information flow in the Ras-cAMP signaling network. <i>Genome Res</i> 2006;16(4):520–6.	doi:10.1101/gr.4473506 PMID 16533914 PMC1457029	GSE2927	–	20	–	–	–	1	1
200	Chitikila 2002 Chitikila C, Huisinga KL, Irvin JD, Basehoar AD, Pugh BF. Interplay of TBP inhibitors in global transcriptional control. <i>Mol Cell</i> 2002;10(4):871–82.	doi:10.1016/s1097-2765(02)00683-4 PMID 12419230	none	–	20	–	–	–	1	2
201	Garcia-Martinez 2016 García-Martínez J, Troulé K, Chávez S, Pérez-Ortín JE. Growth rate controls mRNA turnover in steady and non-steady states. <i>RNA Biol</i> 2016;13(12):1175–1181.	doi:10.1080/15476286.2016.1236171 PMID 27648972 PMC5207384	GSE72356	–	20	–	–	–	1	2
202	Jin 2007 Jin Y, Rodriguez AM, Stanton JD, Kitazono AA, Wyrick JJ. Simultaneous mutation of methylated lysine residues in histone H3 causes enhanced gene silencing, cell cycle defects, and cell lethality in <i>Saccharomyces cerevisiae</i> . <i>Mol Cell Biol</i> 2007;27(19):6832–41.	doi:10.1128/MCB.00745-07 PMID 17664279 PMC2099221	GSE6331	–	20	–	–	–	1	1
203	Oromendia 2012 Oromendia AB, Dodgson SE, Amon A. Aneuploidy causes proteotoxic stress in yeast. <i>Genes Dev</i> 2012;26(24):2696–708.	doi:10.1101/gad.207407.112 PMID 23222101 PMC3533075	GSE40073	–	20	–	–	–	1	2
204	Parra 2006 Parra MA, Kerr D, Fahy D, Pouchnik DJ, Wyrick JJ. Deciphering the roles of the histone H2B N-terminal domain in genome-wide transcription. <i>Mol Cell Biol</i> 2006;26(10):3842–52.	doi:10.1128/MCB.26.10.3842-3852.2006 PMID 16648479 PMC1489011	GSE3806	–	20	–	–	–	1	1
205	Shivaswamy 2008 Shivaswamy S, Iyer VR. Stress-dependent dynamics of global chromatin remodeling in yeast: dual role for SWI/SNF in the heat shock stress response. <i>Mol Cell Biol</i> 2008;28(7):2221–34.	doi:10.1128/MCB.01659-07 PMID 18212068 PMC2268435	GSE7665	–	19	6	1	–	1	2
206	Lewis 2012 Lewis JA, Gasch AP. Natural variation in the yeast glucose-signaling network reveals a new role for the Mig3p transcription factor. <i>G3 (Bethesda)</i> 2012;2(12):1607–12.	doi:10.1534/g3.112.004127 PMID 23275883 PMC3516482	GSE40153	–	19	–	–	–	1	2
207	Urban 2007 Urban J, Soulard A, Huber A, et al., Loewith R. Sch9 is a major target of TORC1 in <i>Saccharomyces cerevisiae</i> . <i>Mol Cell</i> 2007;26(5):663–74.	doi:10.1016/j.molcel.2007.04.020 PMID 17560372	GSE7660	–	19	–	–	–	3	2
208	Busti 2012 Busti S, Gotti L, Balestrieri C, et al., Vanoni M. Overexpression of Far1, a cyclin-dependent kinase inhibitor, induces a large transcriptional reprogramming in which RNA synthesis senses Far1 in a Sfp1-mediated way. <i>Biotechnol Adv</i> 2012;30(1):185–201.	doi:10.1016/j.biotechadv.2011.09.007 PMID 21964263	GSE31143	–	18	6	1	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
209	Sabet 2004 Sabet N, Volo S, Yu C, Madigan JP, Morse RH. Genome-wide analysis of the relationship between transcriptional regulation by Rpd3p and the histone H3 and H4 amino termini in budding yeast. <i>Mol Cell Biol</i> 2004;24(20):8823–33.	doi:10.1128/MCB.24.20.8823-8833.2004 PMID 15456858 PMC517902	none	–	18	2	1	–	1	2
210	Aguilera 2005 Aguilera J, Petit T, de Winde JH, Pronk JT. Physiological and genome-wide transcriptional responses of <i>Saccharomyces cerevisiae</i> to high carbon dioxide concentrations. <i>FEMS Yeast Res</i> 2005;5(6–7):579–93.	doi:10.1016/j.femsyr.2004.09.009 PMID 15780657	GSE8900	–	18	–	–	–	1	1
211	Barreto 2012 Barreto L, Canadell D, Valverde-Saubí D, Casamayor A, Ariño J. The short-term response of yeast to potassium starvation. <i>Environ Microbiol</i> 2012;14(11):3026–42.	doi:10.1111/j.1462-2920.2012.02887.x PMID 23039231	GSE24712	–	18	–	–	–	1	2
212	Bernstein 2000 Bernstein BE, Tong JK, Schreiber SL. Genomewide studies of histone deacetylase function in yeast. <i>Proc Natl Acad Sci U S A</i> 2000;97(25):13708–13.	doi:10.1073/pnas.250477697 PMID 11095743 PMC17640	none	–	18	–	–	3	3	2
213	Chiu 2011 Chiu J, Tactacan CM, Tan SX, Lin RC, Wouters MA, Dawes IW. Cell cycle sensing of oxidative stress in <i>Saccharomyces cerevisiae</i> by oxidation of a specific cysteine residue in the transcription factor Swi6p. <i>J Biol Chem</i> 2011;286(7):5204–14.	doi:10.1074/jbc.M110.172973 PMID 21147769 PMC3037633	GSE18334	–	18	–	–	–	1	1
214	Cook 2013 Cook R, Hennell JR, Lee S, et al., Sucher NJ. The <i>Saccharomyces cerevisiae</i> transcriptome as a mirror of phytochemical variation in complex extracts of <i>Equisetum arvense</i> from America, China, Europe and India. <i>BMC Genomics</i> 2013;14:445.	doi:10.1186/1471-2164-14-445 PMID 23826764 PMC3720287	GSE24888	–	18	–	–	–	1	1
215	Fry 2006 Fry RC, DeMott MS, Cosgrove JP, Begley TJ, Samson LD, Dedon PC. The DNA-damage signature in <i>Saccharomyces cerevisiae</i> is associated with single-strand breaks in DNA. <i>BMC Genomics</i> 2006;7:313.	doi:10.1186/1471-2164-7-313 PMID 17163986 PMC1764021	GSE5301	–	18	–	–	–	1	1
216	Gamero 2014 Gamero A, Belloch C, Ibáñez C, Querol A. Molecular analysis of the genes involved in aroma synthesis in the species <i>S. cerevisiae</i> , <i>S. kudriavzevii</i> and <i>S. bayanus</i> var. <i>uvarum</i> in winemaking conditions. <i>PLoS One</i> 2014;9(5):e97626.	doi:10.1371/journal.pone.0097626 PMID 24854353 PMC4031168	GSE30778	–	18	–	–	–	1	2
217	García 2009 García R, Rodríguez-Peña JM, Bermejo C, Nombela C, Arroyo J. The high osmotic response and cell wall integrity pathways cooperate to regulate transcriptional responses to zymolyase-induced cell wall stress in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2009;284(16):10901–11.	doi:10.1074/jbc.M808693200 PMID 19234305 PMC2667776	GSE12684	–	18	–	–	–	1	2
218	Garrido-Godino 2016 Garrido-Godino AI, García-López MC, García-Martínez J, et al., Navarro F. Rpb1 foot mutations demonstrate a major role of Rpb4 in mRNA stability during stress situations in yeast. <i>Biochim Biophys Acta</i> 2016;1859(5):731–43.	doi:10.1016/j.bbagr.2016.03.008 PMID 27001033	GSE65283	–	18	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
219	Geijer 2012 Geijer C, Pirkov I, Vongsangnak W, et al., Hohmann S. Time course gene expression profiling of yeast spore germination reveals a network of transcription factors orchestrating the global response. <i>BMC Genomics</i> 2012;13:554.	doi:10.1186/1471-2164-13-554 PMID 23066959 PMC3577491	GSE29960	–	18	–	–	–	1	2
220	Liu 2013 Liu Z, Hou J, Martínez JL, Petranovic D, Nielsen J. Correlation of cell growth and heterologous protein production by <i>Saccharomyces cerevisiae</i> . <i>Appl Microbiol Biotechnol</i> 2013;97(20):8955–62.	doi:10.1007/s00253-013-4715-2 PMID 23392765	GSE40934	–	18	–	–	–	1	1
221	Liu 2013 Liu Z, Österlund T, Hou J, Petranovic D, Nielsen J. Anaerobic α -amylase production and secretion with fumarate as the final electron acceptor in <i>Saccharomyces cerevisiae</i> . <i>Appl Environ Microbiol</i> 2013;79(9):2962–7.	doi:10.1128/AEM.03207-12 PMID 23435897 PMC3623130	GSE38848	–	18	–	–	–	1	1
222	Park 2011 Park EH, Zhang F, Warringer J, Sunnerhagen P, Hinnebusch AG. Depletion of eIF4G from yeast cells narrows the range of translational efficiencies genome-wide. <i>BMC Genomics</i> 2011;12:68.	doi:10.1186/1471-2164-12-68 PMID 21269496 PMC3042410	GSE25721	–	18	–	–	–	1	1
223	Rodríguez-Lombardero 2014 Rodríguez-Lombardero S, Vizoso-Vázquez Á, Lombardía LJ, Becerra M, González-Siso MI, Cerdán ME. Sky1 regulates the expression of sulfur metabolism genes in response to cisplatin. <i>Microbiology (Reading)</i> 2014;160(Pt 7):1357–1368.	doi:10.1099/mic.0.078402-0 PMID 24763424 PMC4076870	GSE41094	–	18	–	–	–	1	1
224	Sanz 2012 Sanz AB, García R, Rodríguez-Peña JM, et al., Arroyo J. Chromatin remodeling by the SWI/SNF complex is essential for transcription mediated by the yeast cell wall integrity MAPK pathway. <i>Mol Biol Cell</i> 2012;23(14):2805–17.	doi:10.1091/mbc.E12-04-0278 PMID 22621902 PMC3395667	GSE31176	–	18	–	–	–	1	1
225	Scherens 2006 Scherens B, Feller A, Vierendeels F, Messenguy F, Dubois E. Identification of direct and indirect targets of the Gln3 and Gat1 activators by transcriptional profiling in response to nitrogen availability in the short and long term. <i>FEMS Yeast Res</i> 2006;6(5):777–91.	doi:10.1111/j.1567-1364.2006.00060.x PMID 16879428	GSE2891	–	18	–	–	–	1	2
226	Stanley 2010 Stanley D, Chambers PJ, Stanley GA, Borneman A, Fraser S. Transcriptional changes associated with ethanol tolerance in <i>Saccharomyces cerevisiae</i> . <i>Appl Microbiol Biotechnol</i> 2010;88(1):231–9.	doi:10.1007/s00253-010-2760-7 PMID 20661734	GSE21120	–	18	–	–	–	1	2
227	Stovicek 2014 Štoviček V, Váchová L, Begany M, Wilkinson D, Palková Z. Global changes in gene expression associated with phenotypic switching of wild yeast. <i>BMC Genomics</i> 2014;15:136.	doi:10.1186/1471-2164-15-136 PMID 24533484 PMC3930820	GSE40625	–	18	–	–	–	1	2
228	Suzuki 2012 Suzuki T, Iwahashi Y. Comprehensive gene expression analysis of type B trichothecenes. <i>J Agric Food Chem</i> 2012;60(37):9519–27.	doi:10.1021/jf3020975 PMID 22897823	GSE36954	–	18	–	–	–	1	1
229	Tyo 2012 Tyo KE, Liu Z, Petranovic D, Nielsen J. Imbalance of heterologous protein folding and disulfide bond formation rates yields runaway oxidative stress. <i>BMC Biol</i> 2012;10:16.	doi:10.1186/1741-7007-10-16 PMID 22380681 PMC3310788	GSE27062	–	18	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
230	Voordeckers 2012 Voordeckers K, De Maeyer D, van der Zande E, et al., Verstrepen KJ. Identification of a complex genetic network underlying <i>Saccharomyces cerevisiae</i> colony morphology. <i>Mol Microbiol</i> 2012;86(1):225–39.	doi:10.1111/j.1365-2958.2012.08192.x PMID 22882838 PMC3470922	GSE36151	–	18	–	–	–	1	1
231	Auld 2006 Auld KL, Brown CR, Casolari JM, Komili S, Silver PA. Genomic association of the proteasome demonstrates overlapping gene regulatory activity with transcription factor substrates. <i>Mol Cell</i> 2006;21(6):861–71.	doi:10.1016/j.molcel.2006.02.020 PMID 16543154	GSE4272	–	17	–	–	–	1	2
232	Chasman 2014 Chasman D, Ho YH, Berry DB, et al., Gasch AP. Pathway connectivity and signaling coordination in the yeast stress-activated signaling network. <i>Mol Syst Biol</i> 2014;10(11):759.	doi:10.15252/msb.20145120 PMID 25411400 PMC4299600	GSE56995 GSE60700	–	17	–	–	–	2	2
233	Cho 1998 Cho RJ, Campbell MJ, Winzeler EA, et al., Davis RW. A genome-wide transcriptional analysis of the mitotic cell cycle. <i>Mol Cell</i> 1998;2(1):65–73.	doi:10.1016/s1097-2765(00)80114-8 PMID 9702192	none	–	17	–	–	–	1	2
234	Clark 2002 Clark TA, Sugnet CW, Ares M Jr. Genomewide analysis of mRNA processing in yeast using splicing-specific microarrays. <i>Science</i> 2002;296(5569):907–10.	doi:10.1126/science.1069415 PMID 11988574	none	–	17	–	–	–	1	2
235	Gibson 2008 Gibson BR, Lawrence SJ, Boulton CA, et al., Smart KA. The oxidative stress response of a lager brewing yeast strain during industrial propagation and fermentation. <i>FEMS Yeast Res</i> 2008;8(4):574–85.	doi:10.1111/j.1567-1364.2008.00371.x PMID 18373683	GSE9423	–	17	–	–	–	1	1
236	Goossens 2015 Goossens KV, Ielasi FS, Nookaew I, et al., Willaert RG. Molecular mechanism of flocculation self-recognition in yeast and its role in mating and survival. <i>mBio</i> 2015;6(2).	doi:10.1128/mBio.00427-15 PMID 25873380 PMC4453552	GSE64468	–	17	–	–	–	1	1
237	Haugen 2004 Haugen AC, Kelley R, Collins JB, et al., Van Houten B. Integrating phenotypic and expression profiles to map arsenic-response networks. <i>Genome Biol</i> 2004;5(12):R95.	doi:10.1186/gb-2004-5-12-r95 PMID 15575969 PMC545798	none	–	17	–	–	–	2	2
238	Hebly 2014 Hebly M, de Ridder D, de Hulster EA, de la Torre Cortes P, Pronk JT, Daran-Lapujade P. Physiological and transcriptional responses of anaerobic chemostat cultures of <i>Saccharomyces cerevisiae</i> subjected to diurnal temperature cycles. <i>Appl Environ Microbiol</i> 2014;80(14):4433–49.	doi:10.1128/AEM.00785-14 PMID 24814792 PMC4068694	GSE55372	–	17	–	–	–	1	1
239	Klim 2021 Klim J, Zielenkiewicz U, Skoneczny M, Skoneczna A, Kurlandzka A, Kaczanowski S. Genetic interaction network has a very limited impact on the evolutionary trajectories in continuous culture-grown populations of yeast. <i>BMC Ecol Evol</i> 2021;21(1):99.	doi:10.1186/s12862-021-01830-9 PMID 34039270 PMC8157726	GSE167397	–	17	–	–	–	1	2
240	Lopez 2000 López MC, Baker HV. Understanding the growth phenotype of the yeast <i>gcr1</i> mutant in terms of global genomic expression patterns. <i>J Bacteriol</i> 2000;182(17):4970–8.	doi:10.1128/JB.182.17.4970-4978.2000 PMID 10940042 PMC111378	none	–	17	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
241	Reinke 2006 Reinke A, Chen JC, Aronova S, Powers T. Caffeine targets TOR complex I and provides evidence for a regulatory link between the FRB and kinase domains of Tor1p. <i>J Biol Chem</i> 2006;281(42):31616–26.	doi:10.1074/jbc.M603107200 PMID 16923813	GSE4584 GSE4586	–	17	–	–	–	3	2
242	Hughes Hallett 2014 Hughes Hallett JE, Luo X, Capaldi AP. State transitions in the TORC1 signaling pathway and information processing in <i>Saccharomyces cerevisiae</i> . <i>Genetics</i> 2014;198(2):773–86.	doi:10.1534/genetics.114.168369 PMID 25085507 PMC4196627	GSE58992	–	16	1	1	–	1	2
243	Amit 2010 Amit M, Weisberg SJ, Nadler-Holly M, et al., Horovitz A. Equivalent mutations in the eight subunits of the chaperonin CCT produce dramatically different cellular and gene expression phenotypes. <i>J Mol Biol</i> 2010;401(3):532–43.	doi:10.1016/j.jmb.2010.06.037 PMID 20600117	GSE19511	–	16	–	–	–	1	1
244	Barbosa 2015 Barbosa C, Mendes-Faia A, Lage P, Mira NP, Mendes-Ferreira A. Genomic expression program of <i>Saccharomyces cerevisiae</i> along a mixed-culture wine fermentation with <i>Hanseniaspora guilliermondii</i> . <i>Microb Cell Fact</i> 2015;14:124.	doi:10.1186/s12934-015-0318-1 PMID 26314747 PMC4552253	GSE66521	–	16	–	–	–	1	1
245	Brice 2014 Brice C, Sanchez I, Tesnière C, Blondin B. Assessing the mechanisms responsible for differences between nitrogen requirements of <i>saccharomyces cerevisiae</i> wine yeasts in alcoholic fermentation. <i>Appl Environ Microbiol</i> 2014;80(4):1330–9.	doi:10.1128/AEM.03856-13 PMID 24334661 PMC3911072	GSE50626	–	16	–	–	–	1	1
246	Casado 2011 Casado C, González A, Platara M, Ruiz A, Ariño J. The role of the protein kinase A pathway in the response to alkaline pH stress in yeast. <i>Biochem J</i> 2011;438(3):523–33.	doi:10.1042/BJ20110607 PMID 21749328 PMC3253439	GSE27925	–	16	–	–	–	1	2
247	Dörfel 2016 Dörfel MJ, Fang H, Crain J, Klingener M, Weiser J, Lyon GJ. Proteomic and genomic characterization of a yeast model for Ogden syndrome. <i>Yeast</i> 2017;34(1):19–37.	doi:10.1002/yea.3211 PMID 27668839 PMC5248646	GSE86358	–	16	–	–	–	1	2
248	Geijer 2013 Geijer C, Medrala-Klein D, Petelenz-Kurziel E, et al., Hohmann S. Initiation of the transcriptional response to hyperosmotic shock correlates with the potential for volume recovery. <i>FEBS J</i> 2013;280(16):3854–67.	doi:10.1111/febs.12382 PMID 23758973	GSE29716	–	16	–	–	–	1	2
249	Kresnowati 2006 Kresnowati MT, van Winden WA, Almering MJ, et al., Daran JM. When transcriptome meets metabolome: fast cellular responses of yeast to sudden relief of glucose limitation. <i>Mol Syst Biol</i> 2006;2:49.	doi:10.1038/msb4100083 PMID 16969341 PMC1681515	GSE3821	–	16	–	–	–	1	1
250	Levy 2011 Levy S, Kafri M, Carmi M, Barkai N. The competitive advantage of a dual-transporter system. <i>Science</i> 2011;334(6061):1408–12.	doi:10.1126/science.1207154 PMID 22158820 PMC4104933	GSE32067	–	16	–	–	–	1	2
251	Ouyang 2011 Ouyang X, Tran QT, Goodwin S, Wible RS, Sutter CH, Sutter TR. Yap1 activation by H ₂ O ₂ or thiol-reactive chemicals elicits distinct adaptive gene responses. <i>Free Radic Biol Med</i> 2011;50(1):1–13.	doi:10.1016/j.freeradbiomed.2010.10.697 PMID 20971184	GSE19213	–	16	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
252	Reiter 2013 Reiter W, Klopff E, De Wever V, et al., Ammerer G. Yeast protein phosphatase 2A-Cdc55 regulates the transcriptional response to hyperosmolarity stress by regulating Msn2 and Msn4 chromatin recruitment. <i>Mol Cell Biol</i> 2013;33(5):1057–72.	doi:10.1128/MCB.00834-12 PMID 23275436 PMC3623084	GSE38565	–	16	–	–	–	1	1
253	Renaud-Young 2015 Renaud-Young M, Lloyd DC, Chatfield-Reed K, George I, Chua G, Cobb J. The NuA4 complex promotes translesion synthesis (TLS)-mediated DNA damage tolerance. <i>Genetics</i> 2015;199(4):1065–76.	doi:10.1534/genetics.115.174490 PMID 25701288 PMC4391564	GSE66176	–	16	–	–	–	1	2
254	Robinson 2015 Robinson DG, Wang JY, Storey JD. A nested parallel experiment demonstrates differences in intensity-dependence between RNA-seq and microarrays. <i>Nucleic Acids Res</i> 2015;43(20):e131.	doi:10.1093/nar/gkv636 PMID 26130709 PMC4787771	GSE65365	–	16	–	–	–	1	2
255	Schiavone 2017 Schiavone M, Déjean S, Sieczkowski N, Castex M, Dague E, François JM. Integration of Biochemical, Biophysical and Transcriptomics Data for Investigating the Structural and Nanomechanical Properties of the Yeast Cell Wall. <i>Front Microbiol</i> 2017;8:1806.	doi:10.3389/fmicb.2017.01806 PMID 29085340 PMC5649194	GSE103392	–	16	–	–	–	1	1
256	Venkatasubrahmanyam 2007 Venkatasubrahmanyam S, Hwang WW, Meneghini MD, Tong AH, Madhani HD. Genome-wide, as opposed to local, antisilencing is mediated redundantly by the euchromatic factors Set1 and H2A.Z. <i>Proc Natl Acad Sci U S A</i> 2007;104(42):16609–14.	doi:10.1073/pnas.0700914104 PMID 17925448 PMC2034229	GSE4826	–	16	–	–	–	1	2
257	Vos 2015 Vos T, de la Torre Cortés P, van Gulik WM, Pronk JT, Daran-Lapujade P. Growth-rate dependency of de novo resveratrol production in chemostat cultures of an engineered <i>Saccharomyces cerevisiae</i> strain. <i>Microb Cell Fact</i> 2015;14:133.	doi:10.1186/s12934-015-0321-6 PMID 26369953 PMC4570684	GSE65942	–	16	–	–	–	1	1
258	Warfield 2017 Warfield L, Ramachandran S, Baptista T, Devys D, Tora L, Hahn S. Transcription of Nearly All Yeast RNA Polymerase II-Transcribed Genes Is Dependent on Transcription Factor TFIID. <i>Mol Cell</i> 2017;68(1):118–129.e5.	doi:10.1016/j.molcel.2017.08.014 PMID 28918900 PMC5679267	GSE96830	–	16	–	–	–	1	1
259	Wu 2022 Wu Y, Li B, Miao B, Xie C, Tang YQ. <i>Saccharomyces cerevisiae</i> employs complex regulation strategies to tolerate low pH stress during ethanol production. <i>Microb Cell Fact</i> 2022;21(1):247.	doi:10.1186/s12934-022-01974-3 PMID 36419096 PMC9685915	GSE210964	–	16	–	–	–	1	1
260	Santos-Pereira 2014 Santos-Pereira JM, García-Rubio ML, González-Aguilera C, Luna R, Aguilera A. A genome-wide function of THSC/TREX-2 at active genes prevents transcription-replication collisions. <i>Nucleic Acids Res</i> 2014;42(19):12000–14.	doi:10.1093/nar/gku906 PMID 25294824 PMC4231764	GSE56702	–	15	9	3	–	1	1
261	Lanza 2012 Lanza AM, Blazek JJ, Crook NC, Alper HS. Linking yeast Gcn5p catalytic function and gene regulation using a quantitative, graded dominant mutant approach. <i>PLoS One</i> 2012;7(4):e36193.	doi:10.1371/journal.pone.0036193 PMID 22558379 PMC3338614	GSE26923	–	15	3	1	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
262	Batista-Nascimento 2013 Batista-Nascimento L, Toledano MB, Thiele DJ, Rodrigues-Pousada C. Yeast protective response to arsenate involves the repression of the high affinity iron uptake system. <i>Biochim Biophys Acta</i> 2013;1833(5):997–1005.	doi:10.1016/j.bbamcr.2012.12.018 PMID 23295455	GSE33427	–	15	–	–	–	1	1
263	Boer 2007 Boer VM, Tai SL, Vuralhan Z, et al., Daran JM. Transcriptional responses of <i>Saccharomyces cerevisiae</i> to preferred and nonpreferred nitrogen sources in glucose-limited chemostat cultures. <i>FEMS Yeast Res</i> 2007;7(4):604–20.	doi:10.1111/j.1567-1364.2007.00220.x PMID 17419774	GSE6405	–	15	–	–	–	1	1
264	Carmel-Harel 2001 Carmel-Harel O, Stearman R, Gasch AP, Botstein D, Brown PO, Storz G. Role of thioredoxin reductase in the Yap1p-dependent response to oxidative stress in <i>Saccharomyces cerevisiae</i> . <i>Mol Microbiol</i> 2001;39(3):595–605.	doi:10.1046/j.1365-2958.2001.02255.x PMID 11169101	none	–	15	–	–	–	1	2
265	Dymond 2011 Dymond JS, Richardson SM, Coombes CE, et al., Boeke JD. Synthetic chromosome arms function in yeast and generate phenotypic diversity by design. <i>Nature</i> 2011;477(7365):471–6.	doi:10.1038/nature10403 PMID 21918511 PMC3774833	GSE31326	–	15	–	–	–	1	1
266	Felipe-Abrio 2015 Felipe-Abrio I, Lafuente-Barquero J, García-Rubio ML, Aguilera A. RNA polymerase II contributes to preventing transcription-mediated replication fork stalls. <i>EMBO J</i> 2015;34(2):236–50.	doi:10.15252/embj.201488544 PMID 25452497 PMC4337070	GSE55223	–	15	–	–	–	1	1
267	Iwahashi 2007 Iwahashi H, Kitagawa E, Suzuki Y, et al., Iwahashi Y. Evaluation of toxicity of the mycotoxin citrinin using yeast ORF DNA microarray and Oligo DNA microarray. <i>BMC Genomics</i> 2007;8:95.	doi:10.1186/1471-2164-8-95 PMID 17408496 PMC1865386	GSE6118	–	15	–	–	–	1	2
268	Kitagawa 2003 Kitagawa E, Momose Y, Iwahashi H. Correlation of the structures of agricultural fungicides to gene expression in <i>Saccharomyces cerevisiae</i> upon exposure to toxic doses. <i>Environ Sci Technol</i> 2003;37(12):2788–93.	doi:10.1021/es026156b PMID 12854720	GSE9337	–	15	–	–	–	1	2
269	Marsit 2016 Marsit S, Sanchez I, Galeote V, Dequin S. Horizontally acquired oligopeptide transporters favour adaptation of <i>Saccharomyces cerevisiae</i> wine yeast to oenological environment. <i>Environ Microbiol</i> 2016;18(4):1148–61.	doi:10.1111/1462-2920.13117 PMID 26549518	GSE62446	–	15	–	–	–	1	1
270	Roberts 2000 Roberts CJ, Nelson B, Marton MJ, et al., Friend SH. Signaling and circuitry of multiple MAPK pathways revealed by a matrix of global gene expression profiles. <i>Science</i> 2000;287(5454):873–80.	doi:10.1126/science.287.5454.873 PMID 10657304	none	–	15	–	–	–	2	2
271	Rosaleny 2005 Rosaleny LE, Antúnez O, Ruiz-García AB, Pérez-Ortín JE, Tordera V. Yeast HAT1 and HAT2 deletions have different life-span and transcriptome phenotypes. <i>FEBS Lett</i> 2005;579(19):4063–8.	doi:10.1016/j.febslet.2005.06.028 PMID 16023114	GSE2434	–	15	–	–	–	1	1
272	Salusjarvi 2008 Salusjärvi L, Kankainen M, Soliymani R, Pitkänen JP, Penttilä M, Ruohonen L. Regulation of xylose metabolism in recombinant <i>Saccharomyces cerevisiae</i> . <i>Microb Cell Fact</i> 2008;7:18.	doi:10.1186/1475-2859-7-18 PMID 18533012 PMC2435516	GSE12890	–	15	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
273	Sunyer-Figueres 2020 Sunyer-Figueres M, Vázquez J, Mas A, Torija MJ, Beltran G. Transcriptomic Insights into the Effect of Melatonin in <i>Saccharomyces cerevisiae</i> in the Presence and Absence of Oxidative Stress. <i>Antioxidants (Basel)</i> 2020;9(10).	doi:10.3390/antiox9100947 PMID 33019712 PMC7650831	GSE154702	–	15	–	–	–	1	2
274	Tauber 2011 Tauber E, Miller-Fleming L, Mason RP, et al., Giorgini F. Functional gene expression profiling in yeast implicates translational dysfunction in mutant huntingtin toxicity. <i>J Biol Chem</i> 2011;286(1):410–9.	doi:10.1074/jbc.M110.101527 PMID 21044956 PMC3012999	GSE18644	–	15	–	–	–	1	1
275	Arana 2012 Arana ME, Kerns RT, Wharey L, Gerrish KE, Bushel PR, Kunkel TA. Transcriptional responses to loss of RNase H2 in <i>Saccharomyces cerevisiae</i> . <i>DNA Repair (Amst)</i> 2012;11(12):933–41.	doi:10.1016/j.dnarep.2012.09.006 PMID 23079308 PMC3535280	GSE34668	–	14	3	1	–	1	1
276	Saint 2014 Saint M, Sawhney S, Sinha I, et al., Natarajan K. The TAF9 C-terminal conserved region domain is required for SAGA and TFIID promoter occupancy to promote transcriptional activation. <i>Mol Cell Biol</i> 2014;34(9):1547–63.	doi:10.1128/MCB.01060-13 PMID 24550006 PMC3993605	GSE44544	–	14	2	1	–	1	2
277	Bessonov 2013 Bessonov K, Walkey CJ, Shelp BJ, van Vuuren HJ, Chiu D, van der Merwe G. Functional analyses of NSF1 in wine yeast using interconnected correlation clustering and molecular analyses. <i>PLoS One</i> 2013;8(10):e77192.	doi:10.1371/journal.pone.0077192 PMID 24130853 PMC3793944	GSE34117	–	14	–	–	–	1	1
278	Brown 2013 Brown NA, de Castro PA, de Castro Pimentel Figueiredo B, et al., Goldman GH. Transcriptional profiling of Brazilian <i>Saccharomyces cerevisiae</i> strains selected for semi-continuous fermentation of sugarcane must. <i>FEMS Yeast Res</i> 2013;13(3):277–90.	doi:10.1111/1567-1364.12031 PMID 23360418	GSE26618 GSE26617	–	14	–	–	–	2	1
279	Cary 2015 Cary GA, Vinh DB, May P, Kuestner R, Dudley AM. Proteomic Analysis of Dhh1 Complexes Reveals a Role for Hsp40 Chaperone Ydj1 in Yeast P-Body Assembly. <i>G3 (Bethesda)</i> 2015;5(11):2497–511.	doi:10.1534/g3.115.021444 PMID 26392412 PMC4632068	GSE65989	–	14	–	–	–	1	1
280	Gonzalez 2013 González A, Casado C, Ariño J, Casamayor A. Ptc6 is required for proper rapamycin-induced down-regulation of the genes coding for ribosomal and rRNA processing proteins in <i>S. cerevisiae</i> . <i>PLoS One</i> 2013;8(5):e64470.	doi:10.1371/journal.pone.0064470 PMID 23704987 PMC3660562	GSE38260	–	14	–	–	–	1	2
281	Hardwick 1999 Hardwick JS, Kuruvilla FG, Tong JK, Shamji AF, Schreiber SL. Rapamycin-modulated transcription defines the subset of nutrient-sensitive signaling pathways directly controlled by the Tor proteins. <i>Proc Natl Acad Sci U S A</i> 1999;96(26):14866–70.	doi:10.1073/pnas.96.26.14866 PMID 10611304 PMC24739	none	–	14	–	–	–	1	2
282	Irvin 2006 Irvin JD, Pugh BF. Genome-wide transcriptional dependence on TAF1 functional domains. <i>J Biol Chem</i> 2006;281(10):6404–12.	doi:10.1074/jbc.M513776200 PMID 16407318	GSE3006	–	14	–	–	–	1	2
283	Ma 2010 Ma M, Liu ZL. Comparative transcriptome profiling analyses during the lag phase uncover YAP1, PDR1, PDR3, RPN4, and HSF1 as key regulatory genes in genomic adaptation to the lignocellulose derived inhibitor HMF for <i>Saccharomyces cerevisiae</i> . <i>BMC Genomics</i> 2010;11:660.	doi:10.1186/1471-2164-11-660 PMID 21106074 PMC3091778	GSE22939	–	14	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
284	Malagon 2006 Malagon F, Kireeva ML, Shafer BK, Lubkowska L, Kashlev M, Strathern JN. Mutations in the <i>Saccharomyces cerevisiae</i> RPB1 gene conferring hypersensitivity to 6-azauracil. <i>Genetics</i> 2006;172(4):2201–9.	doi:10.1534/genetics.105.052415 PMID 16510790 PMC1456368	GSE2831	–	14	–	–	–	1	1
285	Nakamura 2014 Nakamura T, Yamamoto M, Saito K, Ando A, Shima J. Identification of a gene, FMP21, whose expression levels are involved in thermotolerance in <i>Saccharomyces cerevisiae</i> . <i>AMB Express</i> 2014;4:67.	doi:10.1186/s13568-014-0067-2 PMID 25177541 PMC4143573	GSE33276	–	14	–	–	–	1	1
286	Ro 2008 Ro DK, Ouellet M, Paradise EM, et al., Keasling JD. Induction of multiple pleiotropic drug resistance genes in yeast engineered to produce an increased level of anti-malarial drug precursor, artemisinic acid. <i>BMC Biotechnol</i> 2008;8:83.	doi:10.1186/1472-6750-8-83 PMID 18983675 PMC2588579	GSE11620	–	14	–	–	–	1	2
287	Talarek 2010 Talarek N, Cameroni E, Jaquenoud M, et al., De Virgilio C. Initiation of the TORC1-regulated G0 program requires Igo1/2, which license specific mRNAs to evade degradation via the 5'-3' mRNA decay pathway. <i>Mol Cell</i> 2010;38(3):345–55.	doi:10.1016/j.molcel.2010.02.039 PMID 20471941 PMC2919320	GSE20539	–	14	–	–	–	1	2
288	Verwaal 2010 Verwaal R, Jiang Y, Wang J, et al., van Ooyen AJ. Heterologous carotenoid production in <i>Saccharomyces cerevisiae</i> induces the pleiotropic drug resistance stress response. <i>Yeast</i> 2010;27(12):983–98.	doi:10.1002/yea.1807 PMID 20632327	GSE8451	–	14	–	–	–	1	1
289	Yu 2006 Yu C, Palumbo MJ, Lawrence CE, Morse RH. Contribution of the histone H3 and H4 amino termini to Gcn4p- and Gcn5p-mediated transcription in yeast. <i>J Biol Chem</i> 2006;281(14):9755–64.	doi:10.1074/jbc.M513178200 PMID 16461773	GSE4135	–	14	–	–	–	1	1
290	Aghajan 2010 Aghajan M, Jonai N, Flick K, et al., Huang J. Chemical genetics screen for enhancers of rapamycin identifies a specific inhibitor of an SCF family E3 ubiquitin ligase. <i>Nat Biotechnol</i> 2010;28(7):738–42.	doi:10.1038/nbt.1645 PMID 20581845 PMC2902569	GSE22269	–	13	–	–	–	1	1
291	Bloom-Ackermann 2014 Bloom-Ackermann Z, Navon S, Gingold H, Towers R, Pilpel Y, Dahan O. A comprehensive tRNA deletion library unravels the genetic architecture of the tRNA pool. <i>PLoS Genet</i> 2014;10(1):e1004084.	doi:10.1371/journal.pgen.1004084 PMID 24453985 PMC3894157	GSE47050	–	13	–	–	–	1	1
292	Hinkle 2016 Hinkle KL, Anderson CC, Forkey B, et al., Olsen D. Exposure to the lampricide 3-trifluoromethyl-4-nitrophenol results in increased expression of carbohydrate transporters in <i>Saccharomyces cerevisiae</i> . <i>Environ Toxicol Chem</i> 2016;35(7):1727–32.	doi:10.1002/etc.3314 PMID 26606276 PMC5076018	GSE74259	–	13	–	–	–	1	1
293	Leber 2004 Leber JH, Bernales S, Walter P. IRE1-independent gain control of the unfolded protein response. <i>PLoS Biol</i> 2004;2(8):E235.	doi:10.1371/journal.pbio.0020235 PMID 15314654 PMC509300	none	–	13	–	–	–	1	1
294	Reyes 2014 Reyes LH, Gomez JM, Kao KC. Improving carotenoids production in yeast via adaptive laboratory evolution. <i>Metab Eng</i> 2014;21:26–33.	doi:10.1016/j.ymben.2013.11.002 PMID 24262517	GSE51613	–	13	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
295	de Kok 2011 de Kok S, Yilmaz D, Suir E, Pronk JT, Daran JM, van Maris AJ. Increasing free-energy (ATP) conservation in maltose-grown <i>Saccharomyces cerevisiae</i> by expression of a heterologous maltose phosphorylase. <i>Metab Eng</i> 2011;13(5):518–26.	doi:10.1016/j.ymben.2011.06.001 PMID 21684346	GSE30535	–	13	–	–	–	1	1
296	van den Brink 2008 van den Brink J, Daran-Lapujade P, Pronk JT, de Winde JH. New insights into the <i>Saccharomyces cerevisiae</i> fermentation switch: dynamic transcriptional response to anaerobicity and glucose-excess. <i>BMC Genomics</i> 2008;9:100.	doi:10.1186/1471-2164-9-100 PMID 18304306 PMC2292174	GSE8187	–	13	–	–	–	1	1
297	Carroll 2001 Carroll AS, Bishop AC, DeRisi JL, Shokat KM, O’Shea EK. Chemical inhibition of the Pho85 cyclin-dependent kinase reveals a role in the environmental stress response. <i>Proc Natl Acad Sci U S A</i> 2001;98(22):12578–83.	doi:10.1073/pnas.211195798 PMID 11675494 PMC60096	none	–	12	6	2	–	1	2
298	Navlakha 2012 Navlakha S, Gitter A, Bar-Joseph Z. A network-based approach for predicting missing pathway interactions. <i>PLoS Comput Biol</i> 2012;8(8):e1002640.	doi:10.1371/journal.pcbi.1002640 PMID 22916002 PMC3420932	GSE28213	–	12	4	2	–	1	2
299	Boer 2005 Boer VM, Daran JM, Almering MJ, de Winde JH, Pronk JT. Contribution of the <i>Saccharomyces cerevisiae</i> transcriptional regulator Leu3p to physiology and gene expression in nitrogen- and carbon-limited chemostat cultures. <i>FEMS Yeast Res</i> 2005;5(10):885–97.	doi:10.1016/j.femsyr.2005.04.003 PMID 15949974	none	–	12	6	1	–	1	2
300	Chen 2006 Chen JC, Powers T. Coordinate regulation of multiple and distinct biosynthetic pathways by TOR and PKA kinases in <i>S. cerevisiae</i> . <i>Curr Genet</i> 2006;49(5):281–93.	doi:10.1007/s00294-005-0055-9 PMID 16397762	GSE3160	–	12	4	1	–	1	2
301	Cipollina 2008 Cipollina C, van den Brink J, Daran-Lapujade P, Pronk JT, Vai M, de Winde JH. Revisiting the role of yeast Sfp1 in ribosome biogenesis and cell size control: a chemostat study. <i>Microbiology (Reading)</i> 2008;154(Pt 1):337–346.	doi:10.1099/mic.0.2007/011767-0 PMID 18174152	GSE5238	–	12	6	1	–	1	1
302	Drozdova 2014 Drozdova P, Rogoza T, Radchenko E, Lipaeva P, Mironova L. Transcriptional response to the [ISP(+)] prion of <i>Saccharomyces cerevisiae</i> differs from that induced by the deletion of its structural gene, SFP1. <i>FEMS Yeast Res</i> 2014;14(8):1160–70.	doi:10.1111/1567-1364.12211 PMID 25227157	GSE52189	–	12	4	1	–	1	1
303	Hou 2014 Hou J, Tang H, Liu Z, Österlund T, Nielsen J, Petranovic D. Management of the endoplasmic reticulum stress by activation of the heat shock response in yeast. <i>FEMS Yeast Res</i> 2014;14(3):481–94.	doi:10.1111/1567-1364.12125 PMID 24237754	GSE39311	–	12	6	1	–	1	1
304	Pedersen 2012 Pedersen JM, Fredsoe J, Roedgaard M, et al., Andersen AH. DNA Topoisomerases maintain promoters in a state competent for transcriptional activation in <i>Saccharomyces cerevisiae</i> . <i>PLoS Genet</i> 2012;8(12):e1003128.	doi:10.1371/journal.pgen.1003128 PMID 23284296 PMC3527272	GSE22809	–	12	6	1	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
305	Rodriguez-Colman 2010 Rodriguez-Colman MJ, Reverter-Branchat G, Sorolla MA, Tamarit J, Ros J, Cabiscol E. The forkhead transcription factor Hcm1 promotes mitochondrial biogenesis and stress resistance in yeast. <i>J Biol Chem</i> 2010;285(47):37092–101.	doi:10.1074/jbc.M110.174763 PMID 20847055 PMC2978637	GSE20420	–	12	2	1	–	1	2
306	Rossmann 2011 Rossmann MP, Luo W, Tsaponina O, Chabes A, Stillman B. A common telomeric gene silencing assay is affected by nucleotide metabolism. <i>Mol Cell</i> 2011;42(1):127–36.	doi:10.1016/j.molcel.2011.03.007 PMID 21474074 PMC3086572	GSE27222	–	12	6	1	–	1	1
307	Abbott 2007 Abbott DA, Knijnenburg TA, de Poorter LM, Reinders MJ, Pronk JT, van Maris AJ. Generic and specific transcriptional responses to different weak organic acids in anaerobic chemostat cultures of <i>Saccharomyces cerevisiae</i> . <i>FEMS Yeast Res</i> 2007;7(6):819–33.	doi:10.1111/j.1567-1364.2007.00242.x PMID 17484738	GSE5926	–	12	–	–	–	1	1
308	Abbott 2008 Abbott DA, Suir E, van Maris AJ, Pronk JT. Physiological and transcriptional responses to high concentrations of lactic acid in anaerobic chemostat cultures of <i>Saccharomyces cerevisiae</i> . <i>Appl Environ Microbiol</i> 2008;74(18):5759–68.	doi:10.1128/AEM.01030-08 PMID 18676708 PMC2547041	GSE10066	–	12	–	–	–	1	1
309	Albulescu 2012 Albulescu LO, Sabet N, Gudipati M, et al., Pleiss JA. A quantitative, high-throughput reverse genetic screen reveals novel connections between Pre-mRNA splicing and 5' and 3' end transcript determinants. <i>PLoS Genet</i> 2012;8(3):e1002530.	doi:10.1371/journal.pgen.1002530 PMID 22479188 PMC3315463	GSE34330	–	12	–	–	–	1	2
310	Alhaj Sulaiman 2022 Alhaj Sulaiman A, Ali R, Aouida M, Moovarkumudalvan B, Ramotar D. The histone H2B Arg95 residue links the pheromone response pathway to rapamycin-induced G(1) arrest in yeast. <i>Sci Rep</i> 2022;12(1):10023.	doi:10.1038/s41598-022-14053-9 PMID 35705668 PMC9200821	GSE201413	–	12	–	–	–	1	1
311	Alper 2006 Alper H, Moxley J, Nevoigt E, Fink GR, Stephanopoulos G. Engineering yeast transcription machinery for improved ethanol tolerance and production. <i>Science</i> 2006;314(5805):1565–8.	doi:10.1126/science.1131969 PMID 17158319	GSE5185	–	12	–	–	–	1	1
312	Ask 2013 Ask M, Bettiga M, Duraiswamy VR, Olsson L. Pulsed addition of HMF and furfural to batch-grown xylose-utilizing <i>Saccharomyces cerevisiae</i> results in different physiological responses in glucose and xylose consumption phase. <i>Biotechnol Biofuels</i> 2013;6(1):181.	doi:10.1186/1754-6834-6-181 PMID 24341320 PMC3878631	GSE50182	–	12	–	–	–	1	1
313	Belanger 2016 Belanger KD, Larson N, Kahn J, Tkachev D, Ay A. Microarray Analysis of Gene Expression in <i>Saccharomyces cerevisiae</i> kap108Δ Mutants upon Addition of Oxidative Stress. <i>G3 (Bethesda)</i> 2016;6(4):1131–9.	doi:10.1534/g3.116.027011 PMID 26888869 PMC4825647	GSE71068	–	12	–	–	–	1	2
314	Berthelet 2010 Berthelet S, Usher J, Shulist K, et al., Baetz K. Functional genomics analysis of the <i>Saccharomyces cerevisiae</i> iron responsive transcription factor Aft1 reveals iron-independent functions. <i>Genetics</i> 2010;185(3):1111–28.	doi:10.1534/genetics.110.117531 PMID 20439772 PMC2900968	GSE20531	–	12	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
315	Bradley 2009 Bradley PH, Brauer MJ, Rabinowitz JD, Troyanskaya OG. Coordinated concentration changes of transcripts and metabolites in <i>Saccharomyces cerevisiae</i> . <i>PLoS Comput Biol</i> 2009;5(1):e1000270.	doi:10.1371/journal.pcbi.1000270 PMID 19180179 PMC2614473	GSE11754	—	12	—	—	—	2	2
316	Canelas 2010 Canelas AB, Harrison N, Fazio A, et al., Nielsen J. Integrated multilaboratory systems biology reveals differences in protein metabolism between two reference yeast strains. <i>Nat Commun</i> 2010;1:145.	doi:10.1038/ncomms1150 PMID 21266995	GSE24234	—	12	—	—	—	1	1
317	Casamayor 2012 Casamayor A, Serrano R, Platara M, Casado C, Ruiz A, Ariño J. The role of the Snf1 kinase in the adaptive response of <i>Saccharomyces cerevisiae</i> to alkaline pH stress. <i>Biochem J</i> 2012;444(1):39–49.	doi:10.1042/BJ20112099 PMID 22372618	GSE25697	—	12	—	—	—	1	2
318	Celton 2012 Celton M, Sanchez I, Goelzer A, Fromion V, Camarasa C, Dequin S. A comparative transcriptomic, fluxomic and metabolomic analysis of the response of <i>Saccharomyces cerevisiae</i> to increases in NADPH oxidation. <i>BMC Genomics</i> 2012;13:317.	doi:10.1186/1471-2164-13-317 PMID 22805527 PMC3431268	GSE34808 GSE34809	—	12	—	—	—	2	2
319	Chirichella 2017 Chirichella M, Lisi S, Fantini M, et al., Cattaneo A. Post-translational selective intracellular silencing of acetylated proteins with de novo selected intrabodies. <i>Nat Methods</i> 2017;14(3):279–282.	doi:10.1038/nmeth.4144 PMID 28092690	GSE86954	—	12	—	—	—	1	1
320	Daran-Lapujade 2004 Daran-Lapujade P, Jansen ML, Daran JM, van Gulik W, de Winde JH, Pronk JT. Role of transcriptional regulation in controlling fluxes in central carbon metabolism of <i>Saccharomyces cerevisiae</i> . A chemostat culture study. <i>J Biol Chem</i> 2004;279(10):9125–38.	doi:10.1074/jbc.M309578200 PMID 14630934	GSE8895	—	12	—	—	—	1	1
321	De Nadal 2004 De Nadal E, Zapater M, Alepuz PM, Sumoy L, Mas G, Posas F. The MAPK Hog1 recruits Rpd3 histone deacetylase to activate osmoreponsive genes. <i>Nature</i> 2004;427(6972):370–4.	doi:10.1038/nature02258 PMID 14737171	none	—	12	—	—	—	1	2
322	De Nicola 2007 De Nicola R, Hazelwood LA, De Hulster EA, et al., Daran-Lapujade P. Physiological and transcriptional responses of <i>Saccharomyces cerevisiae</i> to zinc limitation in chemostat cultures. <i>Appl Environ Microbiol</i> 2007;73(23):7680–92.	doi:10.1128/AEM.01445-07 PMID 17933919 PMC2168061	GSE8089 GSE8035	—	12	—	—	—	2	1
323	Erlich 2008 Erlich RL, Fry RC, Begley TJ, Dae DL, Lahue RS, Samson LD. Anc1, a protein associated with multiple transcription complexes, is involved in postreplication repair pathway in <i>S. cerevisiae</i> . <i>PLoS One</i> 2008;3(11):e3717.	doi:10.1371/journal.pone.0003717 PMID 19005567 PMC2579579	GSE12150	—	12	—	—	—	1	1
324	Farrugia 2019 Farrugia G, Azzopardi M, Saliba C, et al., Balzan R. Aspirin impairs acetyl-coenzyme A metabolism in redox-compromised yeast cells. <i>Sci Rep</i> 2019;9(1):6152.	doi:10.1038/s41598-019-39489-4 PMID 30992471 PMC6468118	GSE115660	—	12	—	—	—	1	1
325	Gamero 2015 Gamero A, Belloch C, Querol A. Genomic and transcriptomic analysis of aroma synthesis in two hybrids between <i>Saccharomyces cerevisiae</i> and <i>S. kudriavzevii</i> in winemaking conditions. <i>Microb Cell Fact</i> 2015;14:128.	doi:10.1186/s12934-015-0314-5 PMID 26336982 PMC4558966	GSE30779	—	12	—	—	—	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
326	Gardner 2005 Gardner RG, Nelson ZW, Gottschling DE. Degradation-mediated protein quality control in the nucleus. <i>Cell</i> 2005;120(6):803–15.	doi:10.1016/j.cell.2005.01.016 PMID 15797381	GSE2159	–	12	–	–	–	1	2
327	Gil 2011 Gil FN, Gonçalves AC, Jacinto MJ, Becker JD, Viegas CA. Transcriptional profiling in <i>Saccharomyces cerevisiae</i> relevant for predicting alachlor mechanisms of toxicity. <i>Environ Toxicol Chem</i> 2011;30(11):2506–18.	doi:10.1002/etc.640 PMID 21842488	GSE28677	–	12	–	–	–	1	1
328	Hasunuma 2014 Hasunuma T, Ismail KSK, Nambu Y, Kondo A. Co-expression of TAL1 and ADH1 in recombinant xylose-fermenting <i>Saccharomyces cerevisiae</i> improves ethanol production from lignocellulosic hydrolysates in the presence of furfural. <i>J Biosci Bioeng</i> 2014;117(2):165–169.	doi:10.1016/j.jbiosc.2013.07.007 PMID 23916856	GSE45273	–	12	–	–	–	1	2
329	Hong 2011 Hong KK, Vongsangnak W, Vemuri GN, Nielsen J. Unravelling evolutionary strategies of yeast for improving galactose utilization through integrated systems level analysis. <i>Proc Natl Acad Sci U S A</i> 2011;108(29):12179–84.	doi:10.1073/pnas.1103219108 PMID 21715660 PMC3141923	GSE27185	–	12	–	–	–	1	1
330	Kozak 2014 Kozak BU, van Rossum HM, Benjamin KR, et al., van Maris AJ. Replacement of the <i>Saccharomyces cerevisiae</i> acetyl-CoA synthetases by alternative pathways for cytosolic acetyl-CoA synthesis. <i>Metab Eng</i> 2014;21:46–59.	doi:10.1016/j.ymben.2013.11.005 PMID 24269999	GSE47983	–	12	–	–	–	1	1
331	Larsson 2013 Larsson M, Uvell H, Sandström J, Rydén P, Selth LA, Björklund S. Functional studies of the yeast med5, med15 and med16 mediator tail subunits. <i>PLoS One</i> 2013;8(8):e73137.	doi:10.1371/journal.pone.0073137 PMID 23991176 PMC3750046	GSE47712	–	12	–	–	–	1	1
332	Lee 2008 Lee YL, Lee CK. Transcriptional response according to strength of calorie restriction in <i>Saccharomyces cerevisiae</i> . <i>Mol Cells</i> 2008;26(3):299–307.	PMID 18679056	GSE9217	–	12	–	–	–	1	1
333	Lin 2012 Lin L, Chamberlain L, Zhu LJ, Green MR. Analysis of Gal4-directed transcription activation using Tra1 mutants selectively defective for interaction with Gal4. <i>Proc Natl Acad Sci U S A</i> 2012;109(6):1997–2002.	doi:10.1073/pnas.1116340109 PMID 22308403 PMC3277556	GSE31389 GSE31390	–	12	–	–	–	2	1
334	Martin 2004 Martin DE, Demougin P, Hall MN, Bellis M. Rank Difference Analysis of Microarrays (RDAM), a novel approach to statistical analysis of microarray expression profiling data. <i>BMC Bioinformatics</i> 2004;5:148.	doi:10.1186/1471-2105-5-148 PMID 15476558 PMC526220	none	–	12	–	–	–	1	2
335	Matsumoto 2005 Matsumoto R, Akama K, Rakwal R, Iwahashi H. The stress response against denatured proteins in the deletion of cytosolic chaperones SSA1/2 is different from heat-shock response in <i>Saccharomyces cerevisiae</i> . <i>BMC Genomics</i> 2005;6:141.	doi:10.1186/1471-2164-6-141 PMID 16209719 PMC1262714	GSE3316	–	12	–	–	–	1	2
336	Miller-Fleming 2014 Miller-Fleming L, Antas P, Pais TF, Smalley JL, Giorgini F, Outeiro TF. Yeast DJ-1 superfamily members are required for diauxic-shift reprogramming and cell survival in stationary phase. <i>Proc Natl Acad Sci U S A</i> 2014;111(19):7012–7.	doi:10.1073/pnas.1319221111 PMID 24706893 PMC4024926	GSE56124	–	12	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
337	Olesen 2002 Olesen K, Felding T, Gjermansen C, Hansen J. The dynamics of the <i>Saccharomyces carlsbergensis</i> brewing yeast transcriptome during a production-scale lager beer fermentation. <i>FEMS Yeast Res</i> 2002;2(4):563–73.	doi:10.1111/j.1567-1364.2002.tb00123.x PMID 12702272	none	–	12	–	–	–	1	2
338	Park 2013 Park D, Lee Y, Bhupindersingh G, Iyer VR. Widespread misinterpretable ChIP-seq bias in yeast. <i>PLoS One</i> 2013;8(12):e83506.	doi:10.1371/journal.pone.0083506 PMID 24349523 PMC3857294	GSE51376	–	12	–	–	–	1	1
339	Parra 2007 Parra MA, Wyrick JJ. Regulation of gene transcription by the histone H2A N-terminal domain. <i>Mol Cell Biol</i> 2007;27(21):7641–8.	doi:10.1128/MCB.00742-07 PMID 17724083 PMC2169066	GSE7338 GSE7337	–	12	–	–	–	2	1
340	Perez-Valle 2010 Pérez-Valle J, Rothe J, Primo C, et al., Yenush L. Hal4 and Hal5 protein kinases are required for general control of carbon and nitrogen uptake and metabolism. <i>Eukaryot Cell</i> 2010;9(12):1881–90.	doi:10.1128/EC.00184-10 PMID 20952580 PMC3008272	GSE22976	–	12	–	–	–	1	2
341	Rachfall 2013 Rachfall N, Schmitt K, Bandau S, et al., Braus GH. RACK1/Asc1p, a ribosomal node in cellular signaling. <i>Mol Cell Proteomics</i> 2013;12(1):87–105.	doi:10.1074/mcp.M112.017277 PMID 23071099 PMC3536911	GSE34286	–	12	–	–	–	1	2
342	Robertson 2000 Robertson LS, Causton HC, Young RA, Fink GR. The yeast A kinases differentially regulate iron uptake and respiratory function. <i>Proc Natl Acad Sci U S A</i> 2000;97(11):5984–8.	doi:10.1073/pnas.100113397 PMID 10811893 PMC18545	none	–	12	–	–	–	1	2
343	Schuller 2013 Schuller A, Auffermann G, Zoschke K, Schmidt U, Ostermann K, Rödel G. Overexpression of ctr1Δ300, a high-affinity copper transporter with deletion of the cytosolic C-terminus in <i>Saccharomyces cerevisiae</i> under excess copper, leads to disruption of transition metal homeostasis and transcriptional remodelling of cellular processes. <i>Yeast</i> 2013;30(5):201–18.	doi:10.1002/yea.2953 PMID 23576094	GSE44043	–	12	–	–	–	1	1
344	Sheehan 2007 Sheehan KB, McInnerney K, Purevdorj-Gage B, Altenburg SD, Hyman LE. Yeast genomic expression patterns in response to low-shear modeled microgravity. <i>BMC Genomics</i> 2007;8:3.	doi:10.1186/1471-2164-8-3 PMID 17201921 PMC1774566	GSE4136	–	12	–	–	–	1	1
345	Simons 2006 Simons V, Morrissey JP, Latijnhouwers M, et al., Osbourn A. Dual effects of plant steroidal alkaloids on <i>Saccharomyces cerevisiae</i> . <i>Antimicrob Agents Chemother</i> 2006;50(8):2732–40.	doi:10.1128/AAC.00289-06 PMID 16870766 PMC1538658	GSE4669	–	12	–	–	–	1	1
346	Spedale 2012 Spedale G, Meddens CA, Koster MJ, et al., Pijnappel WW. Tight cooperation between Mot1p and NC2β in regulating genome-wide transcription, repression of transcription following heat shock induction and genetic interaction with SAGA. <i>Nucleic Acids Res</i> 2012;40(3):996–1008.	doi:10.1093/nar/gkr784 PMID 21976730 PMC3273795	GSE30148	–	12	–	–	–	1	2
347	Sudarsanam 2000 Sudarsanam P, Iyer VR, Brown PO, Winston F. Whole-genome expression analysis of snf/swi mutants of <i>Saccharomyces cerevisiae</i> . <i>Proc Natl Acad Sci U S A</i> 2000;97(7):3364–9.	doi:10.1073/pnas.97.7.3364 PMID 10725359 PMC16245	none	–	12	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
348	Suzuki 2011 Suzuki T, Iwahashi Y. Gene expression profiles of yeast <i>Saccharomyces cerevisiae</i> sod1 caused by patulin toxicity and evaluation of recovery potential of ascorbic acid. <i>J Agric Food Chem</i> 2011;59(13):7145–54.	doi:10.1021/jf104938p PMID 21648421	GSE26171	–	12	–	–	–	1	1
349	Tai 2007 Tai SL, Daran-Lapujade P, Walsh MC, Pronk JT, Daran JM. Acclimation of <i>Saccharomyces cerevisiae</i> to low temperature: a chemostat-based transcriptome analysis. <i>Mol Biol Cell</i> 2007;18(12):5100–12.	doi:10.1091/mbc.e07-02-0131 PMID 17928405 PMC2096585	GSE6190	–	12	–	–	–	1	1
350	Takagi 2005 Takagi Y, Masuda CA, Chang WH, et al., Kornberg RD. Ubiquitin ligase activity of TFIID and the transcriptional response to DNA damage. <i>Mol Cell</i> 2005;18(2):237–43.	doi:10.1016/j.molcel.2005.03.007 PMID 15837426	none	–	12	–	–	–	1	2
351	Vizoso-Vazquez 2012 Vizoso-Vázquez A, Lamas-Maceiras M, Becerra M, González-Siso MI, Rodríguez-Belmonte E, Cerdán ME. Ixr1p and the control of the <i>Saccharomyces cerevisiae</i> hypoxic response. <i>Appl Microbiol Biotechnol</i> 2012;94(1):173–84.	doi:10.1007/s00253-011-3785-2 PMID 22189861	GSE30046	–	12	–	–	–	1	2
352	Wanichthanarak 2015 Wanichthanarak K, Wongtosrad N, Petranovic D. Genome-wide expression analyses of the stationary phase model of ageing in yeast. <i>Mech Ageing Dev</i> 2015;149:65–74.	doi:10.1016/j.mad.2015.05.008 PMID 26079307	GSE55508	–	12	–	–	–	1	1
353	Yarragudi 2007 Yarragudi A, Parfrey LW, Morse RH. Genome-wide analysis of transcriptional dependence and probable target sites for Abf1 and Rap1 in <i>Saccharomyces cerevisiae</i> . <i>Nucleic Acids Res</i> 2007;35(1):193–202.	doi:10.1093/nar/gkl1059 PMID 17158163 PMC1802568	GSE6073	–	12	–	–	–	1	1
354	Zeng 2017 Zeng WY, Tang YQ, Gou M, Sun ZY, Xia ZY, Kida K. Comparative transcriptomes reveal novel evolutionary strategies adopted by <i>Saccharomyces cerevisiae</i> with improved xylose utilization capability. <i>Appl Microbiol Biotechnol</i> 2017;101(4):1753–1767.	doi:10.1007/s00253-016-8046-y PMID 28004152	GSE80822	–	12	–	–	–	1	1
355	Wan 2011 Wan Y, Chen W, Xing J, et al., Saleem RA. Transcriptome profiling reveals a novel role for trichostatin A in antagonizing histone chaperone Chz1 mediated telomere anti-silencing. <i>FEBS Lett</i> 2011;585(15):2519–25.	doi:10.1016/j.febslet.2011.06.036 PMID 21763693	GSE30228	–	11	10	5	–	1	2
356	Penheiter 2005 Penheiter KL, Washburn TM, Porter SE, Hoffman MG, Jaehning JA. A posttranscriptional role for the yeast Paf1-RNA polymerase II complex is revealed by identification of primary targets. <i>Mol Cell</i> 2005;20(2):213–23.	doi:10.1016/j.molcel.2005.08.023 PMID 16246724	GSE3200	–	11	3	2	–	2	1
357	Guan 2012 Guan Q, Haroon S, Bravo DG, Will JL, Gasch AP. Cellular memory of acquired stress resistance in <i>Saccharomyces cerevisiae</i> . <i>Genetics</i> 2012;192(2):495–505.	doi:10.1534/genetics.112.143016 PMID 22851651 PMC3454879	GSE32196	–	11	6	1	–	1	2
358	Bisschops 2014 Bisschops MM, Zwartjens P, Keuter SG, Pronk JT, Daran-Lapujade P. To divide or not to divide: a key role of Rim15 in calorie-restricted yeast cultures. <i>Biochim Biophys Acta</i> 2014;1843(5):1020–30.	doi:10.1016/j.bbamcr.2014.01.026 PMID 24487068	GSE46853	–	11	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
359	Boender 2011 Boender LG, van Maris AJ, de Hulster EA, et al., Daran-Lapujade P. Cellular responses of <i>Saccharomyces cerevisiae</i> at near-zero growth rates: transcriptome analysis of anaerobic retentostat cultures. <i>FEMS Yeast Res</i> 2011;11(8):603–20.	doi:10.1111/j.1567-1364.2011.00750.x PMID 22093745 PMC3498732	GSE22574	–	11	–	–	–	1	1
360	Bulik 2003 Bulik DA, Olczak M, Lucero HA, Osmond BC, Robbins PW, Specht CA. Chitin synthesis in <i>Saccharomyces cerevisiae</i> in response to supplementation of growth medium with glucosamine and cell wall stress. <i>Eukaryot Cell</i> 2003;2(5):886–900.	doi:10.1128/EC.2.5.886-900.2003 PMID 14555471 PMC219353	none	–	11	–	–	–	1	2
361	Cohen 2002 Cohen BA, Pilpel Y, Mitra RD, Church GM. Discrimination between paralogs using microarray analysis: application to the Yap1p and Yap2p transcriptional networks. <i>Mol Biol Cell</i> 2002;13(5):1608–14.	doi:10.1091/mbc.01-10-0472 PMID 12006656 PMC111130	none	–	11	–	–	–	1	2
362	Epstein 2001 Epstein CB, Waddle JA, Hale W 4th, et al., Butow RA. Genome-wide responses to mitochondrial dysfunction. <i>Mol Biol Cell</i> 2001;12(2):297–308.	doi:10.1091/mbc.12.2.297 PMID 11179416 PMC30944	none	–	11	–	–	–	1	2
363	Gaillard 2014 Gaillard H, Aguilera A. Cleavage factor I links transcription termination to DNA damage response and genome integrity maintenance in <i>Saccharomyces cerevisiae</i> . <i>PLoS Genet</i> 2014;10(3):e1004203.	doi:10.1371/journal.pgen.1004203 PMID 24603480 PMC3945788	GSE50947	–	11	–	–	–	1	1
364	Galitski 1999 Galitski T, Saldanha AJ, Styles CA, Lander ES, Fink GR. Ploidy regulation of gene expression. <i>Science</i> 1999;285(5425):251–4.	doi:10.1126/science.285.5425.251 PMID 10398601	none	–	11	–	–	–	1	2
365	Macadangdang 2014 Macadangdang BR, Oberai A, Spektor T, et al., Kurdistani SK. Evolution of histone 2A for chromatin compaction in eukaryotes. <i>Elife</i> 2014;3.	doi:10.7554/eLife.02792 PMID 24939988 PMC4098067	GSE50440	–	11	–	–	–	1	1
366	Madhani 1999 Madhani HD, Galitski T, Lander ES, Fink GR. Effectors of a developmental mitogen-activated protein kinase cascade revealed by expression signatures of signaling mutants. <i>Proc Natl Acad Sci U S A</i> 1999;96(22):12530–5.	doi:10.1073/pnas.96.22.12530 PMID 10535956 PMC22972	none	–	11	–	–	–	1	2
367	Romero 2019 Romero AM, Ramos-Alonso L, Montellá-Manuel S, et al., Puig S. A genome-wide transcriptional study reveals that iron deficiency inhibits the yeast TORC1 pathway. <i>Biochim Biophys Acta Gene Regul Mech</i> 2019;1862(9):194414.	doi:10.1016/j.bbagr.2019.194414 PMID 31394264	GSE127875	–	11	–	–	–	1	1
368	Moxley 2009 Moxley JF, Jewett MC, Antoniewicz MR, et al., Stephanopoulos G. Linking high-resolution metabolic flux phenotypes and transcriptional regulation in yeast modulated by the global regulator Gcn4p. <i>Proc Natl Acad Sci U S A</i> 2009;106(16):6477–82.	doi:10.1073/pnas.0811091106 PMID 19346491 PMC2672541	GSE4709 GSE7369	–	10	4	4	–	2	1, 2
369	Dong 2007 Dong S, Li C, Zenklusen D, Singer RH, Jacobson A, He F. YRA1 autoregulation requires nuclear export and cytoplasmic Edc3p-mediated degradation of its pre-mRNA. <i>Mol Cell</i> 2007;25(4):559–73.	doi:10.1016/j.molcel.2007.01.012 PMID 17317628 PMC1858660	GSE6647	–	10	5	1	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
370	Lu 2022 Lu PYT, Kirlin AC, Aristizabal MJ, et al., Kobor MS. A balancing act: interactions within NuA4/TIP60 regulate picNuA4 function in <i>Saccharomyces cerevisiae</i> and humans. <i>Genetics</i> 2022;222(3).	doi:10.1093/genetics/iyac136 PMID 36066422 PMC9630986	GSE137261	–	10	2	1	–	1	2
371	Bao 2021 Bao X, Koorengevel MC, Groot Koerkamp MJA, et al., de Kroon AIPM. Shortening of membrane lipid acyl chains compensates for phosphatidylcholine deficiency in choline-auxotroph yeast. <i>EMBO J</i> 2021;40(20):e107966.	doi:10.15252/embj.2021107966 PMID 34520050 PMC8521299	GSE75725	–	10	–	–	–	1	2
372	Boender 2011 Boender LG, Almering MJ, Dijk M, et al., Daran-Lapujade P. Extreme calorie restriction and energy source starvation in <i>Saccharomyces cerevisiae</i> represent distinct physiological states. <i>Biochim Biophys Acta</i> 2011;1813(12):2133–44.	doi:10.1016/j.bbamcr.2011.07.008 PMID 21803078	GSE22602	–	10	–	–	–	1	1
373	Celaj 2017 Celaj A, Schlecht U, Smith JD, et al., St Onge RP. Quantitative analysis of protein interaction network dynamics in yeast. <i>Mol Syst Biol</i> 2017;13(7):934.	doi:10.15252/msb.20177532 PMID 28705884 PMC5527849	GSE72423	–	10	–	–	–	1	1
374	Chan 2008 Chan K, Roth MB. Anoxia-induced suspended animation in budding yeast as an experimental paradigm for studying oxygen-regulated gene expression. <i>Eukaryot Cell</i> 2008;7(10):1795–808.	doi:10.1128/EC.00160-08 PMID 18708563 PMC2568055	GSE12004	–	10	–	–	–	2	2
375	Garcia-Martinez 2021 García-Martínez J, Pérez-Martínez ME, Pérez-Ortín JE, Alepuz P. Recruitment of Xrn1 to stress-induced genes allows efficient transcription by controlling RNA polymerase II backtracking. <i>RNA Biol</i> 2021;18(10):1458–1474.	doi:10.1080/15476286.2020.1857521 PMID 33258404 PMC8489925	GSE151736	–	10	–	–	–	1	2
376	Garcia 2004 García R, Bermejo C, Grau C, et al., Arroyo J. The global transcriptional response to transient cell wall damage in <i>Saccharomyces cerevisiae</i> and its regulation by the cell integrity signaling pathway. <i>J Biol Chem</i> 2004;279(15):15183–95.	doi:10.1074/jbc.M312954200 PMID 14739279	GSE961 GSE960 GSE964	–	10	–	–	–	3	2
377	Gonzalez 2006 González A, Ruiz A, Serrano R, Ariño J, Casamayor A. Transcriptional profiling of the protein phosphatase 2C family in yeast provides insights into the unique functional roles of Ptc1. <i>J Biol Chem</i> 2006;281(46):35057–69.	doi:10.1074/jbc.M607919200 PMID 16973600	GSE6801	–	10	–	–	–	1	2
378	Huang 2004 Huang J, Zhu H, Haggarty SJ, et al., Schreiber SL. Finding new components of the target of rapamycin (TOR) signaling network through chemical genetics and proteome chips. <i>Proc Natl Acad Sci U S A</i> 2004;101(47):16594–9.	doi:10.1073/pnas.0407117101 PMID 15539461 PMC527135	none	–	10	–	–	–	1	2
379	Landry 2007 Landry CR, Lemos B, Rifkin SA, Dickinson WJ, Hartl DL. Genetic properties influencing the evolvability of gene expression. <i>Science</i> 2007;317(5834):118–21.	doi:10.1126/science.1140247 PMID 17525304	GSE7537	–	10	–	–	–	1	2
380	Livas 2011 Livas D, Almering MJ, Daran JM, Pronk JT, Gancedo JM. Transcriptional responses to glucose in <i>Saccharomyces cerevisiae</i> strains lacking a functional protein kinase A. <i>BMC Genomics</i> 2011;12:405.	doi:10.1186/1471-2164-12-405 PMID 21827659 PMC3166949	GSE27541	–	10	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
381	Parreiras 2011 Parreiras LS, Kohn LM, Anderson JB. Cellular effects and epistasis among three determinants of adaptation in experimental populations of <i>Saccharomyces cerevisiae</i> . <i>Eukaryot Cell</i> 2011;10(10):1348–56.	doi:10.1128/EC.05083-11 PMID 21856932 PMC3187067	GSE29353	–	10	–	–	–	1	1
382	Prinz 2004 Prinz S, Avila-Campillo I, Aldridge C, et al., Galitski T. Control of yeast filamentous-form growth by modules in an integrated molecular network. <i>Genome Res</i> 2004;14(3):380–90.	doi:10.1101/gr.2020604 PMID 14993204 PMC353223	none	–	10	–	–	–	1	2
383	Rollero 2016 Rollero S, Mouret JR, Sanchez I, et al., Dequin S. Key role of lipid management in nitrogen and aroma metabolism in an evolved wine yeast strain. <i>Microb Cell Fact</i> 2016;15:32.	doi:10.1186/s12934-016-0434-6 PMID 26861624 PMC4748530	GSE68354	–	10	–	–	–	1	1
384	Travers 2000 Travers KJ, Patil CK, Wodicka L, Lockhart DJ, Weissman JS, Walter P. Functional and genomic analyses reveal an essential coordination between the unfolded protein response and ER-associated degradation. <i>Cell</i> 2000;101(3):249–58.	doi:10.1016/s0092-8674(00)80835-1 PMID 10847680	none	–	10	–	–	1	1	2
385	Xue-Franzen 2013 Xue-Franzen Y, Henriksson J, Bürglin TR, Wright AP. Distinct roles of the Gcn5 histone acetyltransferase revealed during transient stress-induced reprogramming of the genome. <i>BMC Genomics</i> 2013;14:479.	doi:10.1186/1471-2164-14-479 PMID 23865462 PMC3723427	GSE36599	–	10	–	–	–	1	1
386	Zhang 2015 Zhang Y, Liu ZL, Song M. ChiNet uncovers rewired transcription subnetworks in tolerant yeast for advanced biofuels conversion. <i>Nucleic Acids Res</i> 2015;43(9):4393–407.	doi:10.1093/nar/gkv358 PMID 25897127 PMC4482087	GSE50492	–	10	–	–	–	1	2
387	van Leeuwen 2011 van Leeuwen JS, Vermeulen NP, Vos JC. Involvement of the pleiotropic drug resistance response, protein kinase C signaling, and altered zinc homeostasis in resistance of <i>Saccharomyces cerevisiae</i> to diclofenac. <i>Appl Environ Microbiol</i> 2011;77(17):5973–80.	doi:10.1128/AEM.00253-11 PMID 21724882 PMC3165418	GSE29331	–	10	–	–	–	1	1
388	Choi 2017 Choi KM, Hong SJ, van Deursen JM, Kim S, Kim KH, Lee CK. Caloric Restriction and Rapamycin Differentially Alter Energy Metabolism in Yeast. <i>J Gerontol A Biol Sci Med Sci</i> 2017;73(1):29–38.	doi:10.1093/gerona/glx024 PMID 28329151	GSE82324	–	9	–	–	–	1	1
389	Daran-Lapujade 2007 Daran-Lapujade P, Rossell S, van Gulik WM, et al., Bakker BM. The fluxes through glycolytic enzymes in <i>Saccharomyces cerevisiae</i> are predominantly regulated at posttranscriptional levels. <i>Proc Natl Acad Sci U S A</i> 2007;104(40):15753–8.	doi:10.1073/pnas.0707476104 PMID 17898166 PMC2000426	GSE9232	–	9	–	–	–	1	1
390	Dominissini 2012 Dominissini D, Moshitch-Moshkovitz S, Schwartz S, et al., Rechavi G. Topology of the human and mouse m6A RNA methylomes revealed by m6A-seq. <i>Nature</i> 2012;485(7397):201–6.	doi:10.1038/nature11112 PMID 22575960	GSE36958	–	9	–	–	–	1	1
391	Gil 2014 Gil FN, Becker JD, Viegas CA. Potential mechanisms underlying response to effects of the fungicide pyrimethanil from gene expression profiling in <i>Saccharomyces cerevisiae</i> . <i>J Agric Food Chem</i> 2014;62(23):5237–47.	doi:10.1021/jf5007775 PMID 24835131	GSE53126	–	9	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
392	Herrera-Moyano 2014 Herrera-Moyano E, Mergui X, García-Rubio ML, Barroso S, Aguilera A. The yeast and human FACT chromatin-reorganizing complexes solve R-loop-mediated transcription-replication conflicts. <i>Genes Dev</i> 2014;28(7):735–48.	doi:10.1101/gad.234070.113 PMID 24636987 PMC4015491	GSE54340	–	9	–	–	–	1	1
393	Holbein 2009 Holbein S, Wengi A, Decourty L, Freimoser FM, Jacquier A, Dichtl B. Cordycepin interferes with 3' end formation in yeast independently of its potential to terminate RNA chain elongation. <i>RNA</i> 2009;15(5):837–49.	doi:10.1261/rna.1458909 PMID 19324962 PMC2673080	GSE14619	–	9	–	–	–	1	2
394	Iwahashi 2007 Iwahashi H, Ishidou E, Kitagawa E, Momose Y. Combined cadmium and thiuram show synergistic toxicity and induce mitochondrial petite mutants. <i>Environ Sci Technol</i> 2007;41(22):7941–6.	doi:10.1021/es071313y PMID 18075112	GSE8718	–	9	–	–	–	1	2
395	Kim 2009 Kim HS, Huh J, Fay JC. Dissecting the pleiotropic consequences of a quantitative trait nucleotide. <i>FEMS Yeast Res</i> 2009;9(5):713–22.	doi:10.1111/j.1567-1364.2009.00516.x PMID 19456872 PMC2739680	GSE14102	–	9	–	–	–	1	2
396	Kitagawa 2002 Kitagawa E, Takahashi J, Momose Y, Iwahashi H. Effects of the pesticide thiuram: genome-wide screening of indicator genes by yeast DNA microarray. <i>Environ Sci Technol</i> 2002;36(18):3908–15.	doi:10.1021/es015705v PMID 12269742	GSE9336	–	9	–	–	–	1	2
397	Kvitek 2008 Kvitek DJ, Will JL, Gasch AP. Variations in stress sensitivity and genomic expression in diverse <i>S. cerevisiae</i> isolates. <i>PLoS Genet</i> 2008;4(10):e1000223.	doi:10.1371/journal.pgen.1000223 PMID 18927628 PMC2562515	GSE10269	–	9	–	–	–	1	2
398	Lopez-Garcia 2010 López-García B, Gandía M, Muñoz A, Carmona L, Marcos JF. A genomic approach highlights common and diverse effects and determinants of susceptibility on the yeast <i>Saccharomyces cerevisiae</i> exposed to distinct antimicrobial peptides. <i>BMC Microbiol</i> 2010;10:289.	doi:10.1186/1471-2180-10-289 PMID 21078184 PMC2996382	GSE25279	–	9	–	–	–	1	1
399	Lyons 2000 Lyons TJ, Gasch AP, Gaither LA, Botstein D, Brown PO, Eide DJ. Genome-wide characterization of the Zap1p zinc-responsive regulon in yeast. <i>Proc Natl Acad Sci U S A</i> 2000;97(14):7957–62.	doi:10.1073/pnas.97.14.7957 PMID 10884426 PMC16652	none	–	9	–	–	–	1	2
400	Niles 2012 Niles BJ, Mogri H, Hill A, Vlahakis A, Powers T. Plasma membrane recruitment and activation of the AGC kinase Ypk1 is mediated by target of rapamycin complex 2 (TORC2) and its effector proteins Slm1 and Slm2. <i>Proc Natl Acad Sci U S A</i> 2012;109(5):1536–41.	doi:10.1073/pnas.1117563109 PMID 22307609 PMC3277121	GSE33185	–	9	–	–	–	1	2
401	Puig 2008 Puig S, Vergara SV, Thiele DJ. Cooperation of two mRNA-binding proteins drives metabolic adaptation to iron deficiency. <i>Cell Metab</i> 2008;7(6):555–64.	doi:10.1016/j.cmet.2008.04.010 PMID 18522836 PMC2459314	GSE11236	–	9	–	–	–	1	1
402	Rojas 2008 Rojas M, Casado M, Portugal J, Piña B. Selective inhibition of yeast regulons by daunorubicin: a transcriptome-wide analysis. <i>BMC Genomics</i> 2008;9:358.	doi:10.1186/1471-2164-9-358 PMID 18667070 PMC2536678	GSE19351	–	9	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
403	Ruenwai 2011 Ruenwai R, Neiss A, Laoteng K, et al., Nielsen J. Heterologous production of polyunsaturated fatty acids in <i>Saccharomyces cerevisiae</i> causes a global transcriptional response resulting in reduced proteasomal activity and increased oxidative stress. <i>Biotechnol J</i> 2011;6(3):343–56.	doi:10.1002/biot.201000316 PMID 21184438	GSE20155	–	9	–	–	–	1	1
404	Wisselink 2010 Wisselink HW, Cipollina C, Oud B, et al., van Maris AJ. Metabolome, transcriptome and metabolic flux analysis of arabinose fermentation by engineered <i>Saccharomyces cerevisiae</i> . <i>Metab Eng</i> 2010;12(6):537–51.	doi:10.1016/j.ymben.2010.08.003 PMID 20816840	GSE27539	–	9	–	–	–	1	1
405	Zhu 2008 Zhu J, Zhang B, Smith EN, et al., Schadt EE. Integrating large-scale functional genomic data to dissect the complexity of yeast regulatory networks. <i>Nat Genet</i> 2008;40(7):854–61.	doi:10.1038/ng.167 PMID 18552845 PMC2573859	GSE11111	–	9	–	–	–	1	2
406	Dowell 2010 Dowell NL, Sperling AS, Mason MJ, Johnson RC. Chromatin-dependent binding of the <i>S. cerevisiae</i> HMGB protein Nhp6A affects nucleosome dynamics and transcription. <i>Genes Dev</i> 2010;24(18):2031–42.	doi:10.1101/gad.1948910 PMID 20844014 PMC2939365	GSE23607	–	8	8	2	–	1	2
407	Kugou 2007 Kugou K, Sasanuma H, Matsumoto K, Shirahige K, Ohta K. Mre11 mediates gene regulation in yeast spore development. <i>Genes Genet Syst</i> 2007;82(1):21–33.	doi:10.1266/ggs.82.21 PMID 17396017	GSE6620	–	8	4	2	–	1	1
408	Angus-Hill 2001 Angus-Hill ML, Schlichter A, Roberts D, Erdjument-Bromage H, Tempst P, Cairns BR. A Rsc3/Rsc30 zinc cluster dimer reveals novel roles for the chromatin remodeler RSC in gene expression and cell cycle control. <i>Mol Cell</i> 2001;7(4):741–51.	doi:10.1016/s1097-2765(01)00219-2 PMID 11336698	none	–	8	3	1	–	1	2
409	Klopf 2017 Klopf E, Schmidt HA, Clauder-Münster S, Steinmetz LM, Schüller C. INO80 represses osmostress induced gene expression by resetting promoter proximal nucleosomes. <i>Nucleic Acids Res</i> 2017;45(7):3752–3766.	doi:10.1093/nar/gkw1292 PMID 28025392 PMC5397147	GSE78766	–	8	4	1	–	1	1
410	Liko 2007 Liko D, Slattery MG, Heideman W. Stb3 binds to ribosomal RNA processing element motifs that control transcriptional responses to growth in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2007;282(36):26623–8.	doi:10.1074/jbc.M704762200 PMID 17616518	GSE8379	–	8	4	1	–	1	1
411	Adomas 2010 Adomas AB, Lopez-Giraldez F, Clark TA, Wang Z, Townsend JP. Multi-targeted priming for genome-wide gene expression assays. <i>BMC Genomics</i> 2010;11:477.	doi:10.1186/1471-2164-11-477 PMID 20716356 PMC3091673	GSE22992 GSE22936	–	8	–	–	–	2	2
412	Ahmed 2019 Ahmed K, Carter DE, Lajoie P. Hyperactive TORC1 sensitizes yeast cells to endoplasmic reticulum stress by compromising cell wall integrity. <i>FEBS Lett</i> 2019;593(15):1957–1973.	doi:10.1002/1873-3468.13463 PMID 31144305	GSE129200	–	8	–	–	–	1	1
413	Aparicio 2011 Aparicio F, Aparicio-Sanchis R, Gadea J, Sánchez-Navarro JÁ, Pallás V, Murguía JR. A plant virus movement protein regulates the Gcn2p kinase in budding yeast. <i>PLoS One</i> 2011;6(11):e27409.	doi:10.1371/journal.pone.0027409 PMID 22087310 PMC3210792	GSE26120 GSE26119	–	8	–	–	–	2	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
414	Arribere 2011 Arribere JA, Doudna JA, Gilbert WV. Reconsidering movement of eukaryotic mRNAs between polysomes and P bodies. <i>Mol Cell</i> 2011;44(5):745–58.	doi:10.1016/j.molcel.2011.09.019 PMID 22152478 PMC3240842	GSE31220	–	8	–	–	–	1	2
415	Baerends 2008 Baerends RJ, de Hulster E, Geertman JM, et al., Pronk JT. Engineering and analysis of a <i>Saccharomyces cerevisiae</i> strain that uses formaldehyde as an auxiliary substrate. <i>Appl Environ Microbiol</i> 2008;74(10):3182–8.	doi:10.1128/AEM.02858-07 PMID 18378663 PMC2394911	GSE8902	–	8	–	–	–	1	1
416	Buhligen 2013 Bühlig F, Rüdinger P, Fetzter I, et al., Müller S. Sustainability of industrial yeast serial repitching practice studied by gene expression and correlation analysis. <i>J Biotechnol</i> 2013;168(4):718–28.	doi:10.1016/j.jbiotec.2013.09.008 PMID 24060829	GSE45950	–	8	–	–	–	1	2
417	Cocklin 2011 Cocklin R, Heyen J, Larry T, Tyers M, Goebel M. New insight into the role of the Cdc34 ubiquitin-conjugating enzyme in cell cycle regulation via Ace2 and Sic1. <i>Genetics</i> 2011;187(3):701–15.	doi:10.1534/genetics.110.125302 PMID 21196523 PMC3063666	GSE8453	–	8	–	–	–	1	1
418	Deshpande 2017 Deshpande AA, Bhatia M, Laxman S, Bachhawat AK. Thiol trapping and metabolic redistribution of sulfur metabolites enable cells to overcome cysteine overload. <i>Microb Cell</i> 2017;4(4):112–126.	doi:10.15698/mic2017.04.567 PMID 28435838 PMC5376351	GSE87794	–	8	–	–	–	1	2
419	Dettman 2007 Dettman JR, Sirjusingh C, Kohn LM, Anderson JB. Incipient speciation by divergent adaptation and antagonistic epistasis in yeast. <i>Nature</i> 2007;447(7144):585–8.	doi:10.1038/nature05856 PMID 17538619	GSE6870	–	8	–	–	–	1	2
420	Dori-Bachash 2011 Dori-Bachash M, Shema E, Tirosh I. Coupled evolution of transcription and mRNA degradation. <i>PLoS Biol</i> 2011;9(7):e1001106.	doi:10.1371/journal.pbio.1001106 PMID 21811398 PMC3139634	GSE28849	–	8	–	–	–	1	1
421	Fry 2003 Fry RC, Sambandan TG, Rha C. DNA damage and stress transcripts in <i>Saccharomyces cerevisiae</i> mutant sgs1. <i>Mech Ageing Dev</i> 2003;124(7):839–46.	doi:10.1016/s0047-6374(03)00144-1 PMID 12875747	none	–	8	–	–	–	1	2
422	Ghazal 2005 Ghazal G, Ge D, Gervais-Bird J, Gagnon J, Abou Elela S. Genome-wide prediction and analysis of yeast RNase III-dependent snoRNA processing signals. <i>Mol Cell Biol</i> 2005;25(8):2981–94.	doi:10.1128/MCB.25.8.2981-2994.2005 PMID 15798187 PMC1069626	GSE65111	–	8	–	–	–	1	1
423	Gomar-Alba 2015 Gomar-Alba M, Morcillo-Parra MÁ, Olmo ML. Response of yeast cells to high glucose involves molecular and physiological differences when compared to other osmostress conditions. <i>FEMS Yeast Res</i> 2015;15(5):fov039.	doi:10.1093/femsyr/fov039 PMID 26048894	GSE59659 GSE68816	–	8	–	–	–	2	2
424	Haitani 2012 Haitani Y, Tanaka K, Yamamoto M, et al., Shima J. Identification of an acetate-tolerant strain of <i>Saccharomyces cerevisiae</i> and characterization by gene expression analysis. <i>J Biosci Bioeng</i> 2012;114(6):648–51.	doi:10.1016/j.jbiosc.2012.07.002 PMID 22841865	GSE36914	–	8	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
425	Iwahashi 2005 Iwahashi H, Odani M, Ishidou E, Kitagawa E. Adaptation of <i>Saccharomyces cerevisiae</i> to high hydrostatic pressure causing growth inhibition. <i>FEBS Lett</i> 2005;579(13):2847–52.	doi:10.1016/j.febslet.2005.03.100 PMID 15876434	GSE2526	–	8	–	–	–	1	2
426	James 2007 James N, Landrieux E, Collart MA. A SAGA-independent function of SPT3 mediates transcriptional deregulation in a mutant of the Ccr4-not complex in <i>Saccharomyces cerevisiae</i> . <i>Genetics</i> 2007;177(1):123–35.	doi:10.1534/genetics.107.076299 PMID 17660549 PMC2013695	GSE9432	–	8	–	–	–	1	1
427	Khakhina 2015 Khakhina S, Johnson SS, Manoharlal R, et al., Moye-Rowley WS. Control of Plasma Membrane Permeability by ABC Transporters. <i>Eukaryot Cell</i> 2015;14(5):442–53.	doi:10.1128/EC.00021-15 PMID 25724885 PMC4421010	GSE65264	–	8	–	–	–	1	2
428	Lee 2000 Lee SE, Pelliccioli A, Demeter J, et al., Haber JE. Arrest, adaptation, and recovery following a chromosome double-strand break in <i>Saccharomyces cerevisiae</i> . <i>Cold Spring Harb Symp Quant Biol</i> 2000;65:303–14.	doi:10.1101/sqb.2000.65.303 PMID 12760044	none	–	8	–	–	–	1	2
429	Lien 2019 Lien PTK, Viet NTM, Mizuno T, Suda Y, Irie K. Pop2 phosphorylation at S39 contributes to the glucose repression of stress response genes, HSP12 and HSP26. <i>PLoS One</i> 2019;14(4):e0215064.	doi:10.1371/journal.pone.0215064 PMID 30973945 PMC6459547	GSE124908	–	8	–	–	–	1	1
430	Ling 2013 Ling H, Chen B, Kang A, Lee JM, Chang MW. Transcriptome response to alkane biofuels in <i>Saccharomyces cerevisiae</i> : identification of efflux pumps involved in alkane tolerance. <i>Biotechnol Biofuels</i> 2013;6(1):95.	doi:10.1186/1754-6834-6-95 PMID 23826995 PMC3717029	GSE38653	–	8	–	–	–	1	2
431	Mith 2015 Mith O, Benhamdi A, Castillo T, et al., Marquès L. The antifungal plant defensin AhPDF1.1b is a beneficial factor involved in adaptive response to zinc overload when it is expressed in yeast cells. <i>Microbiologyopen</i> 2015;4(3):409–22.	doi:10.1002/mbo3.248 PMID 25755096 PMC4475384	GSE63395	–	8	–	–	–	1	1
432	Navarro-Tapia 2016 Navarro-Tapia E, Nana RK, Querol A, Pérez-Torrado R. Ethanol Cellular Defense Induce Unfolded Protein Response in Yeast. <i>Front Microbiol</i> 2016;7:189.	doi:10.3389/fmicb.2016.00189 PMID 26925053 PMC4757686	GSE44863	–	8	–	–	–	1	2
433	Ogawa 2000 Ogawa N, DeRisi J, Brown PO. New components of a system for phosphate accumulation and polyphosphate metabolism in <i>Saccharomyces cerevisiae</i> revealed by genomic expression analysis. <i>Mol Biol Cell</i> 2000;11(12):4309–21.	doi:10.1091/mbc.11.12.4309 PMID 11102525 PMC15074	none	–	8	–	–	–	2	2
434	Otero 2010 Otero JM, Vongsangnak W, Asadollahi MA, et al., Nielsen J. Whole genome sequencing of <i>Saccharomyces cerevisiae</i> : from genotype to phenotype for improved metabolic engineering applications. <i>BMC Genomics</i> 2010;11:723.	doi:10.1186/1471-2164-11-723 PMID 21176163 PMC3022925	GSE21479	–	8	–	–	–	1	1
435	Pastor-Flores 2015 Pastor-Flores D, Ferrer-Dalmau J, Bahí A, Boleda M, Biondi RM, Casamayor A. Depletion of yeast PDK1 orthologs triggers a stress-like transcriptional response. <i>BMC Genomics</i> 2015;16(1):719.	doi:10.1186/s12864-015-1903-8 PMID 26391581 PMC4578605	GSE32623	–	8	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
436	Plateau 2017 Plateau P, Saveanu C, Lestini R, et al., Lazard M. Exposure to selenomethionine causes selenocysteine misincorporation and protein aggregation in <i>Saccharomyces cerevisiae</i> . <i>Sci Rep</i> 2017;7:44761.	doi:10.1038/srep44761 PMID 28303947 PMC53555996	GSE84340	–	8	–	–	–	1	2
437	Protchenko 2008 Protchenko O, Shakoury-Elizeh M, Keane P, Storey J, Androphy R, Philpott CC. Role of PUG1 in inducible porphyrin and heme transport in <i>Saccharomyces cerevisiae</i> . <i>Eukaryot Cell</i> 2008;7(5):859–71.	doi:10.1128/EC.00414-07 PMID 18326586 PMC2394968	GSE9514	–	8	–	–	–	1	1
438	Sahi 2010 Sahi C, Lee T, Inada M, Pleiss JA, Craig EA. Cwc23, an essential J protein critical for pre-mRNA splicing with a dispensable J domain. <i>Mol Cell Biol</i> 2010;30(1):33–42.	doi:10.1128/MCB.00842-09 PMID 19822657 PMC2798280	GSE18485	–	8	–	–	–	1	2
439	Schrader 2006 Schrader R, Young C, Kozian D, Hoffmann R, Lottspeich F. Temperature-sensitive eIF5A mutant accumulates transcripts targeted to the nonsense-mediated decay pathway. <i>J Biol Chem</i> 2006;281(46):35336–46.	doi:10.1074/jbc.M601460200 PMID 16987817	GSE5290	–	8	–	–	–	1	1
440	Shakoury-Elizeh 2010 Shakoury-Elizeh M, Protchenko O, Berger A, et al., Philpott CC. Metabolic response to iron deficiency in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2010;285(19):14823–33.	doi:10.1074/jbc.M109.091710 PMID 20231268 PMC2863190	GSE19016	–	8	–	–	–	1	1
441	Sheltzer 2012 Sheltzer JM, Torres EM, Dunham MJ, Amon A. Transcriptional consequences of aneuploidy. <i>Proc Natl Acad Sci U S A</i> 2012;109(31):12644–9.	doi:10.1073/pnas.1209227109 PMID 22802626 PMC3411958	GSE35853	–	8	–	–	–	1	2
442	Slavov 2014 Slavov N, Budnik BA, Schwab D, Airoidi EM, van Oudenaarden A. Constant growth rate can be supported by decreasing energy flux and increasing aerobic glycolysis. <i>Cell Rep</i> 2014;7(3):705–14.	doi:10.1016/j.celrep.2014.03.057 PMID 24767987 PMC4049626	GSE56773	–	8	–	–	–	1	2
443	Smith 2002 Smith JJ, Marelli M, Christmas RH, et al., Aitchison JD. Transcriptome profiling to identify genes involved in peroxisome assembly and function. <i>J Cell Biol</i> 2002;158(2):259–71.	doi:10.1083/jcb.200204059 PMID 12135984 PMC2173120	none	–	8	–	–	–	1	2
444	Takahashi 2011 Takahashi YH, Schulze JM, Jackson J, et al., Shilatifard A. Dot1 and histone H3K79 methylation in natural telomeric and HM silencing. <i>Mol Cell</i> 2011;42(1):118–26.	doi:10.1016/j.molcel.2011.03.006 PMID 21474073 PMC3085244	GSE27234	–	8	–	–	–	1	1
445	Tompa 2007 Tompa R, Madhani HD. Histone H3 lysine 36 methylation antagonizes silencing in <i>Saccharomyces cerevisiae</i> independently of the Rpd3S histone deacetylase complex. <i>Genetics</i> 2007;175(2):585–93.	doi:10.1534/genetics.106.067751 PMID 17179083 PMC1800606	GSE4934	–	8	–	–	–	1	2
446	Tu 2011 Tu WY, Huang YC, Liu LF, Chang LH, Tam MF. Rpl12p affects the transcription of the PHO pathway high-affinity inorganic phosphate transporters and repressible phosphatases. <i>Yeast</i> 2011;28(6):481–93.	doi:10.1002/yea.1852 PMID 21469193	GSE22644	–	8	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
447	Williams 2002 Williams RM, Primig M, Washburn BK, et al., Esposito RE. The Ume6 regulon coordinates metabolic and meiotic gene expression in yeast. <i>Proc Natl Acad Sci U S A</i> 2002;99(21):13431–6.	doi:10.1073/pnas.202495299 PMID 12370439 PMC129690	none	–	8	–	–	–	1	2
448	Zeng 2016 Zeng WY, Tang YQ, Gou M, Xia ZY, Kida K. Transcriptomes of a xylose-utilizing industrial flocculating <i>Saccharomyces cerevisiae</i> strain cultured in media containing different sugar sources. <i>AMB Express</i> 2016;6(1):51.	doi:10.1186/s13568-016-0223-y PMID 27485516 PMC4970999	GSE80748	–	8	–	–	–	1	1
449	de Castro 2012 de Castro PA, Savoldi M, Bonatto D, et al., Goldman GH. Transcriptional profiling of <i>Saccharomyces cerevisiae</i> exposed to propolis. <i>BMC Complement Altern Med</i> 2012;12:194.	doi:10.1186/1472-6882-12-194 PMID 23092287 PMC3598864	GSE33971	–	8	–	–	–	1	1
450	Chen 2013 Chen K, Wilson MA, Hirsch C, et al., Dent SY. Stabilization of the promoter nucleosomes in nucleosome-free regions by the yeast Cyc8-Tup1 corepressor. <i>Genome Res</i> 2013;23(2):312–22.	doi:10.1101/gr.141952.112 PMID 23124522 PMC3561872	GSE37466	–	7	4	2	–	1	1
451	Chia 2018 Chia SZ, Lai YW, Yagoub D, et al., Wilkins MR. Knockout of the Hmt1p Arginine Methyltransferase in <i>Saccharomyces cerevisiae</i> Leads to the Dysregulation of Phosphate-associated Genes and Processes. <i>Mol Cell Proteomics</i> 2018;17(12):2462–2479.	doi:10.1074/mcp.RA117.000214 PMID 30206180 PMC6283299	GSE99869	–	7	3	1	–	1	1
452	Bedalov 2001 Bedalov A, Gattabont T, Irvine WP, Gottschling DE, Simon JA. Identification of a small molecule inhibitor of Sir2p. <i>Proc Natl Acad Sci U S A</i> 2001;98(26):15113–8.	doi:10.1073/pnas.261574398 PMID 11752457 PMC64992	none	–	7	–	–	–	1	2
453	Bro 2003 Bro C, Regenber B, Lagniel G, Labarre J, Montero-Lomelí M, Nielsen J. Transcriptional, proteomic, and metabolic responses to lithium in galactose-grown yeast cells. <i>J Biol Chem</i> 2003;278(34):32141–9.	doi:10.1074/jbc.M304478200 PMID 12791685	none	–	7	–	–	–	1	2
454	Cap 2012 Cáp M, Stěpánek L, Harant K, Váchová L, Palková Z. Cell differentiation within a yeast colony: metabolic and regulatory parallels with a tumor-affected organism. <i>Mol Cell</i> 2012;46(4):436–48.	doi:10.1016/j.molcel.2012.04.001 PMID 22560924	GSE35887	–	7	–	–	–	1	2
455	Chu 1998 Chu S, DeRisi J, Eisen M, et al., Herskowitz I. The transcriptional program of sporulation in budding yeast. <i>Science</i> 1998;282(5389):699–705.	doi:10.1126/science.282.5389.699 PMID 9784122	none	–	7	–	–	–	1	2
456	DeRisi 1997 DeRisi JL, Iyer VR, Brown PO. Exploring the metabolic and genetic control of gene expression on a genomic scale. <i>Science</i> 1997;278(5338):680–6.	doi:10.1126/science.278.5338.680 PMID 9381177	none	–	7	–	–	–	1	2
457	Driscoll 2017 Driscoll HE, Murray JM, English EL, Hunter TC, Pivarski K, Dolci ED. Microarray data and gene expression statistics for <i>Saccharomyces cerevisiae</i> exposed to simulated asbestos mine drainage. <i>Data Brief</i> 2017;13:37–45.	doi:10.1016/j.dib.2017.05.021 PMID 28560281 PMC5440250	GSE89875	–	7	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
458	Klockow 2008 Klockow C, Stahl F, Scheper T, Hitzmann B. In vivo regulation of glucose transporter genes at glucose concentrations between 0 and 500 mg/L in a wild type of <i>Saccharomyces cerevisiae</i> . <i>J Biotechnol</i> 2008;135(2):161–7.	doi:10.1016/j.jbiotec.2008.03.009 PMID 18455824	GSE6358	–	7	–	–	–	1	2
459	Lopez-Serra 2014 Lopez-Serra L, Kelly G, Patel H, Stewart A, Uhlmann F. The Scc2-Scc4 complex acts in sister chromatid cohesion and transcriptional regulation by maintaining nucleosome-free regions. <i>Nat Genet</i> 2014;46(10):1147–51.	doi:10.1038/ng.3080 PMID 25173104 PMC4177232	GSE56991	–	7	–	–	–	1	1
460	Marton 1998 Marton MJ, DeRisi JL, Bennett HA, et al., Friend SH. Drug target validation and identification of secondary drug target effects using DNA microarrays. <i>Nat Med</i> 1998;4(11):1293–301.	doi:10.1038/3282 PMID 9809554	none	–	7	–	–	–	1	2
461	Wyrick 1999 Wyrick JJ, Holstege FC, Jennings EG, et al., Young RA. Chromosomal landscape of nucleosome-dependent gene expression and silencing in yeast. <i>Nature</i> 1999;402(6760):418–21.	doi:10.1038/46567 PMID 10586882	none	–	7	–	–	–	1	2
462	Yeang 2005 Yeang CH, Mak HC, McCuine S, Workman C, Jaakkola T, Ideker T. Validation and refinement of gene-regulatory pathways on a network of physical interactions. <i>Genome Biol</i> 2005;6(7):R62.	doi:10.1186/gb-2005-6-7-r62 PMID 15998451 PMC1175993	none	–	7	–	–	–	1	2
463	Van de Vosse 2013 Van de Vosse DW, Wan Y, Lapetina DL, et al., Wozniak RW. A role for the nucleoporin Nup170p in chromatin structure and gene silencing. <i>Cell</i> 2013;152(5):969–83.	doi:10.1016/j.cell.2013.01.049 PMID 23452847 PMC3690833	GSE36793	–	6	3	3	–	1	2
464	Tamaki 2005 Tamaki H, Yun CW, Mizutani T, et al., Kumagai H. Glucose-dependent cell size is regulated by a G protein-coupled receptor system in yeast <i>Saccharomyces cerevisiae</i> . <i>Genes Cells</i> 2005;10(3):193–206.	doi:10.1111/j.1365-2443.2005.00828.x PMID 15743410	GSE970	–	6	4	2	–	1	1
465	Hickman 2007 Hickman MJ, Winston F. Heme levels switch the function of Hap1 of <i>Saccharomyces cerevisiae</i> between transcriptional activator and transcriptional repressor. <i>Mol Cell Biol</i> 2007;27(21):7414–24.	doi:10.1128/MCB.00887-07 PMID 17785431 PMC2169065	GSE8613	–	6	3	1	–	1	2
466	Rodríguez-Navarro 2004 Rodríguez-Navarro S, Fischer T, Luo MJ, et al., Hurt E. Sus1, a functional component of the SAGA histone acetylase complex and the nuclear pore-associated mRNA export machinery. <i>Cell</i> 2004;116(1):75–86.	doi:10.1016/s0092-8674(03)01025-0 PMID 14718168	none	–	6	3	1	–	1	2
467	Santos-Pereira 2013 Santos-Pereira JM, Herrero AB, García-Rubio ML, Marín A, Moreno S, Aguilera A. The Npl3 hnRNP prevents R-loop-mediated transcription-replication conflicts and genome instability. <i>Genes Dev</i> 2013;27(22):2445–58.	doi:10.1101/gad.229880.113 PMID 24240235 PMC3841734	GSE50186	–	6	3	1	–	1	1
468	Agarwal 2008 Agarwal AK, Xu T, Jacob MR, et al., Clark AM. Role of heme in the antifungal activity of the azaoxaporphine alkaloid sampangine. <i>Eukaryot Cell</i> 2008;7(2):387–400.	doi:10.1128/EC.00323-07 PMID 18156292 PMC2238161	GSE10073	–	6	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
469	Angell 2006 Angell S, Bench BJ, Williams H, Watanabe CM. Pyocyanin isolated from a marine microbial population: synergistic production between two distinct bacterial species and mode of action. <i>Chem Biol</i> 2006;13(12):1349–59.	doi:10.1016/j.chembiol.2006.10.012 PMID 17185230	GSE6185	–	6	–	–	–	1	1
470	Backhus 2001 Backhus LE, DeRisi J, Bisson LF. Functional genomic analysis of a commercial wine strain of <i>Saccharomyces cerevisiae</i> under differing nitrogen conditions. <i>FEMS Yeast Res</i> 2001;1(2):111–25.	doi:10.1111/j.1567-1364.2001.tb00022.x PMID 12702356	GSE4887	–	6	–	–	–	1	2
471	Brown 2008 Brown KM, Landry CR, Hartl DL, Cavalieri D. Cascading transcriptional effects of a naturally occurring frameshift mutation in <i>Saccharomyces cerevisiae</i> . <i>Mol Ecol</i> 2008;17(12):2985–97.	doi:10.1111/j.1365-294X.2008.03765.x PMID 18422925	GSE9875	–	6	–	–	–	1	2
472	Celona 2011 Celona B, Weiner A, Di Felice F, et al., Agresti A. Substantial histone reduction modulates genomewide nucleosomal occupancy and global transcriptional output. <i>PLoS Biol</i> 2011;9(6):e1001086.	doi:10.1371/journal.pbio.1001086 PMID 21738444 PMC3125158	GSE23711	–	6	–	–	–	1	1
473	Cheung 2008 Cheung V, Chua G, Batada NN, et al., Winston F. Chromatin- and transcription-related factors repress transcription from within coding regions throughout the <i>Saccharomyces cerevisiae</i> genome. <i>PLoS Biol</i> 2008;6(11):e277.	doi:10.1371/journal.pbio.0060277 PMID 18998772 PMC2581627	GSE12272	–	6	–	–	–	1	2
474	Chiasson 2014 Chiasson DM, Loughlin PC, Mazurkiewicz D, et al., Kaiser BN. Soybean SAT1 (Symbiotic Ammonium Transporter 1) encodes a bHLH transcription factor involved in nodule growth and NH ₄ ⁺ transport. <i>Proc Natl Acad Sci U S A</i> 2014;111(13):4814–9.	doi:10.1073/pnas.1312801111 PMID 24707045 PMC3977234	GSE55804	–	6	–	–	–	1	1
475	Falbo 2009 Falbo KB, Alabert C, Katou Y, et al., Shen X. Involvement of a chromatin remodeling complex in damage tolerance during DNA replication. <i>Nat Struct Mol Biol</i> 2009;16(11):1167–72.	doi:10.1038/nsmb.1686 PMID 19855395 PMC2974178	GSE18555	–	6	–	–	–	1	2
476	Fernandez-Acero 2015 Fernández-Acero T, Rodríguez-Escudero I, Molina M, Cid VJ. The yeast cell wall integrity pathway signals from recycling endosomes upon elimination of phosphatidylinositol (4,5)-bisphosphate by mammalian phosphatidylinositol 3-kinase. <i>Cell Signal</i> 2015;27(11):2272–84.	doi:10.1016/j.cellsig.2015.08.004 PMID 26261079	GSE59596	–	6	–	–	–	1	1
477	Flom 2012 Flom GA, Langner E, Johnson JL. Identification of an Hsp90 mutation that selectively disrupts cAMP/PKA signaling in <i>Saccharomyces cerevisiae</i> . <i>Curr Genet</i> 2012;58(3):149–63.	doi:10.1007/s00294-012-0373-7 PMID 22461145	GSE34757	–	6	–	–	–	1	1
478	Gopinath 2014 Gopinath RK, You ST, Chien KY, et al., Leu JY. The Hsp90-dependent proteome is conserved and enriched for hub proteins with high levels of protein-protein connectivity. <i>Genome Biol Evol</i> 2014;6(10):2851–65.	doi:10.1093/gbe/evu226 PMID 25316598 PMC4224352	GSE56186	–	6	–	–	–	1	2
479	Gross 2000 Gross C, Kelleher M, Iyer VR, Brown PO, Winge DR. Identification of the copper regulon in <i>Saccharomyces cerevisiae</i> by DNA microarrays. <i>J Biol Chem</i> 2000;275(41):32310–6.	doi:10.1074/jbc.M005946200 PMID 10922376	none	–	6	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
480	Grund 2008 Grund SE, Fischer T, Cabal GG, Antúnez O, Pérez-Ortín JE, Hurt E. The inner nuclear membrane protein Src1 associates with subtelomeric genes and alters their regulated gene expression. <i>J Cell Biol</i> 2008;182(5):897–910.	doi:10.1083/jcb.200803098 PMID 18762579 PMC25285585	GSE6370	–	6	–	–	–	1	1
481	Hackmann 2014 Hackmann A, Wu H, Schneider UM, Meyer K, Jung K, Krebber H. Quality control of spliced mRNAs requires the shuttling SR proteins Gbp2 and Hrb1. <i>Nat Commun</i> 2014;5:3123.	doi:10.1038/ncomms4123 PMID 24452287	GSE52808	–	6	–	–	–	1	2
482	Hodges 2015 Hodges AJ, Gallegos IJ, Laughery MF, Meas R, Tran L, Wyrick JJ. Histone Sprocket Arginine Residues Are Important for Gene Expression, DNA Repair, and Cell Viability in <i>Saccharomyces cerevisiae</i> . <i>Genetics</i> 2015;200(3):795–806.	doi:10.1534/genetics.115.175885 PMID 25971662 PMC4512544	GSE68159	–	6	–	–	–	1	1
483	Hong 2013 Hong KK, Nielsen J. Adaptively evolved yeast mutants on galactose show trade-offs in carbon utilization on glucose. <i>Metab Eng</i> 2013;16:78–86.	doi:10.1016/j.ymben.2013.01.007 PMID 23376593	GSE39065	–	6	–	–	–	1	1
484	Jansen 2005 Jansen MLA, Diderich JA, Mashego M, et al., Pronk JT. Prolonged selection in aerobic, glucose-limited chemostat cultures of <i>Saccharomyces cerevisiae</i> causes a partial loss of glycolytic capacity. <i>Microbiology (Reading)</i> 2005;151(Pt 5):1657–1669.	doi:10.1099/mic.0.27577-0 PMID 15870473	GSE8898	–	6	–	–	–	1	1
485	Jimenez-Gutierrez 2022 Jiménez-Gutiérrez E, Fernández-Acero T, Alonso-Rodríguez E, Molina M, Martín H. Neomycin Interferes with Phosphatidylinositol-4,5-Bisphosphate at the Yeast Plasma Membrane and Activates the Cell Wall Integrity Pathway. <i>Int J Mol Sci</i> 2022;23(19).	doi:10.3390/ijms231911034 PMID 36232332 PMC9569482	GSE210785	–	6	–	–	–	1	1
486	Jimeno 2011 Jimeno S, Tous C, García-Rubio ML, et al., Aguilera A. New suppressors of THO mutations identify Thp3 (Ypr045c)-Csn12 as a protein complex involved in transcription elongation. <i>Mol Cell Biol</i> 2011;31(4):674–85.	doi:10.1128/MCB.01188-10 PMID 21149575 PMC3028642	GSE18994	–	6	–	–	–	1	1
487	Jin 2004 Jin YS, Laplaza JM, Jeffries TW. <i>Saccharomyces cerevisiae</i> engineered for xylose metabolism exhibits a respiratory response. <i>Appl Environ Microbiol</i> 2004;70(11):6816–25.	doi:10.1128/AEM.70.11.6816-6825.2004 PMID 15528549 PMC525251	none	–	6	–	–	–	1	2
488	Johanson 2006 Johanson K, Allen PL, Gonzalez-Villalobos RA, Baker CB, D’Elia R, Hammond TG. Gene expression and survival changes in <i>Saccharomyces cerevisiae</i> during suspension culture. <i>Biotechnol Bioeng</i> 2006;93(6):1050–9.	doi:10.1002/bit.20810 PMID 16440349	GSE2818	–	6	–	–	–	1	1
489	Joshi 2012 Joshi RS, Piña B, Roca J. Topoisomerase II is required for the production of long Pol II gene transcripts in yeast. <i>Nucleic Acids Res</i> 2012;40(16):7907–15.	doi:10.1093/nar/gks626 PMID 22718977 PMC3439932	GSE18605	–	6	–	–	–	1	2
490	Judeh 2013 Judeh T, Johnson C, Kumar A, Zhu D. TEAK: topology enrichment analysis framework for detecting activated biological subpathways. <i>Nucleic Acids Res</i> 2013;41(3):1425–37.	doi:10.1093/nar/gks1299 PMID 23268448 PMC3561980	GSE40530	–	6	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
491	Kelly 2006 Kelly GT, Liu C, Smith R 3rd, Coleman RS, Watanabe CM. Cellular effects induced by the antitumor agent azinomycin B. <i>Chem Biol</i> 2006;13(5):485–92.	doi:10.1016/j.chembiol.2006.02.015 PMID 16720269	GSE4311	–	6	–	–	–	1	1
492	Kucukgoze 2013 Küçükgoze G, Alkim C, Yılmaz Ü, et al., Çakar ZP. Evolutionary engineering and transcriptomic analysis of nickel-resistant <i>Saccharomyces cerevisiae</i> . <i>FEMS Yeast Res</i> 2013;13(8):731–46.	doi:10.1111/1567-1364.12073 PMID 23992612	GSE50985	–	6	–	–	–	1	2
493	Liang 2013 Liang CY, Wang LC, Lo WS. Dissociation of the H3K36 demethylase Rph1 from chromatin mediates derepression of environmental stress-response genes under genotoxic stress in <i>Saccharomyces cerevisiae</i> . <i>Mol Biol Cell</i> 2013;24(20):3251–62.	doi:10.1091/mbc.E12-11-0820 PMID 23985319 PMC3806659	GSE39950	–	6	–	–	–	1	1
494	Liu 2013 Liu J, Huang X, Withers BR, Blalock E, Liu K, Dickson RC. Reducing sphingolipid synthesis orchestrates global changes to extend yeast lifespan. <i>Aging Cell</i> 2013;12(5):833–41.	doi:10.1111/accel.12107 PMID 23725375 PMC3773046	GSE41362	–	6	–	–	–	1	1
495	Lopez-Malo 2015 López-Malo M, García-Rios E, Melgar B, Sanchez MR, Dunham MJ, Guillamón JM. Evolutionary engineering of a wine yeast strain revealed a key role of inositol and mannoprotein metabolism during low-temperature fermentation. <i>BMC Genomics</i> 2015;16(1):537.	doi:10.1186/s12864-015-1755-2 PMID 26194190 PMC4509780	GSE67428	–	6	–	–	–	1	1
496	Marquina 2012 Marquina M, Queralt E, Casamayor A, Ariño J. Lack of the Glc7 phosphatase regulatory subunit Ypil activates the morphogenetic checkpoint. <i>Int J Biochem Cell Biol</i> 2012;44(11):1862–71.	doi:10.1016/j.biocel.2012.06.029 PMID 22750472	GSE23134	–	6	–	–	–	1	2
497	Mendes 2013 Mendes F, Sieuwerts S, de Hulster E, et al., Daran-Lapujade P. Transcriptome-based characterization of interactions between <i>Saccharomyces cerevisiae</i> and <i>Lactobacillus delbrueckii</i> subsp. <i>bulgaricus</i> in lactose-grown chemostat cocultures. <i>Appl Environ Microbiol</i> 2013;79(19):5949–61.	doi:10.1128/AEM.01115-13 PMID 23872557 PMC3811385	GSE45776	–	6	–	–	–	1	1
498	Moreno-Cermeno 2013 Moreno-Cermeño A, Alsina D, Cabiscol E, Tamarit J, Ros J. Metabolic remodeling in frataxin-deficient yeast is mediated by Cth2 and Adr1. <i>Biochim Biophys Acta</i> 2013;1833(12):3326–3337.	doi:10.1016/j.bbamcr.2013.09.019 PMID 24100161	GSE44871	–	6	–	–	–	1	2
499	Moreno 2019 Moreno AD, González-Fernández C, Ballesteros M, Tomás-Pejó E. Insoluble solids at high concentrations repress yeast's response against stress and increase intracellular ROS levels. <i>Sci Rep</i> 2019;9(1):12236.	doi:10.1038/s41598-019-48733-w PMID 31439886 PMC6706384	GSE115460	–	6	–	–	–	1	1
500	Moreno 2022 Moreno AD, González-Fernández C, Tomás-Pejó E. Insights into cell robustness against lignocellulosic inhibitors and insoluble solids in bioethanol production processes. <i>Sci Rep</i> 2022;12(1):557.	doi:10.1038/s41598-021-04554-4 PMID 35017613 PMC8752620	GSE159167	–	6	–	–	–	1	1
501	Mutiu 2007 Mutiu AI, Hoke SM, Genereaux J, et al., Brandl CJ. Structure/function analysis of the phosphatidylinositol-3-kinase domain of yeast tra1. <i>Genetics</i> 2007;177(1):151–66.	doi:10.1534/genetics.107.074476 PMID 17660562 PMC2013730	GSE6847	–	6	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
502	Nag 2008 Nag R, Gong F, Fahy D, Smerdon MJ. A single amino acid change in histone H4 enhances UV survival and DNA repair in yeast. <i>Nucleic Acids Res</i> 2008;36(11):3857–66.	doi:10.1093/nar/gkn311 PMID 18508805 PMC2441814	GSE11282	–	6	–	–	–	1	1
503	O'Doherty 2013 O'Doherty PJ, Lyons V, Higgins VJ, Rogers PJ, Bailey TD, Wu MJ. Transcriptomic insights into the molecular response of <i>Saccharomyces cerevisiae</i> to linoleic acid hydroperoxide. <i>Free Radic Res</i> 2013;47(12):1054–65.	doi:10.3109/10715762.2013.849344 PMID 24074273	GSE47820	–	6	–	–	–	1	1
504	Pagano 2019 Pagano L, Caldara M, Villani M, Zappettini A, Marmioli N, Marmioli M. In Vivo-In Vitro Comparative Toxicology of Cadmium Sulphide Quantum Dots in the Model Organism <i>Saccharomyces cerevisiae</i> . <i>Nanomaterials (Basel)</i> 2019;9(4).	doi:10.3390/nano9040512 PMID 30986968 PMC6523553	GSE125759	–	6	–	–	–	1	1
505	Pan 2008 Pan Z, Agarwal AK, Xu T, et al., Rimando AM. Identification of molecular pathways affected by pterostilbene, a natural dimethylether analog of resveratrol. <i>BMC Med Genomics</i> 2008;1:7.	doi:10.1186/1755-8794-1-7 PMID 18366703 PMC2330146	GSE10554	–	6	–	–	–	1	1
506	Papini 2010 Papini M, Nookaew I, Scalcinati G, Siewers V, Nielsen J. Phosphoglycerate mutase knock-out mutant <i>Saccharomyces cerevisiae</i> : physiological investigation and transcriptome analysis. <i>Biotechnol J</i> 2010;5(10):1016–27.	doi:10.1002/biot.201000199 PMID 20815084	GSE22122	–	6	–	–	–	1	1
507	Pelechano 2008 Pelechano V, Pérez-Ortín JE. The transcriptional inhibitor thiolutin blocks mRNA degradation in yeast. <i>Yeast</i> 2008;25(2):85–92.	doi:10.1002/yea.1548 PMID 17914747	GSE8629	–	6	–	–	–	1	1
508	Penacho 2012 Penacho V, Blondin B, Valero E, Gonzalez R. Flocculation and transcriptional adaptation to fermentation conditions in a recombinant wine yeast strain defective for KNR4/SMI1. <i>Biotechnol Prog</i> 2012;28(2):327–36.	doi:10.1002/btpr.734 PMID 22065482	GSE26965	–	6	–	–	–	1	2
509	Rodríguez-Quinones 2009 Rodríguez-Quinones JF, Rodríguez-Medina JR. Differential gene expression signatures for cell wall integrity found in chitin synthase II (chs2Delta) and myosin II (myo1Delta) deficient cytokinesis mutants of <i>Saccharomyces cerevisiae</i> . <i>BMC Res Notes</i> 2009;2:87.	doi:10.1186/1756-0500-2-87 PMID 19426543 PMC2684100	GSE5931	–	6	–	–	–	1	2
510	Shakoury-Elizeh 2004 Shakoury-Elizeh M, Tiedeman J, Rashford J, et al., Philpott CC. Transcriptional remodeling in response to iron deprivation in <i>Saccharomyces cerevisiae</i> . <i>Mol Biol Cell</i> 2004;15(3):1233–43.	doi:10.1091/mbc.e03-09-0642 PMID 14668481 PMC363115	none	–	6	–	–	–	1	2
511	Sprouse 2009 Sprouse RO, Wells MN, Auble DT. TATA-binding protein variants that bypass the requirement for Mot1 in vivo. <i>J Biol Chem</i> 2009;284(7):4525–35.	doi:10.1074/jbc.M808951200 PMID 19098311 PMC2640957	GSE12371	–	6	–	–	–	1	2
512	Stanley 2010 Stanley D, Fraser S, Stanley GA, Chambers PJ. Retrotransposon expression in ethanol-stressed <i>Saccharomyces cerevisiae</i> . <i>Appl Microbiol Biotechnol</i> 2010;87(4):1447–54.	doi:10.1007/s00253-010-2562-y PMID 20393705	GSE20590	–	6	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
513	Stefanini 2010 Stefanini I, Trabocchi A, Marchi E, Guarna A, Cavalieri D. A systems biology approach to dissection of the effects of small bicyclic peptidomimetics on a panel of <i>saccharomyces cerevisiae</i> mutants. <i>J Biol Chem</i> 2010;285(30):23477–85.	doi:10.1074/jbc.M110.125153 PMID 20501664 PMC2906338	GSE19331	–	6	–	–	–	1	2
514	Suzuki 2013 Suzuki T, Iwahashi Y. RNA preparation of <i>Saccharomyces cerevisiae</i> using the digestion method may give misleading results. <i>Appl Biochem Biotechnol</i> 2013;169(5):1620–32.	doi:10.1007/s12010-012-0051-8 PMID 23325148 PMC3593083	GSE36955	–	6	–	–	–	1	1
515	Suzuki 2015 Suzuki T, Iwahashi Y. Low toxicity of deoxynivalenol-3-glucoside in microbial cells. <i>Toxins (Basel)</i> 2015;7(1):187–200.	doi:10.3390/toxins7010187 PMID 25609182 PMC4303822	GSE63663	–	6	–	–	–	1	1
516	Tanaka 2006 Tanaka F, Ando A, Nakamura T, Takagi H, Shima J. Functional genomic analysis of commercial baker's yeast during initial stages of model dough-fermentation. <i>Food Microbiol</i> 2006;23(8):717–28.	doi:10.1016/j.fm.2006.02.003 PMID 16943074	GSE3043	–	6	–	–	–	1	1
517	Tarrant 2016 Tarrant DJ, Stirpe M, Rowe M, Howard MJ, von der Haar T, Gourlay CW. Inappropriate expression of the translation elongation factor 1A disrupts genome stability and metabolism. <i>J Cell Sci</i> 2016;129(24):4455–4465.	doi:10.1242/jcs.192831 PMID 27807005 PMC5201016	GSE89286	–	6	–	–	–	1	1
518	Tirosh 2008 Tirosh I, Weinberger A, Bezalet D, Kaganovich M, Barkai N. On the relation between promoter divergence and gene expression evolution. <i>Mol Syst Biol</i> 2008;4:159.	doi:10.1038/msb4100198 PMID 18197176 PMC2238714	GSE7525	–	6	–	–	–	1	2
519	Vemuri 2007 Vemuri GN, Eiteman MA, McEwen JE, Olsson L, Nielsen J. Increasing NADH oxidation reduces overflow metabolism in <i>Saccharomyces cerevisiae</i> . <i>Proc Natl Acad Sci U S A</i> 2007;104(7):2402–7.	doi:10.1073/pnas.0607469104 PMID 17287356 PMC1892921	GSE6277	–	6	–	–	–	1	1
520	Wyers 2005 Wyers F, Rougemaille M, Badis G, et al., Jacquier A. Cryptic pol II transcripts are degraded by a nuclear quality control pathway involving a new poly(A) polymerase. <i>Cell</i> 2005;121(5):725–37.	doi:10.1016/j.cell.2005.04.030 PMID 15935759	GSE2579	–	6	–	–	–	1	1
521	Xu 2010 Xu T, Shively CA, Jin R, et al., Kumar A. A profile of differentially abundant proteins at the yeast cell periphery during pseudohyphal growth. <i>J Biol Chem</i> 2010;285(20):15476–15488.	doi:10.1074/jbc.M110.114926 PMID 20228058 PMC2865295	GSE20351	–	6	–	–	–	1	1
522	Ye 2005 Ye J, Ai X, Eugeni EE, et al., Parthun MR. Histone H4 lysine 91 acetylation a core domain modification associated with chromatin assembly. <i>Mol Cell</i> 2005;18(1):123–30.	doi:10.1016/j.molcel.2005.02.031 PMID 15808514 PMC2855496	GSE2349	–	6	–	–	–	1	1
523	Zhu 2015 Zhu X, Chen L, Carlsten JO, et al., Gustafsson CM. Mediator tail subunits can form amyloid-like aggregates in vivo and affect stress response in yeast. <i>Nucleic Acids Res</i> 2015;43(15):7306–14.	doi:10.1093/nar/gkv629 PMID 26138482 PMC4551914	GSE65666	–	6	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
524	de Kok 2012 de Kok S, Nijkamp JF, Oud B, et al., van Maris AJ. Laboratory evolution of new lactate transporter genes in a <i>jen1Δ</i> mutant of <i>Saccharomyces cerevisiae</i> and their identification as ADY2 alleles by whole-genome resequencing and transcriptome analysis. <i>FEMS Yeast Res</i> 2012;12(3):359–374.	doi:10.1111/j.1567-1364.2012.00787.x PMID 22257278	GSE31634	–	6	–	–	–	1	1
525	Baetz 2001 Baetz K, Moffat J, Haynes J, Chang M, Andrews B. Transcriptional coregulation by the cell integrity mitogen-activated protein kinase Slt2 and the cell cycle regulator Swi4. <i>Mol Cell Biol</i> 2001;21(19):6515–28.	doi:10.1128/MCB.21.19.6515-6528.2001 PMID 11533240 PMC99798	none	–	5	5	4	–	1	2
526	Lu 2005 Lu L, Roberts GG, Oszust C, Hudson AP. The YJR127C/ZMS1 gene product is involved in glycerol-based respiratory growth of the yeast <i>Saccharomyces cerevisiae</i> . <i>Curr Genet</i> 2005;48(4):235–46.	doi:10.1007/s00294-005-0023-4 PMID 16208474	GSE29530	–	5	2	1	–	1	1
527	Aow 2013 Aow JS, Xue X, Run JQ, Lim GF, Goh WS, Clarke ND. Differential binding of the related transcription factors Pho4 and Cbf1 can tune the sensitivity of promoters to different levels of an induction signal. <i>Nucleic Acids Res</i> 2013;41(9):4877–87.	doi:10.1093/nar/gkt210 PMID 23558744 PMC3643608	GSE26770	–	5	–	–	–	1	1
528	Jansen 2004 Jansen ML, Daran-Lapujade P, de Winde JH, Piper MD, Pronk JT. Prolonged maltose-limited cultivation of <i>Saccharomyces cerevisiae</i> selects for cells with improved maltose affinity and hypersensitivity. <i>Appl Environ Microbiol</i> 2004;70(4):1956–63.	doi:10.1128/AEM.70.4.1956-1963.2004 PMID 15066785 PMC383169	GSE8897	–	5	–	–	–	1	1
529	Jones 2003 Jones DL, Petty J, Hoyle DC, et al., Stateva LI. Transcriptome profiling of a <i>Saccharomyces cerevisiae</i> mutant with a constitutively activated Ras/cAMP pathway. <i>Physiol Genomics</i> 2003;16(1):107–18.	doi:10.1152/physiolgenomics.00139.2003 PMID 14570984	none	–	5	–	–	–	1	2
530	Oberstrass 2006 Oberstrass FC, Lee A, Stefl R, Janis M, Chanfreau G, Allain FH. Shape-specific recognition in the structure of the Vts1p SAM domain with RNA. <i>Nat Struct Mol Biol</i> 2006;13(2):160–7.	doi:10.1038/nsmb1038 PMID 16429156	GSE3859	–	5	–	–	–	1	1
531	Roberts 2006 Roberts GG, Hudson AP. Transcriptome profiling of <i>Saccharomyces cerevisiae</i> during a transition from fermentative to glycerol-based respiratory growth reveals extensive metabolic and structural remodeling. <i>Mol Genet Genomics</i> 2006;276(2):170–86.	doi:10.1007/s00438-006-0133-9 PMID 16741729	none	–	5	–	–	–	1	2
532	Yasokawa 2010 Yasokawa D, Murata S, Iwahashi Y, et al., Iwahashi H. Toxicity of methanol and formaldehyde towards <i>Saccharomyces cerevisiae</i> as assessed by DNA microarray analysis. <i>Appl Biochem Biotechnol</i> 2010;160(6):1685–98.	doi:10.1007/s12010-009-8684-y PMID 19499198	GSE9231	–	5	–	–	–	1	2
533	van Attikum 2004 van Attikum H, Fritsch O, Hohn B, Gasser SM. Recruitment of the INO80 complex by H2A phosphorylation links ATP-dependent chromatin remodeling with DNA double-strand break repair. <i>Cell</i> 2004;119(6):777–88.	doi:10.1016/j.cell.2004.11.033 PMID 15607975	none	–	5	–	–	–	2	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
534	Orlandi 2004 Orlandi I, Bettiga M, Alberghina L, Vai M. Transcriptional profiling of ubp10 null mutant reveals altered subtelomeric gene expression and insurgence of oxidative stress response. <i>J Biol Chem</i> 2004;279(8):6414–25.	doi:10.1074/jbc.M306464200 PMID 14623890	none	–	4	2	2	–	1	2
535	Sen 2011 Sen P, Ghosh S, Pugh BF, Bartholomew B. A new, highly conserved domain in Swi2/Snf2 is required for SWI/SNF remodeling. <i>Nucleic Acids Res</i> 2011;39(21):9155–66.	doi:10.1093/nar/gkr622 PMID 21835776 PMC3241646	GSE30675	–	4	1	1	–	1	2
536	Arigo 2006 Arigo JT, Eyler DE, Carroll KL, Corden JL. Termination of cryptic unstable transcripts is directed by yeast RNA-binding proteins Nrd1 and Nab3. <i>Mol Cell</i> 2006;23(6):841–51.	doi:10.1016/j.molcel.2006.07.024 PMID 16973436	GSE4657	–	4	–	–	–	1	1
537	Blasco-Moreno 2019 Blasco-Moreno B, de Campos-Mata L, Böttcher R, et al., Díez J. The exonuclease Xrn1 activates transcription and translation of mRNAs encoding membrane proteins. <i>Nat Commun</i> 2019;10(1):1298.	doi:10.1038/s41467-019-09199-6 PMID 30899024 PMC6428865	GSE123326	–	4	–	–	–	1	1
538	Boccazzi 2006 Boccazzi P, Zhang Z, Kurosawa K, et al., Sinskey A.J. Differential gene expression profiles and real-time measurements of growth parameters in <i>Saccharomyces cerevisiae</i> grown in microliter-scale bioreactors equipped with internal stirring. <i>Biotechnol Prog</i> 2006;22(3):710–7.	doi:10.1021/bp0504288 PMID 16739953	GSE2605	–	4	–	–	–	1	1
539	Cai 2013 Cai Y, Fitcher B. Effects of the yeast RNA-binding protein Whi3 on the half-life and abundance of CLN3 mRNA and other targets. <i>PLoS One</i> 2013;8(12):e84630.	doi:10.1371/journal.pone.0084630 PMID 24386402 PMC3875557	GSE51784	–	4	–	–	–	1	2
540	Chattopadhyay 2022 Chattopadhyay S, Garcia-Martinez J, Haimovich G, et al., Choder M. RNA-controlled nucleocytoplasmic shuttling of mRNA decay factors regulates mRNA synthesis and a novel mRNA decay pathway. <i>Nat Commun</i> 2022;13(1):7184.	doi:10.1038/s41467-022-34417-z PMID 36418294 PMC9684461	GSE204991	–	4	–	–	–	1	1
541	Chujo 2015 Chujo M, Yoshida S, Ota A, Murata K, Kawai S. Acquisition of the ability to assimilate mannitol by <i>Saccharomyces cerevisiae</i> through dysfunction of the general corepressor Tup1-Cyc8. <i>Appl Environ Microbiol</i> 2015;81(1):9–16.	doi:10.1128/AEM.02906-14 PMID 25304510 PMC4272704	GSE55755	–	4	–	–	–	1	1
542	Crisp 2006 Crisp RJ, Adkins EM, Kimmel E, Kaplan J. Recruitment of Tup1p and Cti6p regulates heme-deficient expression of Aft1p target genes. <i>EMBO J</i> 2006;25(3):512–21.	doi:10.1038/sj.emboj.7600961 PMID 16437160 PMC1383541	GSE3470	–	4	–	–	–	1	1
543	Demae 2007 Demae M, Murata Y, Hisano M, Haitani Y, Shima J, Takagi H. Overexpression of two transcriptional factors, Kin28 and Pog1, suppresses the stress sensitivity caused by the <i>rsp5</i> mutation in <i>Saccharomyces cerevisiae</i> . <i>FEMS Microbiol Lett</i> 2007;277(1):70–8.	doi:10.1111/j.1574-6968.2007.00947.x PMID 17986087	GSE8729	–	4	–	–	–	1	1
544	Edskes 2011 Edskes HK, Engel A, McCann LM, Brachmann A, Tsai HF, Wickner RB. Prion-forming ability of Ure2 of yeasts is not evolutionarily conserved. <i>Genetics</i> 2011;188(1):81–90.	doi:10.1534/genetics.111.127217 PMID 21368275 PMC3120146	GSE26611	–	4	–	–	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
545	Ferea 1999 Ferea TL, Botstein D, Brown PO, Rosenzweig RF. Systematic changes in gene expression patterns following adaptive evolution in yeast. <i>Proc Natl Acad Sci U S A</i> 1999;96(17):9721–6.	doi:10.1073/pnas.96.17.9721 PMID 10449761 PMC22277	none	–	4	–	–	–	1	2
546	Golla 2015 Golla U, Bandi G, Tomar RS. Molecular cytotoxicity mechanisms of allyl alcohol (acrolein) in budding yeast. <i>Chem Res Toxicol</i> 2015;28(6):1246–64.	doi:10.1021/acs.chemrestox.5b00071 PMID 25919230	GSE62399	–	4	–	–	–	1	1
547	Golla 2016 Golla U, Joseph D, Tomar RS. Combined Transcriptomics and Chemical-Genetics Reveal Molecular Mode of Action of Valproic acid, an Anticancer Molecule using Budding Yeast Model. <i>Sci Rep</i> 2016;6:35322.	doi:10.1038/srep35322 PMID 27734932 PMC5062167	GSE62400	–	4	–	–	–	1	1
548	Golla 2017 Golla U, Swagatika S, Chauhan S, Tomar RS. A systematic assessment of chemical, genetic, and epigenetic factors influencing the activity of anticancer drug KP1019 (FFC14A). <i>Oncotarget</i> 2017;8(58):98426–98454.	doi:10.18632/oncotarget.21416 PMID 29228701 PMC5716741	GSE76985	–	4	–	–	–	1	1
549	Hashikawa 2006 Hashikawa N, Mizukami Y, Imazu H, Sakurai H. Mutated yeast heat shock transcription factor activates transcription independently of hyperphosphorylation. <i>J Biol Chem</i> 2006;281(7):3936–42.	doi:10.1074/jbc.M510827200 PMID 16361698	GSE3361	–	4	–	–	–	1	1
550	Hasin 2014 Hasin N, Cusack SA, Ali SS, Fitzpatrick DA, Jones GW. Global transcript and phenotypic analysis of yeast cells expressing Ssa1, Ssa2, Ssa3 or Ssa4 as sole source of cytosolic Hsp70-Ssa chaperone activity. <i>BMC Genomics</i> 2014;15(1):194.	doi:10.1186/1471-2164-15-194 PMID 24628813 PMC4022180	GSE32433	–	4	–	–	–	1	1
551	Keller 2001 Keller G, Ray E, Brown PO, Winge DR. Haa1, a protein homologous to the copper-regulated transcription factor Ace1, is a novel transcriptional activator. <i>J Biol Chem</i> 2001;276(42):38697–702.	doi:10.1074/jbc.M107131200 PMID 11504737	none	–	4	–	–	–	1	2
552	Lopez-Serra 2013 Lopez-Serra L, Lengronne A, Borges V, Kelly G, Uhlmann F. Budding yeast Wapl controls sister chromatid cohesion maintenance and chromosome condensation. <i>Curr Biol</i> 2013;23(1):64–9.	doi:10.1016/j.cub.2012.11.030 PMID 23219725	GSE42458	–	4	–	–	–	1	1
553	Menacho-Marquez 2007 Menacho-Marquez M, Perez-Valle J, Ariño J, Gadea J, Murguía JR. Gcn2p regulates a G1/S cell cycle checkpoint in response to DNA damage. <i>Cell Cycle</i> 2007;6(18):2302–5.	doi:10.4161/cc.6.18.4668 PMID 17890903	GSE8111	–	4	–	–	–	1	2
554	Miyake 2004 Miyake T, Reese J, Loch CM, Auble DT, Li R. Genome-wide analysis of ARS (autonomously replicating sequence) binding factor 1 (Abf1p)-mediated transcriptional regulation in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2004;279(33):34865–72.	doi:10.1074/jbc.M405156200 PMID 15192094	none	–	4	–	–	–	1	2
555	Nasuno 2014 Nasuno R, Aitoku M, Manago Y, Nishimura A, Sasano Y, Takagi H. Nitric oxide-mediated antioxidative mechanism in yeast through the activation of the transcription factor Mac1. <i>PLoS One</i> 2014;9(11):e113788.	doi:10.1371/journal.pone.0113788 PMID 25423296 PMC4244153	GSE61836	–	4	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
556	Ogawa 2016 Ogawa T, Tsubakiyama R, Kanai M, et al., Mizunuma M. Stimulating S-adenosyl-l-methionine synthesis extends lifespan via activation of AMPK. <i>Proc Natl Acad Sci U S A</i> 2016;113(42):11913–11918.	doi:10.1073/pnas.1604047113 PMID 27698120 PMC5081624	GSE76206	–	4	–	–	–	1	1
557	Protchenko 2001 Protchenko O, Ferea T, Rashford J, et al., Philpott CC. Three cell wall mannoproteins facilitate the uptake of iron in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2001;276(52):49244–50.	doi:10.1074/jbc.M109220200 PMID 11673473	none	–	4	–	–	–	1	2
558	Ragni 2011 Ragni E, Piberger H, Neupert C, et al., Strahl S. The genetic interaction network of CCW12, a <i>Saccharomyces cerevisiae</i> gene required for cell wall integrity during budding and formation of mating projections. <i>BMC Genomics</i> 2011;12:107.	doi:10.1186/1471-2164-12-107 PMID 21320323 PMC3049148	GSE22649	–	4	–	–	–	1	1
559	Rivera-Ruiz 2010 Rivera-Ruiz ME, Rodríguez-Quinones JF, Akamine P, Rodríguez-Medina JR. Post-transcriptional regulation in the <i>myo1Δ</i> mutant of <i>Saccharomyces cerevisiae</i> . <i>BMC Genomics</i> 2010;11:690.	doi:10.1186/1471-2164-11-690 PMID 21126371 PMC3017085	GSE20203	–	4	–	–	–	1	2
560	Romagnoli 2014 Romagnoli G, Knijnenburg TA, Liti G, Louis EJ, Pronk JT, Daran JM. Deletion of the <i>Saccharomyces cerevisiae</i> ARO8 gene, encoding an aromatic amino acid transaminase, enhances phenylethanol production from glucose. <i>Yeast</i> 2015;32(1):29–45.	doi:10.1002/yea.3015 PMID 24733517	GSE52256	–	4	–	–	–	1	1
561	Rupwate 2012 Rupwate SD, Rupwate PS, Rajasekharan R. Regulation of lipid biosynthesis by phosphatidylinositol-specific phospholipase C through the transcriptional repression of upstream activating sequence inositol containing genes. <i>FEBS Lett</i> 2012;586(10):1555–60.	doi:10.1016/j.febslet.2012.04.022 PMID 22673525	GSE35677	–	4	–	–	–	1	1
562	Shirra 2008 Shirra MK, McCartney RR, Zhang C, Shokat KM, Schmidt MC, Arndt KM. A chemical genomics study identifies Snf1 as a repressor of GCN4 translation. <i>J Biol Chem</i> 2008;283(51):35889–98.	doi:10.1074/jbc.M805325200 PMID 18955495 PMC2602918	GSE12061	–	4	–	–	–	1	1
563	Tesniere 2013 Tesnière C, Delobel P, Pradal M, Blondin B. Impact of nutrient imbalance on wine alcoholic fermentations: nitrogen excess enhances yeast cell death in lipid-limited must. <i>PLoS One</i> 2013;8(4):e61645.	doi:10.1371/journal.pone.0061645 PMID 23658613 PMC3637302	GSE42027	–	4	–	–	–	1	1
564	Thierry 2015 Thierry A, Khanna V, Créno S, et al., Dujon B. Macrotene chromosomes provide insights to a new mechanism of high-order gene amplification in eukaryotes. <i>Nat Commun</i> 2015;6:6154.	doi:10.1038/ncomms7154 PMID 25635677 PMC4317496	GSE64431	–	4	–	–	–	1	2
565	Trott 2008 Trott A, West JD, Klaić L, et al., Morano KA. Activation of heat shock and antioxidant responses by the natural product celastrol: transcriptional signatures of a thiol-targeted molecule. <i>Mol Biol Cell</i> 2008;19(3):1104–12.	doi:10.1091/mbc.e07-10-1004 PMID 18199679 PMC2262981	GSE5608	–	4	–	–	–	1	2
566	Tsang 2014 Tsang CK, Liu Y, Thomas J, Zhang Y, Zheng XF. Superoxide dismutase 1 acts as a nuclear transcription factor to regulate oxidative stress resistance. <i>Nat Commun</i> 2014;5:3446.	doi:10.1038/ncomms4446 PMID 24647101 PMC4678626	GSE55081	–	4	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
567	Wang 2014 Wang Y, Chen H, Yu O. A plant malonyl-CoA synthetase enhances lipid content and polyketide yield in yeast cells. <i>Appl Microbiol Biotechnol</i> 2014;98(12):5435–47.	doi:10.1007/s00253-014-5612-z PMID 24682482	GSE52701	–	4	–	–	–	1	2
568	Wiesenberger 2007 Wiesenberger G, Steinleitner K, Malli R, et al., Stadler JA. Mg2+ deprivation elicits rapid Ca2+ uptake and activates Ca2+/calcineurin signaling in <i>Saccharomyces cerevisiae</i> . <i>Eukaryot Cell</i> 2007;6(4):592–9.	doi:10.1128/EC.00382-06 PMID 17337637 PMC1865649	GSE6687	–	4	–	–	–	1	2
569	Wu 2008 Wu CY, Bird AJ, Chung LM, Newton MA, Winge DR, Eide DJ. Differential control of Zap1-regulated genes in response to zinc deficiency in <i>Saccharomyces cerevisiae</i> . <i>BMC Genomics</i> 2008;9:370.	doi:10.1186/1471-2164-9-370 PMID 18673560 PMC2535606	GSE11983	–	4	–	–	–	1	2
570	Yamamoto 2005 Yamamoto A, Mizukami Y, Sakurai H. Identification of a novel class of target genes and a novel type of binding sequence of heat shock transcription factor in <i>Saccharomyces cerevisiae</i> . <i>J Biol Chem</i> 2005;280(12):11911–9.	doi:10.1074/jbc.M411256200 PMID 15647283	none	–	4	–	–	–	1	2
571	Yun 2000 Yun CW, Ferea T, Rashford J, et al., Philpott CC. Desferrioxamine-mediated iron uptake in <i>Saccharomyces cerevisiae</i> . Evidence for two pathways of iron uptake. <i>J Biol Chem</i> 2000;275(14):10709–15.	doi:10.1074/jbc.275.14.10709 PMID 10744769	none	–	4	–	–	–	1	2
572	Zhu 2017 Zhu X, Zou S, Li Y, Liang Y. Transcriptomic analysis of <i>Saccharomyces cerevisiae</i> upon honokiol treatment. <i>Res Microbiol</i> 2017;168(7):626–635.	doi:10.1016/j.resmic.2017.04.007 PMID 28499955	GSE94945	–	4	–	–	–	1	1
573	Byeon 2013 Byeon B, Wang W, Barski A, et al., Wu WH. The ATP-dependent chromatin remodeling enzyme Fun30 represses transcription by sliding promoter-proximal nucleosomes. <i>J Biol Chem</i> 2013;288(32):23182–93.	doi:10.1074/jbc.M113.471979 PMID 23779104 PMC3743490	GSE48570	–	3	2	2	–	1	2
574	Kamata 2023 Kamata K, Ayano T, Oki M. Spt3 and Spt8 Are Involved in the Formation of a Silencing Boundary by Interacting with TATA-Binding Protein. <i>Biomolecules</i> 2023;13(4).	doi:10.3390/biom13040619 PMID 37189367 PMC10136053	GSE220290	–	3	2	2	–	1	1
575	Uluışik 2011 Uluışik I, Kaya A, Fomenko DE, et al., Koc A. Boron stress activates the general amino acid control mechanism and inhibits protein synthesis. <i>PLoS One</i> 2011;6(11):e27772.	doi:10.1371/journal.pone.0027772 PMID 22114689 PMC3219688	GSE29887	–	3	2	2	–	1	2
576	Levy 2008 Levy A, Eyal M, Hershkovits G, Salmon-Divon M, Klutstein M, Katcoff DJ. Yeast linker histone Hho1p is required for efficient RNA polymerase I processivity and transcriptional silencing at the ribosomal DNA. <i>Proc Natl Acad Sci U S A</i> 2008;105(33):11703–8.	doi:10.1073/pnas.0709403105 PMID 18687885 PMC2575252	GSE9095	–	3	2	1	–	1	2
577	Abe 2007 Abe F. Induction of DAN/TIR yeast cell wall mannoprotein genes in response to high hydrostatic pressure and low temperature. <i>FEBS Lett</i> 2007;581(25):4993–8.	doi:10.1016/j.febslet.2007.09.039 PMID 17910955	GSE9136	–	3	–	–	–	1	1

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
578	Burrill 2011 Burrill DR, Silver PA. Synthetic circuit identifies subpopulations with sustained memory of DNA damage. <i>Genes Dev</i> 2011;25(5):434–9.	doi:10.1101/gad.1994911 PMID 21363961 PMC3049284	GSE36210	–	3	–	–	–	1	2
579	Canadell 2015 Canadell D, González A, Casado C, Ariño J. Functional interactions between potassium and phosphate homeostasis in <i>Saccharomyces cerevisiae</i> . <i>Mol Microbiol</i> 2015;95(3):555–72.	doi:10.1111/mmi.12886 PMID 25425491	GSE58985	–	3	–	–	–	1	2
580	Gregori 2008 Gregori C, Schüller C, Frohner IE, Ammerer G, Kuchler K. Weak organic acids trigger conformational changes of the yeast transcription factor War1 in vivo to elicit stress adaptation. <i>J Biol Chem</i> 2008;283(37):25752–25764.	doi:10.1074/jbc.M803095200 PMID 18621731	GSE8399	–	3	–	–	–	1	2
581	Kamei 2014 Kamei Y, Tamada Y, Nakayama Y, Fukusaki E, Mukai Y. Changes in transcription and metabolism during the early stage of replicative cellular senescence in budding yeast. <i>J Biol Chem</i> 2014;289(46):32081–32093.	doi:10.1074/jbc.M114.600528 PMID 25294875 PMC4231685	GSE59797	–	3	–	–	–	1	2
582	Kuhn 2001 Kuhn KM, DeRisi JL, Brown PO, Sarnow P. Global and specific translational regulation in the genomic response of <i>Saccharomyces cerevisiae</i> to a rapid transfer from a fermentable to a nonfermentable carbon source. <i>Mol Cell Biol</i> 2001;21(3):916–27.	doi:10.1128/MCB.21.3.916-927. 2001 PMID 11154278 PMC86682	none	–	3	–	–	–	1	2
583	Mizuguchi 2004 Mizuguchi G, Shen X, Landry J, Wu WH, Sen S, Wu C. ATP-driven exchange of histone H2AZ variant catalyzed by SWR1 chromatin remodeling complex. <i>Science</i> 2004;303(5656):343–8.	doi:10.1126/science.1090701 PMID 14645854	none	–	3	–	–	–	1	2
584	Nadal-Ribelles 2012 Nadal-Ribelles M, Conde N, Flores O, et al., Posas F. Hog1 bypasses stress-mediated down-regulation of transcription by RNA polymerase II redistribution and chromatin remodeling. <i>Genome Biol</i> 2012;13(11):R106.	doi:10.1186/gb-2012-13-11-r106 PMID 23158682 PMC3580498	GSE41451	–	3	–	–	–	1	2
585	Orozco 2012 Orozco H, Matallana E, Aranda A. Oxidative stress tolerance, adenylate cyclase, and autophagy are key players in the chronological life span of <i>Saccharomyces cerevisiae</i> during winemaking. <i>Appl Environ Microbiol</i> 2012;78(8):2748–57.	doi:10.1128/AEM.07261-11 PMID 22327582 PMC3318821	GSE33068	–	3	–	–	–	1	2
586	Prunuske 2012 Prunuske AJ, Waltner JK, Kuhn P, Gu B, Craig EA. Role for the molecular chaperones Zuo1 and Ssz1 in quorum sensing via activation of the transcription factor Pdr1. <i>Proc Natl Acad Sci U S A</i> 2012;109(2):472–7.	doi:10.1073/pnas.1119184109 PMID 22203981 PMC3258599	GSE31693	–	3	–	–	–	1	1
587	Yadav 2016 Yadav KK, Rajasekharan R. Microarray data analyses of yeast RNA Pol I subunit RPA12 deletion strain. <i>Genom Data</i> 2016;8:104–5.	doi:10.1016/j.gdata.2016.04.011 PMID 27222810 PMC4872367	GSE68731	–	3	–	–	–	1	1
588	Costelloe 2012 Costelloe T, Louge R, Tomimatsu N, et al., Llorente B. The yeast Fun30 and human SMARCA1 chromatin remodellers promote DNA end resection. <i>Nature</i> 2012;489(7417):581–4.	doi:10.1038/nature11353 PMID 22960744 PMC3493121	GSE38735	–	2	2	1	–	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
589	Radman-Livaja 2012 Radman-Livaja M, Quan TK, Valenzuela L, et al., Hartzog GA. A key role for Chd1 in histone H3 dynamics at the 3' ends of long genes in yeast. <i>PLoS Genet</i> 2012;8(7):e1002811.	doi:10.1371/journal.pgen.1002811 PMID 22807688 PMC3395613	GSE38496	—	2	2	1	—	1	2
590	Cullen 2004 Cullen PJ, Sabbagh W Jr, Graham E, et al., Sprague GF Jr. A signaling mucin at the head of the Cdc42- and MAPK-dependent filamentous growth pathway in yeast. <i>Genes Dev</i> 2004;18(14):1695–708.	doi:10.1101/gad.1178604 PMID 15256499 PMC478191	none	—	2	—	—	—	1	2
591	Hoffman 2015 Hoffman EA, McCulley A, Haarer B, Arnak R, Feng W. Break-seq reveals hydroxyurea-induced chromosome fragility as a result of unscheduled conflict between DNA replication and transcription. <i>Genome Res</i> 2015;25(3):402–12.	doi:10.1101/gr.180497.114 PMID 25609572 PMC4352882	GSE64446	—	2	—	—	—	1	1
592	Licursi 2014 Licursi V, Salvi C, De Cesare V, et al., Negri R. The COP9 signalosome is involved in the regulation of lipid metabolism and of transition metals uptake in <i>Saccharomyces cerevisiae</i> . <i>FEBS J</i> 2014;281(1):175–90.	doi:10.1111/febs.12584 PMID 24164706	GSE51563	—	2	—	—	—	1	2
593	Merrill 2014 Merrill AE, Hebert AS, MacGilvray ME, et al., Coon JJ. NeuCode labels for relative protein quantification. <i>Mol Cell Proteomics</i> 2014;13(9):2503–12.	doi:10.1074/mcp.M114.040287 PMID 24938287 PMC4159665	GSE53549	—	2	—	—	—	1	2
594	Noble 2015 Noble J, Sanchez I, Blondin B. Identification of new <i>Saccharomyces cerevisiae</i> variants of the MET2 and SKP2 genes controlling the sulfur assimilation pathway and the production of undesirable sulfur compounds during alcoholic fermentation. <i>Microb Cell Fact</i> 2015;14:68.	doi:10.1186/s12934-015-0245-1 PMID 25947166 PMC4432976	GSE55083	—	2	—	—	—	1	2
595	Ochi 2012 Ochi K, Nishizawa T, Inaoka T, et al., Ozeki Y. Heterologous expression of a plant RelA-SpoT homologue results in increased stress tolerance in <i>Saccharomyces cerevisiae</i> by accumulation of the bacterial alarmone ppGpp. <i>Microbiology (Reading)</i> 2012;158(Pt 8):2213–2224.	doi:10.1099/mic.0.057638-0 PMID 22679107	GSE38239	—	2	—	—	—	1	1
596	Takahashi 2006 Takahashi H, McCaffery JM, Irizarry RA, Boeke JD. Nucleocytosolic acetyl-coenzyme a synthetase is required for histone acetylation and global transcription. <i>Mol Cell</i> 2006;23(2):207–17.	doi:10.1016/j.molcel.2006.05.040 PMID 16857587	GSE2806	—	2	—	—	—	1	1
597	Wuttke 2012 Wuttke D, Connor R, Vora C, et al., de Magalhães JP. Dissecting the gene network of dietary restriction to identify evolutionarily conserved pathways and new functional genes. <i>PLoS Genet</i> 2012;8(8):e1002834.	doi:10.1371/journal.pgen.1002834 PMID 22912585 PMC3415404	GSE38635	—	2	—	—	—	1	1
598	Yeh 2017 Yeh CS, Chang SL, Chen JH, et al., Chang TH. The conserved AU dinucleotide at the 5' end of nascent U1 snRNA is optimized for the interaction with nuclear cap-binding-complex. <i>Nucleic Acids Res</i> 2017;45(16):9679–9693.	doi:10.1093/nar/gkx608 PMID 28934473 PMC5766165	GSE97015	—	2	—	—	—	1	2
599	Davis 2006 Davis CA, Ares M Jr. Accumulation of unstable promoter-associated transcripts upon loss of the nuclear exosome subunit Rrp6p in <i>Saccharomyces cerevisiae</i> . <i>Proc Natl Acad Sci U S A</i> 2006;103(9):3262–7.	doi:10.1073/pnas.0507783103 PMID 16484372 PMC1413877	GSE3813	—	1	1	1	—	1	2

Table 5, continued

#	Publication	Links	GEO	new	cond.	KO cond.	KO genes	named	PCL	ch.
600	Chang 2013 Chang M, Kang HJ, Baek IJ, Kang CM, Park YS, Yun CW. Kdx1 regulates RCK1 gene expression by interacting with Rlm1 in <i>Saccharomyces cerevisiae</i> . <i>Biochem Biophys Res Commun</i> 2013;435(3):350–5.	doi:10.1016/j.bbrc.2013.04.083 PMID 23660188	GSE45692	–	1	–	–	–	1	2
601	Chang 2013 Chang M, Kang HJ, Baek IJ, Kang CM, Park YS, Yun CW. Rck1 up-regulates Hog1 activity by down-regulating Slt2 activity in <i>Saccharomyces cerevisiae</i> . <i>Biochem Biophys Res Commun</i> 2013;440(1):119–24.	doi:10.1016/j.bbrc.2013.09.045 PMID 24051094	GSE49340	–	1	–	–	–	1	2
602	Tai 2017 Tai A, Kamei Y, Mukai Y. The forkhead-like transcription factor (Fhl1p) maintains yeast replicative lifespan by regulating ribonucleotide reductase 1 (RNR1) gene transcription. <i>Biochem Biophys Res Commun</i> 2017;488(1):218–223.	doi:10.1016/j.bbrc.2017.05.038 PMID 28495531	GSE95591	–	1	–	–	–	1	2
603	Yamane-Sando 2014 Yamane-Sando Y, Shimobayashi E, Shimobayashi M, Kozutsumi Y, Oka S, Takematsu H. Fpk1/2 kinases regulate cellular sphingoid long-chain base abundance and alter cellular resistance to LCB elevation or depletion. <i>Microbiologyopen</i> 2014;3(2):196–212.	doi:10.1002/mbo3.160 PMID 24510621 PMC3996568	GSE42083	–	1	–	–	–	1	1